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Design of a Memory System for Sparse Data Processing

Présentée par :

Valentin ISAAC--CHASSANDE

Direction de thèse :

Frédéric ROUSSEAU PROFESSEUR DES UNIVERSITES, Grenoble INP - UGA	Directeur de thèse
Adrian EVANS CEA	Co-encadrant de thèse

Rapporteurs :

Thèse soutenue publiquement le , devant le jury composé de :

Valentin ISAAC--CHASSANDE
Université Grenoble Alpes
CEA Grenoble (DRT/LIST/DSCIN/LSTA)
TIMA Laboratory (SLS)
ORCID : 0009-0007-9842-5777
valentin.isaacchassande@univ-grenoble-alpes.fr
valentin.isaacchassande@cea.fr

THESIS

DESIGN OF A MEMORY SUB-SYSTEM FOR SPARSE DATA PROCESSING

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Thesis Adviser: Pr. Frédéric ROUSSEAU
Thesis Co-Supervisors: Dr. Adrian EVANS
Dr. Yves DURAND

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Acknowledgements

TODO

"Work. Or school. This is for the money. But what is anything else, for that matter, but required means of enhancing your bare living – better sleep and better food – like spending decades just to turn a little knob higher on a pathetic radio called existence."

Karl Kristian Flores, The Goodbye Song.

Résumé

TODO

Abstract

TODO

List of Publications

This thesis is based on the following publications:

I Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. Dedicated Hardware Accelerators for Processing of Sparse Matrices and Vectors: A Survey. *ACM Trans. Archit. Code Optim.* 21, 2, Article 27 (Feb. 2024), 26 pages. Reference [31]

II Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. SpDCache: Region-Based Reduction Cache for Outer-Product Sparse Matrix Kernels. In *2024 IEEE 35th International Conference on Application-specific Systems, Architectures and Processors (ASAP)*. IEEE Computer Society, Los Alamitos, CA, USA, 3–7. Reference [32]

III **TODO**

IV **add patent?**

The papers will be referred to in the thesis using their Roman numerals.

Summary

Notations

$$\forall (a,b) \in \mathbb{Z}^2, [\![a,b]\!] = [a,b] \cap \mathbb{Z}$$

nnz, M, A, b and c, $\overline{nnz_r}$, d_r ...

TODO

Glossary

band Refers to an area of a matrix that is concentrated around the diagonal. 9, 10, 17, 18, 38, 39, 41, 42, 43, 46, 47, 48, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 74, 76, 83, 90, 92, 93, 106

banded Refers to a matrix structure where the non-zero elements are concentrated around the diagonal. 17, 18, 19, 28, 31, 33, 34, 38, 39, 41, 46, 47, 48, 50, 53, 55, 58, 60, 62, 74, 77, 83, 84, 106

bandwidth With this typography, refers to the size of the band of a matrix. 17

common rank Refers to a rank that is common between two data-entries, when computing a kernel. 20, 21, 25, 28

dense format Is the uncompressed format where all zeros and non-zeros are stored, without any use of metadata. 18, 25, 27

eviction Is a cache event where a valid cache entry is flushed and freed. 35, 36, 37, 38, 39, 40, 41, 42, 44, 45, 46, 49, 50, 52, 53, 54, 55, 56, 58, 59, 62, 66, 67, 68, 70, 71, 72, 80, 81, 83, 91, 92, 93, 106

fiber Is the generic term to describe a one-dimensional stream of data or metadata. It corresponds to a row or column in the case of a two-dimensional matrix. A stream of indices is a *fiber* of metadata. 24, 28

gather Refers to reading data from memory. We specifically use this term to categorize data transfers that are irregular. 26, 31

hit Is a cache event where a given address does match one of the valid cache entries. 35, 36, 37, 40, 41, 52, 58, 66, 67, 68, 70, 71, 72, 77, 79, 81, 83, 91, 92, 93

ineffectual operation Is an operation that leads to a neutral result, 0 for an addition or 1 for a multiplication. 24

intersection Refers to the process of finding non-zero partial sums, when computing a matrix multiplication. 20, 21, 23, 24, 25, 26, 28, 30, 31, 109

irregular access See “random” access. 10, 15, 16, 31, 34, 35, 48, 49, 79

irregular data Refers to data that has low spatial and temporal locality, thus being more likely to generate irregular accesses. 31, 36

metadata Is data that provides information about other data, such as index arrays of sparse formats. 9, 10, 18, 19, 24, 26, 99

miss Is a cache event where a given address does not match any of the valid cache entries. 25, 30, 35, 36, 37, 40, 41, 52, 66, 67, 70, 71, 72, 77, 91

partial sum Refers to the intermediary result before updating the output, when computing a matrix multiplication. 9, 21, 24, 25, 26

'perfectly' banded Refers to a matrix structure where all the non-zero elements are concentrated within a delimitated area around the diagonal (the band). 17, 48, 51, 53, 55, 59, 83, 84

pipeline Is a sequence of processing stages where instructions flow through each stage in order. 64, 66, 67, 69, 70, 71, 72, 86, 87, 89, 90, 106

"random" access Is a memory access that occurs on an arbitrary memory address (not necessarily conscutive in memory with previous and next accesses). 9, 10, 16

"random" update Is a "random" access where the access is a read, modify and write on a given address. It is similar to a "random" *reduction*. 25

rank Refers to a dimension of the kernel. 9, 18, 20, 21, 22, 23, 26

reduction Refers to the process of updating an entry. 10, 21, 25, 26, 27, 30, 31, 33, 34, 35, 36, 37, 39, 40, 41, 43, 45, 46, 57, 64, 65, 66, 67, 69, 72, 73, 77, 81, 83, 86, 87, 89, 90, 91, 92, 93, 106, 109

reduction cache Is a data-cache supporting *reduction* operations. 36, 37, 38, 39, 41, 42, 43, 44, 46, 47, 48, 49, 50, 52, 53, 55, 56, 57, 60, 62, 67, 68, 69, 72, 73, 86

scatter Refers to updating data in memory. We specifically use this term to categorize data transfers that are irregular. 26, 31

sequential accesses Are a group of memory accesses (from a data array, for example) that occur on conscutive memory addresses, in order through time. 18, 35

set Refers to a fixed number of cache-lines, a subdivision related to set-associative caches. 10, 35, 37, 39, 42, 44, 45, 48, 49, 51, 52, 58, 67, 68, 69, 73, 85, 89

sparse format Is a compression format for sparse matrices, which avoids storing zeros by using metadata. 10, 18, 19, 22, 23, 24, 25, 31, 37, 99

tag Refers to cache metadata that allows identification of cache-line memory addresses. 35, 42

tile Refers to a sub-region of a *tiling* process. 22, 23, 30

tiling Is a method consisting in splitting data into several independent sub-regions to be processed separately. 10, 22, 23, 26, 28, 30, 31, 99, 106, 109

way Refers to the number of cache-lines that a *set* contains. 35, 37, 42, 48, 58

Acronyms

BaCSC *Banded Compressed Sparse Column*. 19, 31, 74, 76, 79, 80, 81, 82, 83, 93, 106, 110

BCSR *Blocked Compressed Sparse Row*. 19

CAM Content Addressable Memory. 68, 70, 71, 72

CMO Cache Management Operation. 67, 69

COO *COOrdinate*. 18, 19, 27, 31, 99, 106

CSC *Compressed Sparse Column*. 18, 19, 24, 25, 27, 28, 31, 37, 49, 74, 76, 77, 79, 80, 81, 82, 90, 91, 93, 99, 106, 109

CSR *Compressed Sparse Row*. 18, 19, 23, 24, 25, 27, 28, 31, 99, 106, 107, 109

DIA *DIAgonal*. 19

DRC *Dense Region Cache*. 39, 40, 41, 42, 43, 44, 45, 46, 53, 55, 58, 59, 60, 62, 64, 67, 68, 69, 70, 71, 72, 73, 74, 77, 79, 81, 83, 85, 86, 90, 91, 92, 93, 106

ELL *ELLPACK*. 19

GC Generic Cache. 42, 43, 44, 67, 73, 74, 76, 77, 79, 80, 81, 82, 83, 84, 85, 89, 106, 107

GRC *Generic Region Cache*. 39, 41, 42, 44, 45, 46, 49, 62, 64, 68, 69, 73, 85, 86, 91, 92, 106

`Gust` *Gustavson*. 20, 21, 22, 23, 26, 28, 29, 30, 31, 108, 109

HYB *HYBbrid*. 19

IBR *In-Band Region*. 39, 40, 41, 46, 69, 70, 71, 74, 90, 106

`IP` *Inner-Product*. 20, 21, 22, 23, 24, 25, 26, 28, 29, 30, 31, 51, 108, 109

L1 First-Level Cache. 91, 92, 93

L2 Second-Level Cache. 91, 92, 93

LLC Last-Level Cache. 91, 92, 93

LRU *Least Recently Used*. 35, 42, 49, 59, 60

MM Matrix-Matrix. 19, 20, 21, 22, 28, 31, 109

MV Matrix-Vector. 19, 20, 21, 22, 23, 28, 31, 109

nop *no-operation*. 66

OBR *Out-of-Band Region*. 39, 40, 41, 46, 69, 71, 72, 74, 90, 106

OP *Outer-Product*. 20, 21, 22, 23, 25, 26, 27, 28, 29, 30, 31, 33, 34, 37, 38, 39, 41, 43, 44, 46, 47, 49, 50, 51, 53, 55, 60, 62, 74, 75, 76, 77, 79, 81, 90, 106, 108, 109, 110

PO *partial output*. 21, 22, 23, 25, 26, 28, 31, 44, 93

RAM Random-Access Memory. 66, 67, 68, 69, 70, 71, 72

RED *Reduction*. 36, 37, 39, 40, 41, 42, 43, 44, 45, 67, 69, 70, 71, 72, 74, 75, 76, 80, 90, 91, 110

RedC Reduction Cache. 42, 43, 44, 45, 67, 73, 74, 76, 77, 79, 80, 81, 82, 83, 84, 85, 89, 90, 107

REG *Region*. 69, 74, 76, 90

RMW *Read-Modify-Write*. 40, 41, 42, 43, 57, 58, 68, 69, 72, 89, 93

RMWQ *Read-Modify-Write Queue*. 40, 41, 68, 72

RTL Register Transfer Language. 46, 64, 65, 67, 69, 72, 73, 79, 80, 83, 84, 85, 86, 87, 89, 106, 107, 108

SpDC SpDCache. 34, 38, 39, 40, 41, 42, 43, 44, 45, 46, 48, 49, 53, 55, 57, 59, 60, 62, 64, 65, 66, 67, 68, 69, 72, 73, 74, 76, 77, 79, 80, 81, 82, 83, 84, 85, 86, 87, 89, 90, 91, 92, 93, 106, 107

SpGEMM General Sparse Matrix-Matrix multiplication. 19

SpGEMV General Sparse Matrix-Vector multiplication. 19

SpMM Sparse Matrix-Matrix multiplication. 19

SpMSpM Sparse Matrix-Sparse Matrix multiplication. 19, 20, 22, 24, 25, 26, 27, 28, 29, 30, 31, 108, 109

SpMSpV Sparse Matrix-Sparse Vector multiplication. 19

SpMV Sparse Matrix-Vector multiplication. 19, 20, 23, 30, 31, 33, 34, 37, 38, 39, 41, 42, 43, 44, 46, 47, 49, 50, 51, 53, 55, 60, 62, 74, 75, 76, 77, 79, 81, 89, 90, 106, 109, 110

SRT *Sparse Region Table*. 39, 40, 41, 42, 43, 44, 45, 46, 60, 62, 64, 68, 69, 70, 71, 72, 73, 77, 79, 83, 85, 86, 91, 92, 93, 106

Introduction

THE rapid growth of data-intensive applications has made sparse and irregular data structures increasingly prevalent in scientific computing, machine learning, and graph analytics. However, efficiently processing sparse matrices remains a fundamental challenge due to their inherent irregularity, which leads to unpredictable memory access patterns, limited opportunities for data reuse, and suboptimal utilization of hardware resources. Existing hardware accelerators have demonstrated partial solutions, yet they often rely on restrictive assumptions, exhibit limited scalability, or fail to fully exploit the memory bandwidth.

This thesis addresses the following central question:

PROBLEM STATEMENT

How can hardware and architectural mechanisms be designed to efficiently support computation with irregular sparse data, in particular random reductions, while ensuring scalability, adaptability, and high performance across diverse workloads?

To tackle this problem, we investigate the limitations of existing algorithms and accelerators, propose a novel cache architecture with support for reduction operations, tailored to sparse data. We then evaluate the behaviour of this specialized cache through modeling, simulation, and hardware prototyping.

This manuscript is structured around several key contributions. We begin with an in-depth discussion of irregular data, with a particular emphasis on sparse matrices (Chapter 1). This includes a study of fundamental sparse matrix multiplication kernels and their associated algorithms, highlighting the main hardware challenges that arise when dealing with sparse data. A high-level analysis of algorithms for sparse data computing is provided, with a focus on basic storage schemes and opportunities for data reuse. Building on this foundation, we review the state-of-the-art in hardware accelerators designed for efficient sparse matrix computing (Chapter 2). This review identifies common characteristics, proposes a classification of existing accelerators and design tendencies, offers a comparative analysis, and identifies the major tendencies across designs. This analysis of the state of the art guided our subsequent architectural choices.

Following this survey, we introduce a novel cache architecture specifically designed to address the challenge of "random" reductions (Chapter 3). The proposed cache integrates reduction support and organizes its memory into multiple regions optimized for denser and

sparser data regions. A Python model was developed to simulate memory transactions in this cache and to compare its behavior with that of a generic cache. To further investigate its performance, we complement this analysis with a probabilistic model, implemented in C, that enables rapid simulation and sizing parameters exploration depending on matrices structure (Chapter 4). At the micro-architectural level, we detail the virtual partitioning of cache memory into multiple configurable regions, which ensures both flexibility and adaptability to diverse workloads (Chapter **??**).

The proposed cache is then evaluated through RTL simulation, which reveals two distinct classes of matrices that exhibit different performance profiles (Chapter **??**). Finally, the manuscript outlines potential future enhancements to improve both performance and scalability, including a multicore extension of the cache architecture and optimizations targeting bandwidth efficiency (Chapter 6).

Contributions of the Thesis

The main contributions of this thesis can be summarized as follows:

1. **Analysis of irregular and sparse data computing** (Chapter 1)
   - Study of fundamental sparse matrix multiplication kernels and algorithms.
   - Identification of the key hardware challenges associated with sparse data processing.
   - High-level analysis of algorithms for sparse data computing, with emphasis on storage schemes and data reuse opportunities.
2. **Survey and classification of state-of-the-art accelerators** (Chapter 2)
   - Comprehensive review of hardware accelerators for sparse matrix computing.
   - Identification of common architectural features and design tendencies.
   - Comparative analysis to explain design choices under varying objectives and constraints.
3. **Design of a cache architecture for random reductions** (Chapter 3)
   - Proposal of a cache with native reduction support, capable of handling both dense and sparse data via multiple cache regions.
   - Development of a Python-based simulation model to analyze memory transactions and benchmark against a generic cache.
4. **Probabilistic modeling and exploration of cache behavior** (Chapter 4)
   - Implementation of a lightweight C-based probabilistic model for rapid simulation.
   - Exploration of cache dimensioning and behavior across diverse workloads.
5. **Micro-architecture of the reduction cache** (Chapter **??**)
   - Introduction of a virtual memory partitioning scheme enabling multiple configurable regions.
   - Emphasis on flexibility and adaptability in cache design.
6. **Evaluation through RTL simulation** (Chapter **??**)
   - Assessment of the proposed cache architecture using RTL-level simulation.
   - Identification of two distinct groups of sparse matrices leading to different performance profiles.
7. **Future directions for performance and scalability** (Chapter 6)
   - Proposal of a multicore version of the cache to extend scalability.
   - Exploration of bandwidth optimization strategies.

Chapter 1

Background

Contents

1.1 Introduction

MODERN computing systems are designed to deliver high performance by exploiting memory hierarchies, parallelism, and increasingly sophisticated hardware features. However, the efficiency of these systems is strongly tied to the regularity of memory access patterns. Applications with predictable, sequential data access can fully benefit from cache locality, prefetching mechanisms, and vectorization. In contrast, applications characterized by irregular accesses, where memory locations are not known in advance or follow non-contiguous patterns, are not able to achieve peak performance.

In domains such as graph analytics, sparse linear algebra, unstructured mesh computations, and database queries, the memory access patterns are often irregular. In these applications, pointer chasing, indirect indexing, and dynamically changing data structures disrupt spatial and temporal locality. As a result, the hardware struggles to hide the latency of memory accesses, leading to poor cache utilization and underutilization of compute ressources.

The impact of irregular accesses extends beyond raw performance. They also complicate algorithm design, hinder scalability on many-core architectures, and increase energy consumption due to repeated off-chip memory requests. Understanding and addressing these

challenges is therefore a central concern in both computer design and application optimization. This chapter provides the necessary background for analyzing irregular accesses, highlighting their sources, effects, and the architectural considerations they expose, focusing on the case of indirect indexing in sparse matrix computation. The content of this chapter is partially based on the content from Papers I and II.

DEFINITION

Irregular accesses, or **"random" accesses**, are sequences of accesses where the sequence can not be easily predicted.

1.2 Sparse Matrices

SPARSE matrices are two dimensional arrays filled with many zeros and stored in compressed formats in memory. These sparse matrices are commonly used in diverse fields such as linear algebra, where sparse matrix multiplication is used to solve linear systems of equations for scientific purposes or simulation, as well as graph analytics where the vertices of a graph are represented as a sparse matrix [11], or finally transformers or other similar AI features where weights and activations can be represented and processed as sparse matrices [18, 21]. In these contexts, matrices are large, with dimensions up to billions of elements, and highly sparse, with non-zero densities down to 10^{-7} [38].

We note matrices with capital letters A, B, C, conversely to vectors a, b, c. Their dimension are denoted using M, N and K. For example, we can define a square matrix A of dimension M, or a matrix B of dimension (K, N). We note nnz the number of non-zero elements of a matrix, and d the associated density. With the example of a matrix $A(M, N)$:

$$d_A = \frac{nnz_A}{M \times N}$$

DEFINITION

A **sparse matrix** is a matrix filled with many zeros.

NOTE

In general, we consider a matrix to be highly sparse when its number of non-zero elements has the same order of magnitude than its dimension, $nnz \sim M$. But there is no intrinsic definition of the frontier between a sparse and a dense matrix. In this manuscript, we consider a matrix as sparse when it needs to be compressed in memory (see Section 1.3).

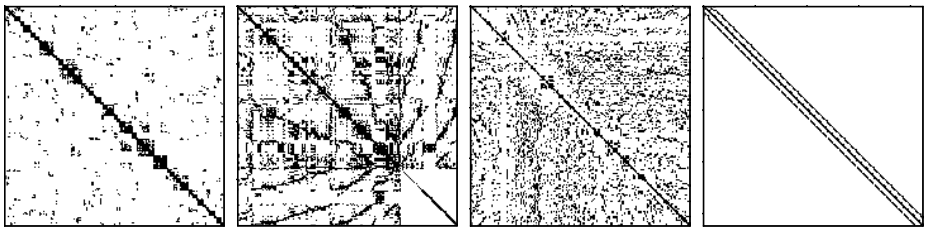
1.2.1 Banded Matrices

Matrices used in scientific computation originate from physical modelling where a system of partial differential equations is discretized, resulting in a large sparse matrix. In these systems, the forces between elements decreases quickly with their distance, creating banded matrices. Moreover, numerical techniques exist to transform matrices into banded matrices [41, 42, 58].

A banded matrix is a matrix which has a dense region around its main diagonal and which is sparser away from the diagonal. To give a few visual examples, Figure 1.1 shows the structure plots for four typical matrices found in the SuiteSparse Matrix Collection [38], having a high-density band along the diagonal. In Figure 1.2, we illustrate a banded, square matrix of dimension M. We define w_B as the width of the banded region, or *bandwidth*, as shown in *red* in the figure. Sometimes it is easier to refer to the half-*bandwidth*, w_{hB}, which respects the relation $w_B = 2 \times w_{hB} - 1$. The average density in the band and out of the band is defined as d_{IB} and d_{OB}, respectively. As in Figure 1.1d, in a 'perfectly' banded matrix, there are no elements outside the band, thus $d_{OB} = 0$.

DEFINITION

A **banded matrix** is a sparse matrix which has a higher density of non-zero elements around the diagonal. A **'perfectly' banded matrix** concentrates all its non-zero elements within a



(a) *roadNet-CA* **(b)** *scircuit* **(c)** *msc10848* **(d)** *raefsky3*

Figure 1.1: Examples of Banded Matrix Structure

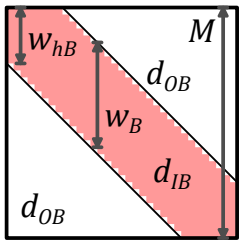


Figure 1.2: Banded Matrix Definition

delimitated region around the diagonal. This region is called the band.

NOTE

The SuiteSparse Matrix Collection [38] is a large, open source data base of sparse matrices from multiple real-world applications, many of which were provided by companies such as Boeing. It gathers around 3,000 matrices of various sizes, densities and structure, which are extensively used for benchmarking (see Chapter 2).

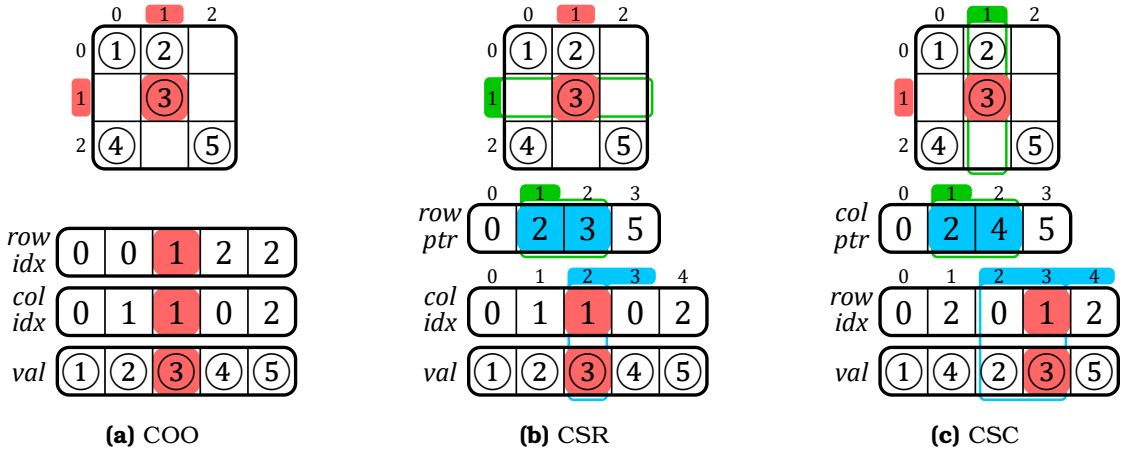
1.3 Sparse Formats

In the absence of a compressed format, matrices are stored in fixed-size two-dimensional arrays. With this dense format, elements can be randomly accessed ($\mathcal{O}(1)$), and sequential accesses are highly efficient. The two dimensions are called ranks which correspond to the rows and columns. For matrices, there are thus two variants of the dense format, depending on which rank is stored sequentially in memory. In a *row-wise* (or *row-major*) storage, the elements within the rows are sequential, and conversely for *column-wise* (or *column-major*).

Sparse matrices contain a very large number of zeros, and to avoid storing these repeated zeros, there exist many compressed formats [6, 7, 14, 35, 49, 55], where certain formats are best-suited for matrices with a specific structure. The main principle of these sparse formats is to avoid storing the zero elements, but this requires additional metadata which gives the position of the non-zero elements.

For example, let us consider the commonly used *COOrdinate* (COO) format that is relatively simple. It consists of three tables of size nnz with only the non-zero elements (val table in Figure 1.3a) and their row and column indices (row and col idx tables). The tuples do not necessarily need to be sorted, which makes it easy to append new entries. However, if the tuples are not sorted, this format is not well suited for sequential traversal or locating a specific matrix entry. This format can be sorted *row-wise* as in the example in Figure 1.3a or *column-wise* if needed, but in this case, it is less efficient than the two other widely used sparse formats *Compressed Sparse Row* (CSR) (Figure 1.3b) and its *column-major* counterpart *Compressed Sparse Column* (CSC) (Figure 1.3c), which are sorted by construction. With CSR, the val table of size nnz stores the non-zero elements sorted *row-wise* and the idx table of size nnz stores the corresponding column indices, similarly to COO. The third table ptr of size $M + 1$ stores the positions of the rows. For example, in Figure 1.3b, the second row of index 1 (*green*) has one non-zero element between positions 2 and 3 (*blue*, 3 excluded) of tables idx and val. The CSC format works similarly but in a *column-wise* manner: the val table of size nnz is sorted *column-wise* and the idx table of size nnz stores column indices. The ptr table

of size $N+1$ stores the positions of the columns. In the example in Figure 1.3c, the second column of index 1 (*green*) has non-zero elements between positions 2 and 4 (*blue*, 4 excluded) of tables idx and val. The ptr table allows CSR and CSC to have a lower memory footprint than COO.



(a) COO **(b)** CSR **(c)** CSC

Figure 1.3: Example of a Sparse Matrix in COO, CSR and CSC Formats

Many others classes of sparse formats exists such as *Blocked Compressed Sparse Row* (BCSR), *DIAgonal* (DIA), *ELLPACK* (ELL) and *HYBbrid* (HYB), which all have their own advantages depending on the matrix structure or the hardware setup. Additional information about these other formats is provided in Appendix A. In general, there are many derivates from the above main categories of sparse formats. Ultimately, we can create any format we want depending on the software and hardware constraints. Later in this manuscript, we propose our own custom sparse format called *Banded Compressed Sparse Column* (BaCSC) that is an extension of CSC for banded matrices.

DEFINITION

A **sparse format** is a memory representation of a sparse matrix which avoids storing zeros by using metadata.

NOTE

The most commonly used sparse formats are COO, CSR and CSC.

CONCLUSION

With the use of sparse formats, it is possible to greatly **decrease the memory footprint** of a sparse matrix while increasing the data dimensions. These formats **can be tailored** based on the needs of the software and hardware.

1.4 Algorithms for Dense and Sparse Matrix Multiplication

THE main matrix multiplication kernels that are of interest are Matrix-Vector (MV) and Matrix-Matrix (MM), which include SpMV and SpMSpV ($A \times b = c$), SpGEMV ($A \times b + d = c$), SpMM and SpMSpM ($A \times B = C$), and SpGEMM ($A \times B + D = C$), with 'Sp' for *sparse* and 'GE' for *general*. SpMV, as well as its variant SpGEMV, are by far the dominant operations in modern algebraic kernels (linear solvers, eigensolvers) which have been explicitly based on them [16,

23, 26, 33, 37, 46, 64]. Since these kernels are ubiquitous, their efficient implementation is primordial. In contrast, the SpMSpM multiplication is less frequent in standard algebraic kernels, but it is extensively used in algebraic graph manipulation [3, 10, 13, 15, 22, 34, 47], as well as in multigrid solvers [2, 9, 40, 48]. The growing importance of large graphs has motivated new research on optimizing SpMSpM applications.

In the following sections, we focus on SpMV and SpMSpM, with the notation $A(M,K) \times b(K) = c(M)$ and $A(M,K) \times B(K,N) = C(M,N)$, respectively. The rank K is the common rank between the two inputs A and B (or b). The existence of different ranks allows several algorithmic possibilities to compute matrix multiplication: the *Inner-Product* (**IP**) algorithm, the *Outer-Product* (**OP**) algorithm and *Gustavson*'s (**Gust**) algorithm [17, 25]. In all cases, the matrix A is traversed sequentially. The three algorithms, however, impose different access patterns for B (or b) and C (or c). Indeed, in many cases, they have to be accessed multiple times, as discussed in the following sections.

NOTE

In this manuscript, we focus on only two kernels, SpMV and SpMSpM. Note that, among MV kernels, SpMV is the most used and we specifically work on this kernel in Chapter **??**. SpMSpM is also widely used and studying this MM kernel is of interest as the memory access patterns in both matrices are irregular.
The *general* ('GE') version of these kernels is not fundamentally different, as the majority of the compute operations are in the matrix multiplication, so going forward, we will ignore the 'GE' variants.

1.4.0.1 Inner-Product Algorithm

The *Inner-Product* algorithm [17] is an intuitive way to calculate a matrix multiplication because it follows the loop order of the mathematical formulas (1.1) and (1.2) for MV and MM, respectively. The common rank K is the innermost loop of the algorithm (see Algorithms 1.1 and 1.2, lines 2 and 3, respectively). The output C (or c) is always updated sequentially, but the B-matrix (or b-vector) will be read entirely M times. In the context of sparse matrices, the reads of B (or b) are conditioned by the non-zero elements of A, which requires indirect accesses. If B (or b) is also sparse, the algorithm needs to perform *intersections* between the non-zero elements in the rows of A and the columns of B (or in the b-vector) (see Section 1.5). In Algorithm 1.2, exchanging M and N (lines 1 and 2) simply causes the output to be updated *column-wise*.

$$\forall i \in [\![0, M-1]\!],\ c_i = \sum_{k=0}^{K-1} A_{i,k} \times b_k \tag{1.1}$$

$$\forall (i,j) \in [\![0, M-1]\!] \times [\![0, N-1]\!],\ C_{i,j} = \sum_{k=0}^{K-1} A_{i,k} \times B_{k,j} \tag{1.2}$$

1.4.0.2 Outer-Product Algorithm

The *Outer-Product* algorithm [17] has the opposite rank loop order than the *Inner-Product*. Indeed, Algorithms 1.3 and 1.4 show that the common rank K is the outermost loop (line 1). It changes nothing mathematically except that the B-matrix (or b-vector) is always read

Algorithm 1.1: Dense Matrix-Vector *Inner-Product* Algorithm

```
1 for i ∈ [[0, M − 1]] do                    // row first
2 |   for k ∈ [[0, K − 1]] do
3 |   |   c_i += A_{i,k} × b_k
```

Algorithm 1.2: Dense Matrix-Matrix *Inner-Product* Algorithm

```
1 for i ∈ [[0, M − 1]] do                    // A-row first
2 |   for j ∈ [[0, N − 1]] do
3 |   |   for k ∈ [[0, K − 1]] do
4 |   |   |   C_{i,j} += A_{i,k} × B_{k,j}
```

sequentially. In the context of sparse matrices, the output C (or c) is updated with indirect accesses. The algorithm needs to perform *reductions* of the partial sums $A_{j,k} \times B_{k,j}$ (or $A_{j,k} \times b_k$), as we show in Section 1.5. To avoid irregularity when updating C (or c), the **OP** algorithm is often separated into two phases. First, during the *multiply* phase, we compute several *partial outputs* (*POs*) consisting of intermediate matrices (or vectors) of partial sums $A_{j,k} \times B_{k,j}$ (or $A_{j,k} \times b_k$), then during the *merge* phase, we accumulate all the *POs* to form the final C-matrix (or c-vector). With this method, **OP** necessitates the generation of at most K *POs* of densities conditioned by the non-zeros elements of both inputs A and B (or b).

Algorithm 1.3: Dense Matrix-Vector *Outer-Product* Algorithm

```
1 for k ∈ [[0, K − 1]] do                    // column first
2 |   for i ∈ [[0, M − 1]] do
3 |   |   c_i += A_{i,k} × b_k
```

Algorithm 1.4: Dense Matrix-Matrix *Outer-Product* Algorithm

```
1 for k ∈ [[0, K − 1]] do // common rank first
2 |   for i ∈ [[0, M − 1]] do
3 |   |   for j ∈ [[0, N − 1]] do
4 |   |   |   C_{i,j} += A_{i,k} × B_{k,j}
```

1.4.0.3 Gustavson's Algorithm

As the MM algorithms have three ranks instead of two for MV, there is a third loop ordering, known as *Gustavson*'s algorithm [25] where the common rank K is the middle-loop (see line 2 in Algorithm 1.5). With this approach, the output is computed row-by-row with *PO*-rows generated and accumulated for each output C-row. With this algorithm, updates of the C-rows are sequential, but elements within the rows are updated with indirect accesses. In the dense case, the B-matrix is entirely read M times and K *PO*-rows, of size N, are generated. In the context of sparse matrices, conversely to the C-rows, the selection of the rows in B is indirect, but the accesses are sequential within a row. Thus, the *intersection* search is only needed for the B-rows and not each element within rows. The *PO*-rows can be accumulated before computing the next C-row. Overall, this algorithm can be seen as a trade-off between **IP** and **OP** for MM kernels, since the indirect accesses are shared between B and C. Again, for this algorithm, exchanging M and N causes the three matrices to be accessed *column-wise* instead of *row-wise*.

Algorithm 1.5: Dense Matrix-Matrix *Gustavson*'s Algorithm

```
1 for i ∈ [[0, M − 1]] do
2 |   for k ∈ [[0, K − 1]] do    // common rank
3 |   |   for j ∈ [[0, N − 1]] do
4 |   |   |   C_{i,j} += A_{i,k} × B_{k,j}
```

1.4.0.4 Comparison of the Algorithms

In Table 1.1, we summarize the main features of **IP**, **OP** and **Gust** in terms of accesses to the matrix A, B (or b) and C (or c). From this high level analysis, we can easily see that the data that is accessed with indirections shifts depending on the algorithm, from the input B (or b) to the output C (or c), with **Gust** being a trade-off between **IP** and **OP** for MM multiplication. To have a better understanding of the consequences on the actual number of memory accesses, we propose a more detailed analysis in Section 1.5.4, for the case of SpMSpM.

NOTE

Note that these algorithms are available in dense as well as in sparse MV and MM multiplications. In the sparse variants, the above algorithms (from 1.1 to 1.5) have the same behavior from a mathematical point-of-view, but need adjustments to traverse a sparse format instead of a dense matrix. Sparse versions of these algorithms are presented in Section 1.5.

Table 1.1: Comparison of Matrix Multiplication Algorithms

	***Inner-Product* (IP)**[1]	***Outer-Product* (OP)**[1]	***Gustavson* (Gust)**[2]
A(M,K) **Accesses**	• *row-wise* • read N times[3,4] • sequential	• *column-wise* • read once • sequential	• *row-wise* • read once • sequential
B(K,N) **Accesses**	• *column-wise* • read M times[5] • not sequential	• *row-wise*[6] • read $M \times d(A)$ times[4,7] • sequential	• *row-wise* • read $M \times d(A)$ times[7] • sequential within rows
C(M,N) **Accesses**	• generated *row-wise*[6] • one sequential traversal	• generated without specific structure • K *POs* produced[8] (each *PO* has $\overline{nnz_c}(A) \times \overline{nnz_r}(B)$ non-zero elements[9])	• generated *row-wise* • $K \times d(A)$ *PO*-rows produced[10] for each row of A (M) ($\overline{nnz_r}(B)$ non-zero elements per *PO*-row)

[1] Considering $N = 1$ in the case of MV multiplication.
[2] Only for MM multiplication.
[3] Or less than N times if B has empty columns.
[4] Read once in the case of MV multiplication.
[5] If B is stored with a sorted sparse format, else it would be read $\overline{nnz_r}(A) \times M = nnz(A)$ times.
[6] *Element-wise* in the case of MV multiplication (there is no row).
[7] Which is equal to $\overline{nnz_c}(A)$, the average number of non-zero elements in the columns of A.
[8] Or less than K times if A has empty columns or B has empty rows.
[9] $\overline{nnz_c}(A)$ in the case of MV multiplication.
[10] Which is equal to $\overline{nnz_r}(A)$, the average number of non-zero elements in the rows of A.

CONCLUSION

There are *three main algorithms* to compute sparse MV and MM kernels, ***Inner-Product* (IP)**, ***Outer-Product* (OP)** and ***Gustavson* (Gust)**. Each presents different characteristics, the main one being the **irregularity of the accesses**: on the **input** matrix B (or vector b) with **IP**; on the **output** matrix C (or vector c) with **OP**; on both with **Gust** but at **row scale**.

1.4.1 Tiling

The *tiling* process [24] consists in partitioning a matrix: a matrix may be divided into *tiles*, non-overlapping sub-matrices. In other words, this method consists of adding ranks to delimit the *tiles*, and thus adding loop iterations in the kernel algorithm.

Tiling methods make it possible to process smaller portions of the matrix and thus to adapt the working set to the size of the local memory [39, 60], particularly when the matrix has a structure with denser regions. *Tiling* can also increase parallelism opportunities. Software optimizations for sparse matrix computation, such as OSKI [56] and Sparse BLAS [16], are based on *tiling*. In previous sections, the study of the **IP**, **OP** and **Gust** algorithms highlighted the influence of rank ordering on the sequentiality and data movement, and thus, adding intermediate ranks with *tiling* breaks the regularity of pure **IP** or **OP** algorithms, requiring additional hardware support (see Section 1.5).

For example, let us consider the A-matrix in Figure 1.4 which is divided into four *tiles* (one level of *tiling*). For a SpMV, two new ranks appear for a total of four ranks, as shown in Algorithm 1.6: M_1 and K_1 allow *tile* selection and M_2 and K_2 allow data access within a *tile*. Thus, we could decide to apply an **OP** on *tiles* and conversely an **IP** within *tiles*, mixing-up the pros and cons of these two algorithms at different rank levels. In the case of the example in Figure 1.4, the upper-left A-*tile* (*red*) is evaluated first, sequentially updating the upper *tile* of the c-vector in an **IP** manner. The lower-left A-*tile* (*yellow*) is then computed the same way. At this point, *intersections* have been done with only the upper b-*tile*, and c has been updated sequentially (first *PO*). The two next A-*tiles* (*green*, then *blue*) will compute *intersections* with the lower b-*tile* and update the entire c-vector sequentially, showing the **OP** characteristic of having *POs* to *merge*, two in this case.

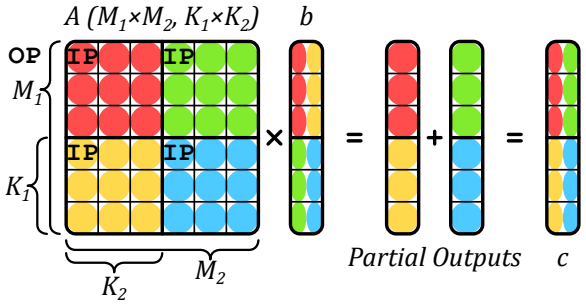


Figure 1.4: An Example of *Tiling* on SpMV

Algorithm 1.6: Matrix-Vector Algorithm with Custom *Tiling* of Figure 1.4

```
1 for k1 ∈ [0, K1 − 1] do
2     for i1 ∈ [0, M1 − 1] do
3         for i2 ∈ [0, M2 − 1] do
4             for k2 ∈ [0, K2 − 1] do
5                 i ← i1 × M2 + i2
6                 k ← k1 × K2 + k2
7                 ci += Ai,k × bk
```

DEFINITION

Tiling is a method for partitioning matrices in non-overlapping sub-matrices, called ***tiles***.

NOTE

Note that in case of *tiling*, the A-matrix is no longer read sequentially, which can be problematic if it is stored in a classical sparse format like CSR that is specifically made for *row-wise* accesses. Custom sparse formats adapted for *tiling* can be used [30], but this requires preprocessing to perform the conversion.

CONCLUSION

Tiling methods are widely used to **distribute workloads** across multicore systems, and to **locally reduce the data dimensions**. Complex *tiling* algorithms with many possible rank ordering **inherit the behaviors of IP and OP**, and can be described as a combinations of both at different *tile* levels.

1.5 Hardware Challenges for Sparse Matrix Computation

In the previous section we presented a brief background on matrix multiplication without considering the underlying hardware. Although the various sparse formats reduce the memory footprint, they have the disadvantage that indirect accesses are required, which creates data-dependencies and reduces the effectiveness of caches. As we have seen, with all three algorithms, at least some of the accesses are irregular. The lack of spatial locality for these accesses reduces the benefit of having caches [43, 62]. In other words, a full cache-line is loaded, but only a few entries are read or modified. Moreover, the matrices of interest are extremely large and even the non-zero elements, and associated metadata, can not generally fit in the cache. Depending on the algorithm, successive accesses to the same element may be too far apart to benefit from temporal locality in the cache.

1.5.1 Inner-Product Algorithm

In an **IP** algorithm, the A-metadata needs to be read to indirectly access the B-matrix (or b-vector), which may itself be stored in a sparse format. In this case, the algorithm needs to check if the non-zero A-indices are present in B (or b). If an index appears in neither list, there is no need to perform the multiplication (ineffectual operation). More generally, this problem applies to any *fiber* and is called the *intersection* search. As a minimum, this requires reading all the metadata for A and B (or b), and identifying those present in both.

In Algorithm 1.7, we present the pseudo-code for an *Inner-Product* (**IP**) SpMSpM with A and B stored in CSR and CSC formats, respectively. The outer-loop (M, line 1) iterates over the rows of A. The middle-loop (N, line 2) iterates over the columns of B. The inner-loop (line 4) iterates over the non-zero elements of the current row (i) of A, and for each index (k_A of position pos_A in the idx array of A, line 5), a search is performed in the metadata of B (line 6), to see if the corresponding index in B is present. Assuming it is ($isMatch$ is true, line 7), the partial sum is computed and then accumulated locally (acc, line 8). This example highlights the need to efficiently search for the *intersection* of indices in the metadata of the inputs.

Different algorithms can be used to address the *intersection* search, depending on whether the elements are sorted. When the elements are sorted, the *intersection* becomes a *merge* and generally requires at most $2 \times nnz - 1$ comparisons (in some cases the number of comparisons can be reduced [5]). In Algorithm 1.7, we use sorted sparse formats (CSR and CSC). Thus, the naïve *intersection* search proposed (line 10) does at most one traversal of each the current row of A (i) and the current column of B (j). Note that computing an *intersection* in software induces the use of conditional loops and branches, which can be costly for classical CPU branch predictors. In case the matrix B (or vector b) is stored in a dense format, the *intersection* search is simplified, since all indices are implicitly present, but the accesses to B (or b) remain indirect. As we see in Chapter 2, certain accelerators provide support for performing the *intersection* operation directly in hardware.

DEFINITION

The ***intersection* search** consists in founding non-zero elements of two sparse arrays that have matching indices. This search is costly in software on classical CPUs, thus being key to improve performance of **IP** algorithm.

Algorithm 1.7: `IP` SpMSpM Performing *Intersection* Searches with CSR and CSC Formats

Data: $A(M, K)$ in CSR: $A.val, A.idx, A.ptr$
Data: $B(K, N)$ in CSC: $B.val, B.idx, B.ptr$
Data: $C(M, N)$ in dense format (for readability)
Result: $C = A \times B$

```
1  for i ∈ [0, M − 1] do                                   // Iterations over the rows of A
2      for j ∈ [0, N − 1] do                               // Iterations over the columns of B
3          acc ← 0; pos_B ← B.ptr_j;
4          for pos_A ∈ [A.ptr_i, A.ptr_{i+1} − 1] do
               // Iterations over the non-zero elements of the i-th row of A
5              k_A ← A.idx_{pos_A};
               // Column-index of the (i, pos_A) non-zero element of A
6              (isMatch, pos_B) ← IntersectionSearch(k_A, j, pos_B, B.idx, B.ptr);
               // Return true and the matching pos_B if idx exists in the j-th column of B
7              if isMatch then                             // If indices did match
8                  acc ← acc + A.val_{pos_A} × B.val_{pos_B};
                   // Local partial sum accumulation
9          C_{i,j} ← acc;                  // Write in memory of the resulting (i, j) element of C
10 function IntersectionSearch(k_A, j, pos_B, B.idx, B.ptr) is
       // Naive intersection search allowing the i-th row of A and the j-th column of B to
          be traversed only once at iteration (i, j)
11     while B.idx_{pos_B} < k_A and pos_B < B.ptr_{j+1} do
12         pos_B ← pos_B + 1;
13     if B.idx_{pos_B} = k_A then                         // Returns true if indices from A and B match
14         return (true, pos_B);
15     return (false, pos_B);
```

1.5.2 Outer-Product Algorithm

In an `OP` algorithm, the inputs A and B (or b) are accessed sequentially, but the updates of the output C (or c) require indirect accesses. The pattern of the accesses, while the output is generated, is irregular and determined by the structure of the inputs. Furthermore, if these "random" updates provoke cache misses where only one word within a cache-line is modified, the bandwidth to external memory is poorly utilized. The `OP` algorithm generates several *partial outputs* (*POs*) that must be accumulated to obtain the final output. We can identify two strategies: *immediate update* and *deferred update*. In the former, the final output is immediately updated as each partial sum is generated. If the output is stored using a sparse format, either the index being updated does not yet exist, and the new partial sum must be inserted. Or, if it exists, a lookup must be done to locate it, the update performed and the result written back. The problem associated with combining the *POs* is called *reduction*.

In Algorithm 1.8, we present two pseudo-code for an *Outer-Product* (`OP`) SpMSpM using *immediate update* (from line 1) and *deferred update* (from line 7) with A and B stored in CSC and CSR formats, respectively. The outer-loop (K, line 1 or 7) iterates over the columns of A and the rows of B (common rank). The middle-loop (line 2 or 9) iterates over the non-zero elements of the current column (k) of A. The inner-loop (line 4 or 11) iterates over the non-zero elements of the current row (k) of B. Because we only iterate over the non-zero values, there is no need for an *intersection* search. The partial sum is then computed (line 6 or 15). Finally, in the *immediate update* version, the partial sum must be accumulated to C (line 6). These accesses are irregular, making it necessary to lookup the value in C, update it, and write it

back. This is a *reduction* operation, performed on irregular data.

To avoid the complexities of these "random" *reductions* with *immediate updates*, the *deferred update* version postpones the accumulation and simply stores the partial sums in memory. One such technique is called *output batching* [36]. With this technique, arrays containing the indices and the partial sum are streamed sequentially to memory (PO_k, lines 13-15). Then, during a second pass (line 17), they are read back and the accumulations are performed, an approach which results in sequential memory accesses during both the first and second passes, with the downside that the volume of data transferred to memory is increased. As we see in Chapter 2, hardware accelerators may combine the *immediate* and *deferred update* techniques to optimize memory bandwidth.

DEFINITION

"Random" *reductions* occur when updating data from memory at irregular addresses, which happen on *immediate update* `OP`. With *deferred update* `OP`, the *reductions* are sequential but large sets of data need to be buffered in memory. Optimizing *reduction* operations is key to improving the performance on `OP` algorithm.

1.5.3 Summary

In Table 1.2, we summarize the advantages and hardware challenges for each algorithm, including `Gust` which presents a trade-off between `IP` and `OP`. The potential for parallelism and data-reuse varies across algorithms, `IP` being the most limited unless *tiling* methods are used. With `OP`, *POs* can easily be generated across a multicore system but at the cost of transferring a large volume of data to memory. With *immediate updates*, `OP` transfers less data but all accesses on the output are "random" *reductions*.

The issues presented with `IP` and `OP` can be generalized to two concepts: *gather* and *scatter*. The *gather* problem exists when the inputs are indirectly accessed, and need to be read from memory using metadata to present the processor with the correct terms for computing the partial sums. Typically, with `IP`, the *gather* problem is more difficult. The *scatter* problem exists when the order in which the output matrix/vector is generated is not sequential and is typically associated with the `OP` algorithm. In the case of *tiling* or with `Gust`, both *gather* and *scatter* must be performed creating a need for efficient *intersection* searches and random *reductions*.

CONCLUSION

There are **two main challenges** to address when computing sparse matrices. When reading irregular data from memory, addressing the ***gather*** problem is key. It consists of processing efficient ***intersection* searches**. When updating irregular data from memory, addressing the ***scatter*** problem is key. It consists of **managing *POs*** in memory or processing efficient **"random" *reductions***. These two challenges are tied to `IP` an `OP`, respectively, and are present in any higher rank algorithms for sparse matrix computing.

1.5.4 SpMSpM Algorithm Analysis

To better understand how the algorithms impact the number and types of memory accesses, we developed, and presented in Figure 1.5, a model based on memory accesses counts for SpMSpM. We identify five memory access patterns: (1) those where a matrix is read only once,

Algorithm 1.8: OP SpMSpM Performing *Reduction* with CSR and CSC Formats

Data: $A(M, K)$ in CSR: $A.val, A.idx, A.ptr$
Data: $B(K, N)$ in CSC: $B.val, B.idx, B.ptr$
Data: $PO_k(M, N)$ for all $k \in [\![0, K-1]\!]$ in COO: $PO_k.val, PO_k.row, PO_k.col$
Data: $C(M, N)$ in dense format (for readability)
Result: $C = A \times B$

```
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * *
// * * * 'Immediate updates' version * * * * * * * * * * * * * * * * * * * * * * *
1  for k ∈ [[0, K−1]] do                          // Iterations over the columns of A and the rows of B
2      for pos_A ∈ [[A.ptr_k, A.ptr_{k+1}]] do
           // Iterations over the non-zero elements of the k-th column of A
3          i_A ← A.idx_{pos_A};                   // Row-index of the (pos_A, k) non-zero element of A
4          for pos_B ∈ [[B.ptr_k, B.ptr_{k+1}]] do
               // Iterations over the non-zero elements of the k-th row of B
5              j_B ← B.idx_{pos_B};
               // Column-index of the (k, pos_B) non-zero element of B
6              C_{i_A,j_B} ← C_{i_A,j_B} + A.val_{pos_A} × B.val_{pos_B};
               // Reduction (to memory) of the partial sum into the (i_A, j_B) element of C
                  (immediate update)
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * *
// * * * 'Deferred updates' version  * * * * * * * * * * * * * * * * * * * * * * *
// Multiply phase
7  for k ∈ [[0, K−1]] do                          // Iterations over the columns of A and the rows of B
8      pos_PO ← 0;
9      for pos_A ∈ [[A.ptr_k, A.ptr_{k+1}]] do
           // Iterations over the non-zero elements of the k-th column of A
10         i_A ← A.idx_{pos_A};                   // Row-index of the (pos_A, k) non-zero element of A
11         for pos_B ∈ [[B.ptr_k, B.ptr_{k+1}]] do
               // Iterations over the non-zero elements of the k-th row of B
12             j_B ← B.idx_{pos_B};
               // Column-index of the (k, pos_B) non-zero element of B
13             PO_k.row_{pos_PO} ← i_A;
14             PO_k.col_{pos_PO} ← j_B;
15             PO_k.val_{pos_PO} ← A.val_{pos_A} × B.val_{pos_B};
               // The k-th partial sum of the (i_A, j_B) element of C is stored in memory in the
                  k-th partial output
16             pos_PO ← pos_PO + 1;
       // Each partial output is generated sorted, row-wise
// Merge phase
17 for k ∈ [[0, K−2]] do                          // Naive serial merge of the K partial outputs
18     PO_{k+1} ← 2DListsMerge(PO_k, PO_{k+1});
// PO_{K-1} is C stored in COO format
19 for pos ∈ [[0, length(PO_{K−1}) − 1]] do
       // Update C (dense) from the fully merged partial outputs
20     i ← PO_{K−1}.row_{pos};
21     j ← PO_{K−1}.col_{pos};
22     C_{i,j} ← PO_{K−1}.val_{pos};                // Write in memory of the (i, j) element of C
```

Table 1.2: Comparison of Advantages and Challenges with Matrix Multiplication Algorithms

		***Inner-Product* (IP)**	***Outer-Product* (OP)**	***Gustavson* (Gust)**
Access Count	**MV**	***A*:** once ***b*:** $nnz(A)$ ($\sim M\times$) ***c*:** once	***A*:** once ***b*:** once ***c*:** $nnz(A)$ ($\sim K\times$)	*N/A*
	MM	***A*:** $N\times$ ***B*:** $M\times$ ***C*:** once	***A*:** once ***B*:** $M \times d(A)\times$ ***C*:** $nnz(A) \times \overline{nnz_r}(B)$ ($\sim K\times$)	***A*:** once ***B*:** $M \times d(A)\times$ ***C*:** $nnz(A) \times \overline{nnz_r}(B)$
Sequential Accesses		Accesses to A and C	Accesses to A and B	Accesses to A and C Accesses to B-rows
Input Format		Accesses to A and B benefit from alternate formats (CSR and CSC)		Allows consistent format for A and B
Parallelism Opportunity		Possible with *tiling*	Generation of *POs* over common rank of A and B	Reading A-rows and updating C-rows
Reuse Opportunity		Potentially B, but limited by its size	An A-column or B-row while computing *POs*	A set of B-rows, based on non-zero elements of A-rows
Challenges		*Intersection* search between non-zero elements in A and B	Storage for *POs* or in-place *merge* and *reduce*	*Merge* and *reduce* of *PO*-rows

thus there is no opportunity for reuse, (2) where the accesses are sequential, but a *fiber* is read multiple times, (3) where *fibers* are read consecutively, but the accesses within a *fiber* are "random", (4) where the *fibers* are accessed "randomly", but within a *fiber* the accesses are sequential, and (5) where the matrix elements are accessed multiple times, and with a "random" access pattern.

For each of the three algorithms, we counted the number of read and write accesses, based on the density of the input matrices, and the local memory storage available ("0D", "1D" and "2D"), as presented in Table 1.3. We refer to a system with no local storage or cache as '0 Dimensional' ("0D"), a system which has local storage that is sufficient to hold a single row or column as '1 Dimensional' ("1D"), and a system which can cache an entire matrix as '2 Dimensional' ("2D"). In Figure 1.5, we plotted the number of memory accesses for SpMSpM, in a logarithmic scale, based on a square matrix with $M = 10,000$. Indeed, the number of non-zero elements in C depends on the structure of the input matrices and in this model, we have assumed the non-zero entries in the input matrices are uniformly, randomly distributed. This corresponds to a *worst case* for sparse matrices (no spatial and temporal locality). If A or B are banded matrices, for example, there would be fewer non-zero elements in C. In each graph, we have separated the accesses into A (*bottom*), B (*middle*) and C (*top*). The color code shows the access pattern, based on the five categories described above.

In the top-row of Figure 1.5 (graphs ①, ② and ③), we start by considering a naïve system which has no local storage ("0D"), thus all data comes from main memory. From the graphs in this row, we clearly see that **IP** requires more accesses at low density, as the A-matrix must be read N times (graph ①). Whereas for **OP** and **Gust**, the A-matrix is read only once (thus the *grey* color). For **OP** and **Gust** the total number of accesses is the same, however, with **OP** (graph ②), the accesses to C are "random" (*red*). With **Gust** (graph ③), the accesses to B are sequential within the rows (*orange*) and the C-rows are generated sequentially (*yellow*), thus **Gust** avoids having any fully "random" accesses (*red*).

Since main memory accesses are the principal bottleneck, systems integrate local storage

Table 1.3: Memory Access Count for Each Matrices of `IP`, `OP` and `Gust` SpMSpM

	Inner-Product (`IP`)	***Outer-Product*** (`OP`)	***Gustavson*** (`Gust`)
$\mathbf{A}(M,K)$	**0D**: $N \times nnz(A)$ **1D-2D**: $nnz(A)$	**0D-1D-2D**: $nnz(A)$	**0D-1D-2D**: $nnz(A)$
$\mathbf{B}(K,N)$	**0D-1D**: $M \times nnz(B)$ **2D**: $nnz(B)$	**0D**: $M \times d(A) \times nnz(B)$ **1D-2D**: $nnz(B)$	**0D-1D**: $M \times d(A) \times nnz(B)$ **2D**: $nnz(B)$
$\mathbf{C}(M,N)$	**0D-1D-2D**: $nnz(C)$ (max: $M \times N$)[1]	**0D-1D**: $2 \times M \times d(A) \times nnz(B)$ **2D**: $nnz(C)$ (max: $M \times N$)[1]	**0D**: $2 \times M \times d(A) \times nnz(B)$ **1D-2D**: $nnz(C)$ (max: $M \times N$)[1]

[1] We can not easily quantify $nnz(C)$ knowing $nnz(A)$ and $nnz(B)$. We thus use a statistic approximation.

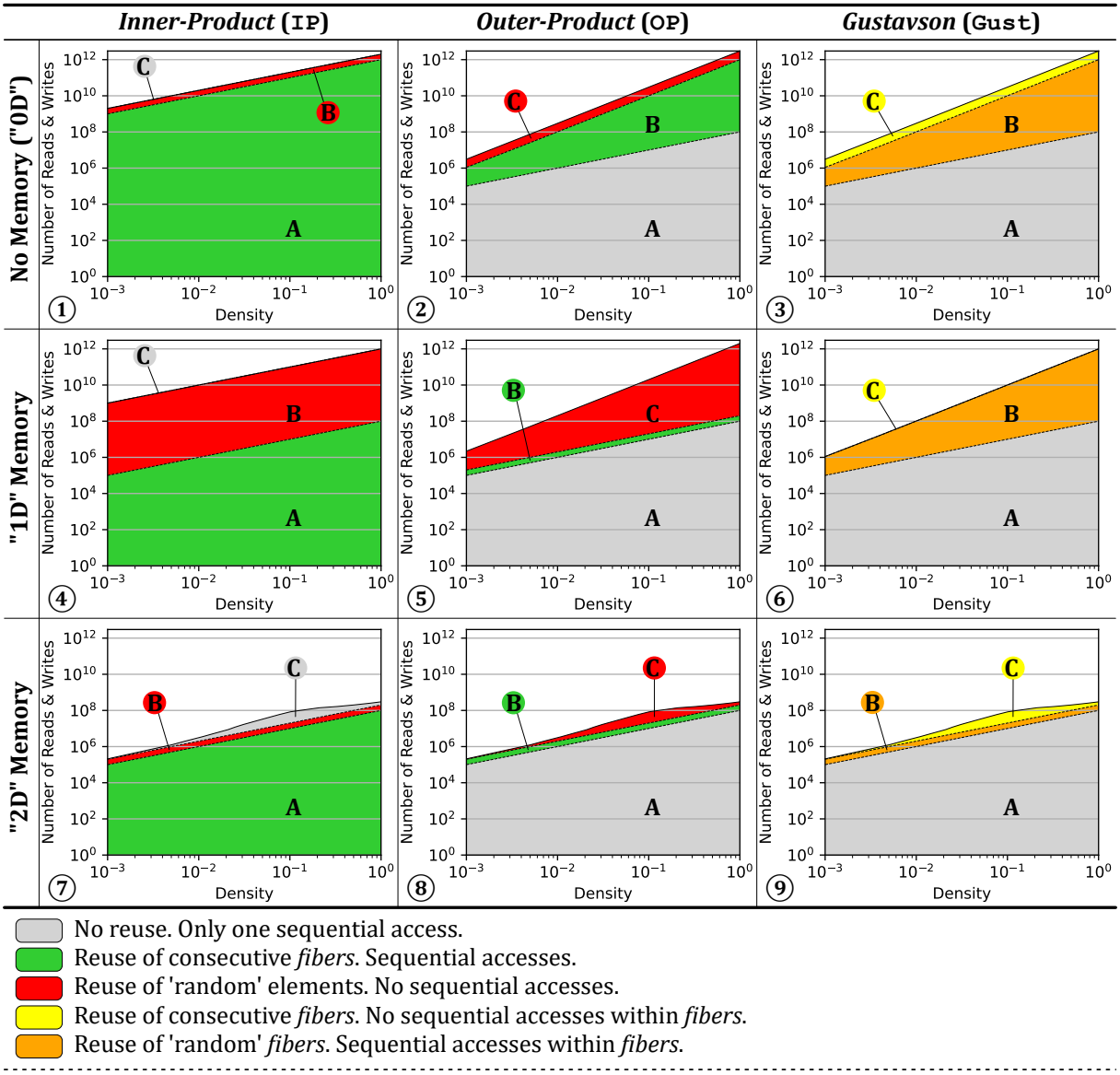


Figure 1.5: Model of Memory Access Counts Depending on Algorithm and Local Memory Size

(such as buffers or caches). Thus, we consider the case of a system with sufficient local memory to store one full row, per matrix (equivalent to few sparse rows, "1D"). We assume this local storage is perfect (no misses) and we re-calculate the number of remaining main memory accesses, as shown in the middle-row of Figure 1.5 (graphs ④, ⑤ and ⑥). The matrices are assumed to be initially stored in main memory, thus they must be accessed at least once. As can be seen, for **IP** (graph ④), the number of A-accesses is reduced, because rows are reused, thus the total number of A-accesses becomes equal to that in **OP** and **Gust**. For **OP** (graph ⑤), the single row buffer greatly reduces the number of B-accesses, however, it does nothing for C (despite the appearance, due to the logarithmic scale). For **Gust** (graph ⑥), the single row buffer is sufficient to cache the *reduction* operations in C, and thus, it now has fewer overall accesses than **OP**. It is also interesting to note that **OP** and **Gust** scale better than **IP** with lower densities, as seen by the shallower slope.

Finally, we have considered the extreme case where an system has a buffer of sufficient size to store an entire matrix ("2D"), which is shown in the bottom-row of Figure 1.5 (graphs ⑦, ⑧ and ⑨). Although this case is not practical for large matrices, through the proper use of *tiling* to locally reduce the dimensions of a sub-matrix, this case can be achieved at the level of a *tile*. In this case, the total number of external accesses is equal for the three algorithms, as each matrix is accessed only once from main memory. For **IP** (graph ⑦), it is interesting to note that such a large buffer is needed to locally address the *intersection* search in B. Similarly for **OP** (graph ⑧), a large storage is required to address the *reductions* in C. For **Gust**, in fact, it is not necessary to buffer the entire B-matrix. Indeed, buffering a few rows (corresponding to $nnz_r(A)$) is sufficient to achieve the number of accesses shown in graph ⑨. This corresponds to an intermediate design point (multiple row buffers for B) not explicitly shown in the figure (between graphs ⑥ and ⑨).

When viewed overall, Figure 1.5 highlights the benefits of **Gust**. We clearly see that **Gust** does not require any fully "random" accesses (*red*). With a single row buffer, it is possible to address the *reductions* in C. And, with a limited number of row buffers, the number of external memory accesses for B can be reduced. Based on this analysis, it is clear that, among the three base algorithms, **Gust** is the best candidate for hardware implementation of SpMSpM.

NOTE

For SpMV, such an analysis does not show significant differences between **IP** and **OP**, as both algorithms require similar number of memory accesses. However, **OP** shows more potential for parallelism and data-reuse as seen in the previous section (1.5.3), thus being preferred for SpMV hardware accelerators.

CONCLUSION

Based on a high level analysis of the memory accesses on SpMSpM, ***Gustavson*'s algorithm** appears as the **best trade-off** to minimize memory traffic with reasonable amount of local storage. The second best option is **OP**, which requires fewer memory accesses than **IP** when the matrices have a low density.

1.6 Conclusion

In this chapter, we established the foundation for understanding the relationship between applications working with irregular data and the underlying challenges in hardware. Irregular accesses are common in sparse computing and cause reduced cache efficiency, limited parallelism, and higher memory latency exposure. We analysed the most representative algorithms for sparse matrix multiplication kernels, and identified the operation that provoke irregular accesses. *Intersections* and *reductions* on irregular data with classical systems are costly, thus they represent the central challenge for efficient computation on sparse data. To improve the performance of these operations, specialized hardware has been proposed as will be seen in the next chapter.

TAKEAWAYS

Sparse matrices: large, highly sparse, structured (such as banded matrices)
Sparse formats: diverse, reduce memory footprint

- COO, the simplest
- CSR and CSC, the mostly used
- BaCSC, our own sparse format for banded matrices

Sparse kernels: sparse MV and MM multiplications (such as SpMV and SpMSpM), generate indirect accesses
Algorithms: *Inner-Product* (`IP`), *Outer-Product* (`OP`) and *Gustavson* (`Gust`), extended with *tiling* methods

- Best potential to reduce memory transfers: `Gust`, followed by `OP` at low density

Problem: irregular accesses, generate high memory transaction latencies
Challenges:

- *gather*: *intersection* searches
- *scatter*: "random" *reductions*, or storing and *merging partial outputs* (*POs*)

Chapter 2

State-of-the-Art

Contents

2.1 Introduction

TODO

Chapter 3

Architectural Presentation of SpDCache

Contents

3.1 Introduction

In the previous chapters, we analyzed the causes and effects of irregular memory accesses in sparse computations and identified the *Outer-Product* (`OP`) algorithm of Sparse Matrix-Vector multiplication (SpMV) ($A \times b = c$) as an important example. The `OP` algorithm generates irregular updates on the output vector c, as each non-zero element of the input matrix A produces updates to distinct, and often distant, location in the output vector c. These "random" *reductions* render stress the memory system.

Conventional cache hierarchies are designed for regular, predictable access streams that exploit spatial and temporal locality. In contrast, `OP` SpMV exhibits poor locality, leading to low cache utilization, frequent evictions, and thus requiring more memory bandwidth.

Previous works, such as PHI [44], SortCache [51], and ASA [66], have proposed *reduction*-aware cache mechanisms to alleviate irregularity by improving accumulation and storage efficiency. However, these solutions do not exploit the structural properties of the processed matrices, and therefore miss optimization opportunities when the matrix has some regularity. This observation motivates our focus on banded sparse matrices, which combine overall sparsity with a dense region near the diagonal. Banded matrices are very common and serve as an initial example to study how dense and sparse regions can be leveraged to enhance performance.

This chapter begins by reviewing the main concepts of generic data-caches, introducing certain terminology and mechanisms which are required to discuss cache behavior under irregular access conditions. It then presents a *reduction*-aware cache architecture, designed to efficiently support "random" *reductions* for the `OP` SpMV kernel. The proposed design, SpDCache (presented in Paper II), extends traditional caching models with a region-based organization and native *reduction* support. Finally, we evaluate its performance through high-level modeling on banded and non-banded matrices, highlighting the influence of data regularity on cache efficiency.

3.2 Generic Data-Caches

MODERN processors and accelerators integrate data-caches to bridge the latency and bandwidth gap between computation units and main memory [28]. A cache is a small, fast memory that stores recently accessed data anticipating that the data will be reused in the near future. Its efficiency relies on two main forms of locality:

- temporal locality, where recently accessed data is likely to be reused within a short interval of time, and
- spatial locality, where data stored near recently accessed addresses is likely to be used next.

By exploiting these properties, caches reduce the average access latency and increase the overall bandwidth efficiency.

A cache is composed of several cache-lines, each grouping contiguous memory words. The cache-line is the minimum granularity of data movement between memory levels. When an access occurs, the cache controller searches for the requested address among the currently stored cache-lines. If the address is found, a hit occurs and the data is served directly to the core. Otherwise, a miss occurs, triggering a memory fetch from a lower cache level or from main memory. If necessary, an existing cache-line is evicted to make room for the new data. Each cache-line is associated with a *tag* that record its address and additional bits to keep track of its state.

Several mapping policies exist. In a direct-mapped cache, each memory block maps to a single cache-line. This approach is simple to implement but results in conflicts and poor utilization. A n-*way set*-associative cache allows each block to be placed in one of n possible cache-lines within a *set*, improving hit rate with only a moderate additional hardware cost. A *fully associative* cache removes these constraints, as any block may occupy any cache-line, but requires more complex *tag* searches. In all cases, when a line must be evicted and multiple candidates exist, a replacement policy such as *Least Recently Used* (*LRU*) or *Random* decides which entry is evicted.

Caches have different policies when writes occur. With a *write-through* policy, every store operation updates both the cache and main memory, maintaining consistency but increasing traffic. A *write-back* policy delays the update until eviction, reducing memory bandwidth at the cost of additional state bits (dirty flags) to maintain consistency.

For workloads dominated by sequential accesses, these mechanisms and policies are highly effective. However, irregular accesses tend to degrade cache efficiency by increasing miss rates and provoking frequent evictions.

Modern processors implement hierarchical cache organizations to balance latency, capacity, and throughput. A typical CPU integrates small L1 caches close to each core, larger L2 caches shared by a group of cores, and a high-capacity L3 cache shared at the chip level. Accelerators and domain-specific architectures often employ local data storage or scratchpad memories serving the same purpose, providing low-latency access to frequently reused data (see Chapter 2).

The generic cache assumes regular read and write operations with predictable memory access patterns. Yet workloads operating on sparse data violate this assumption. In particular, “random” *reductions* often have poor performance because each *reduction* triggers a read and a write of a full cache-line, to simply update one word. Furthermore, the lack of spatial locality means each *reduction* can trigger an eviction, in the worst case.

In the following sections, we introduce cache architectures with native *reduction* support, specifically designed to handle irregular data. These architectures extend the classical cache model with *reduction*-aware mechanisms, improving bandwidth efficiency under irregular access patterns.

CONCLUSION

Generic data-caches are effective when access patterns exhibit spatial and temporal locality. This is not the case for sparse data and their performance is particularly poor for "random" *reduction* operations. This limitation motivates the introduction of ***reduction*-aware cache architectures** which are optimized for such "random" *reductions*.

3.3 Reduction Cache

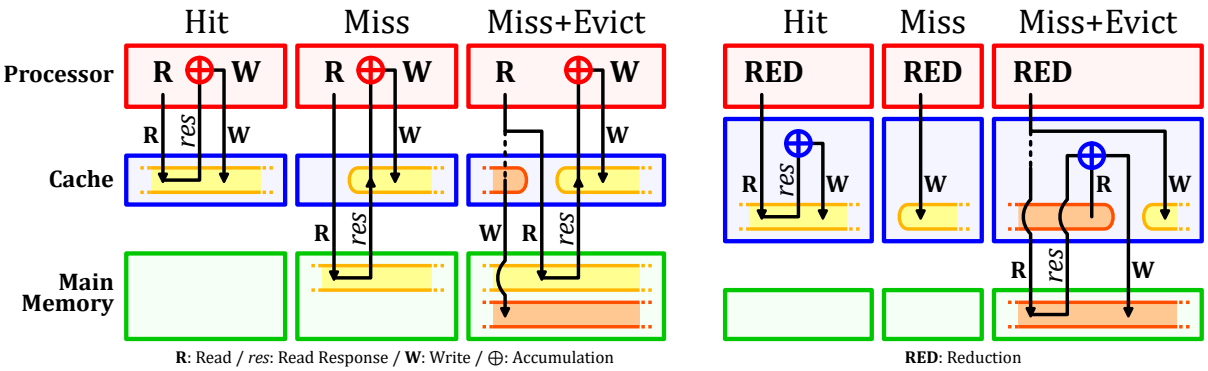
PERFORMING a *reduction* in hardware requires a read from memory, followed by an accumulation and a write back. This operation is denoted $RED(key, value)$ [51] with the *key* being the position in the output vector c. In Figure 3.1a, we show three cases that occur in a generic data-cache when a *reduction* is performed. If the vector entry (c_{key}) hits in the cache, the *update* is performed by the processor (in *red*) with no external memory access (on the left in the figure). When there is a miss, the operation is blocked until the read from main memory completes (in the middle of the figure). Potentially, the miss will also trigger an eviction, which exacerbates the blocking condition, because a cache-line must first be written back, before the read from main memory can be performed (on the right in the figure). In scientific applications, the matrices are extremely large and only a small portion of the vector c can reside in the cache, thus the *miss+evict* case arises frequently.

A *reduction cache*, such as [51] or [44], is a data-cache which accepts *reduction* instructions (*RED*) from the processor and performs accumulations locally (near-memory). In the case of a hit, see Figure 3.1b, the *reduction* takes place in the cache (in *blue*, on the left in the figure). For a miss, the cache allocates a new line with zeros, and simply inserts the value (in the middle of the figure). In this case, the cache-line is inconsistent with main memory. Only when the line needs to be evicted, is a read performed (on the right in the figure). In the cache, the values in the line are accumulated with the main memory data, and the result is written back.

As seen in the figure, a *reduction cache* reduces the number of main memory accesses in the case of a miss. Instead of having the classic ascendant policy where data is stored in the cache to be pulled-up again by the processor latter-on, the *reduction cache* has a descendant policy: data is written/updated in the cache to be pushed-down to memory latter-on. Instead of two memory transactions that can stall the processor, the *reduction cache* only needs a single "fire-and-forget" instruction, meaning it can be issued without waiting for an answer, thus avoiding processor stalls. At the end of the algorithm, the *reduction cache* must be flushed.

DEFINITION

A ***reduction cache*** is a data-cache containing an arithmetic unit and a dedicated support for *reduction* operations. One implementation is to provide the processor with a *RED* instruction which tells the cache to perform the *reduction*. For the processor, this instruction is "fire-and-forget".



(a) Generic Cache (Write-Back) **(b)** Reduction Cache

Figure 3.1: Read (R) and Write (W) Transactions of a *Reduction* Operation in Generic and *Reduction* Caches

CONCLUSION

To improve the processing efficiency of *reduction* operations, a ***reduction cache*** can be used to **avoid unnecessary memory transactions and stalls**, by offloading the accumulation near-memory, and updating the main memory only at eviction.

3.4 Outer-Product SpMV with a Reduction Cache

A widely used sparse format to compress the non-zero elements in a matrix A is CSC [49], where A is a sparse matrix of dimensions (M, N) with a number of non-zero elements nnz. As seen in Chapter 1, it consists of three tables (val, idx, ptr) used to store non-zero values, row indices and column positions. When performing an **OP** SpMV ($A \times b = c$) with this format (Algorithm 3.1), we see that a *RED* is needed to accumulate the output vector c (line 6). Moreover, A is read sequentially while c is updated "randomly". In fact, the *key* is determined via an indirection with an access to idx.

Algorithm 3.1: *Outer-Product* SpMV – Single Threaded

```
1 for n ∈ [0, N − 1] do                          // Loop over the columns of A
2     for pos ∈ [ptr_n, ptr_{n+1} − 1] do
3         key ← idx_pos;                          // Row index
4         val ← val_pos × b_n;                    // val = A_key,n × b_n
5         RED(key, val);                          // c_key += val
```

In memory, A is stored as three dense tables, but conceptually, as we perform the **OP** multiplication, the matrix is traversed from top to bottom, then left to right. A row in the matrix corresponds to a *key*, or equivalently, a position in the output vector c. In Figure 3.2, we show an example of a sparse matrix with a few non-zero values (① - ⑥). As we traverse the matrix, we first perform a *reduction* for the value on row key_{10}, then on row key_{20} and then key_{30}. Each of these results in a *RED* on c. As we continue, we then hit another value ⑤ on the row key_{10} which must be *reduced* with an existing entry ① (as shown on the left of Figure 3.2, *Conceptual View*).

Let us consider how c is stored in a two-*way* *set*-associative cache with 32B lines. We assume that all the *keys* map to set_1. When ① is updated, there is a miss and $line_1$ is allocated

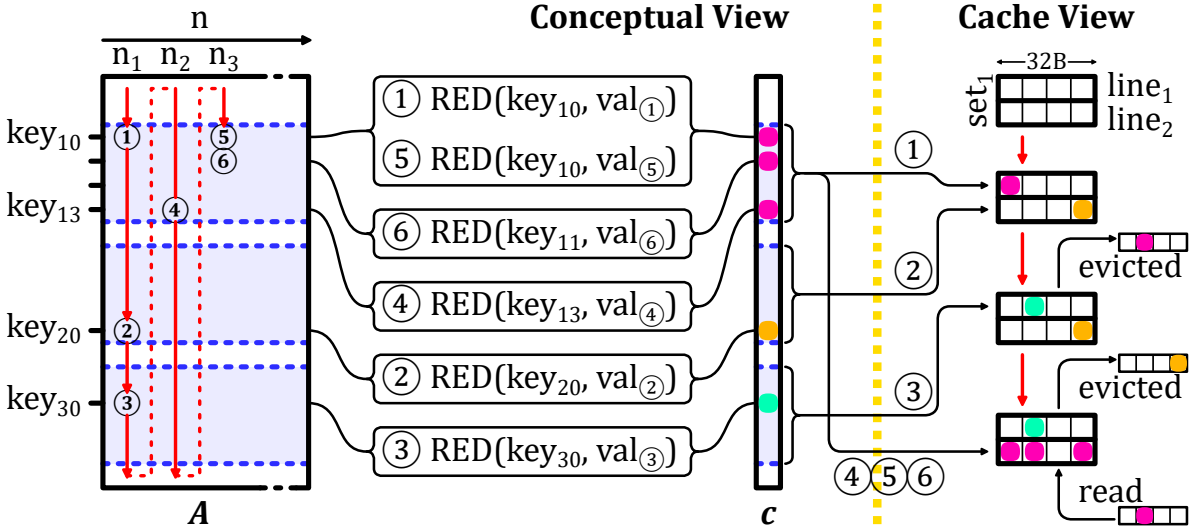


Figure 3.2: Traversal of Non-Zero Elements of Input Matrix for `OP` SpMV (left) and Impact on Cache (right)

for key_{10}. Similarly with ②, $line_2$ is allocated for key_{20}. When key_{30} comes along, an eviction must be performed. We assume $line_1$, holding key_{10}, is evicted. We then come to ④ which has key_{13}, adjacent in memory to key_{10}, and thus stored in the same line. This line must be brought back to process ④, ⑤ and ⑥.

This example illustrates how a conventional cache can be inefficient. First, we note the line holding key_{10} was evicted, although it was frequently accessed after ①. This eviction was triggered by key_{30}, which was only accessed once. Furthermore, for key_{20} and key_{30}, a full line was loaded, only to modify a single entry. A "smarter" cache would have maintained the line with key_{10} to avoid an unnecessary main memory access. Furthermore, this "smarter" cache would not have allocated full lines for key_{20} and key_{30} to only modify a single word.

The globally low local and temporal localities prevent caches, and even *reduction caches*, to take advantage of any reuse opportunity. However, sparse matrices have various types of structure which may have reuse potential in some parts of the matrix. In the example in Figure 3.2, the first rows (with key_{10} to key_{13}) are denser than the rest of the matrix. A "smarter" cache would prefer to focus on these rows, keeping the corresponding cache-line and accumulating every elements before a final eviction. On the contrary, the sparser rows of the matrix would not be kept by the "smarter" cache, thus avoiding thrashing data from denser regions.

Going forward, we focus on the case of banded matrices which have a denser region around the diagonal. For a *reduction cache* to work efficiently with such matrices, the dense band should be prioritized, and the isolated elements out of the band should be processed separately to avoid these sparse elements disturbing the storage of the dense data. This is the main idea of SpDCache (Paper II), which is a *reduction cache* with dedicated memory regions for the in-band and out-of-band of a matrix. We describe its behavior in the following sections.

CONCLUSION

When computing an `OP` **SpMV kernel** on a very **large matrix**, a **conventional *reduction cache* does not provide enormous benefit** because the cache is not large enough to reuse data from one column iteration to the next.

We propose **SpDCache**, a *reduction cache* with dedicated memory regions, **leveraging matrix structure** with dense and sparse regions which correspond to the structure of **banded matrices**.

3.5 Architecture of SpDCache

SpDCache was developed to replace a generic L1 data-cache. Thus, it must function as a generic cache when receiving load and store memory instructions from the processor. Indeed, when processing an `OP` SpMV kernel, the sequential reads to the matrix A and the vector b should take advantage of the generic cache behavior, due to their spatial locality. Only, the output vector c suffers from "random" *reductions* that must be processed with a region-wise *reduction cache*. SpDCache can be extended to a multilevel memory hierarchy, as well as multicore architectures (see Chapter 6). However, our study focuses on a single level cache hierarchy with a single core, to better understand SpDCache behavior.

As shown in Figure 3.3, SpDCache is divided into three regions. The first, *Generic Region Cache* (*GRC*), is for all memory transactions unrelated to the output vector c, including reads of A and b. It is a classic, *set*-associative cache and is required for the processor to operate normally, but is not the focus in this chapter. The other two regions, called *Dense Region Cache* (*DRC*) and *Sparse Region Table* (*SRT*), are able to perform *reductions* on c. The *DRC* works as a line-wise, *set*-associative *reduction cache* to hold high density data in the *In-Band Region* of the matrix (*IBR*, *orange*). The difference with the *GRC* is that it is able to perform *reductions* and it is dedicated to the *banded* region, so that spurious accesses outside of the band do not provoke undesired evictions. Finally, the *SRT* works as an element-wise, fully-associative table whose role is to handle sparse regions, such as the *Out-of-Band Region* in the matrix (*OBR*, *blue*).

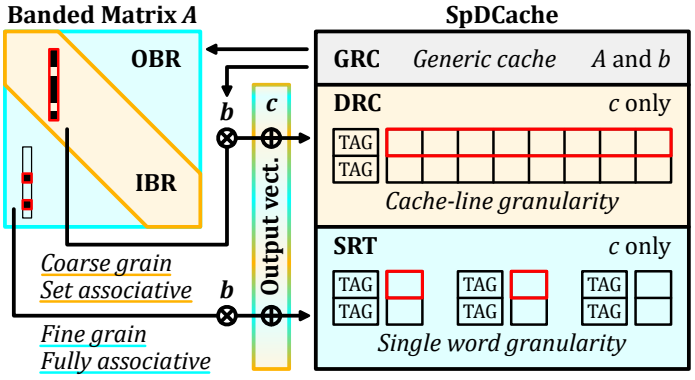


Figure 3.3: Data-Cache Partitioning into Regions (*GRC*, *DRC*, *SRT*)

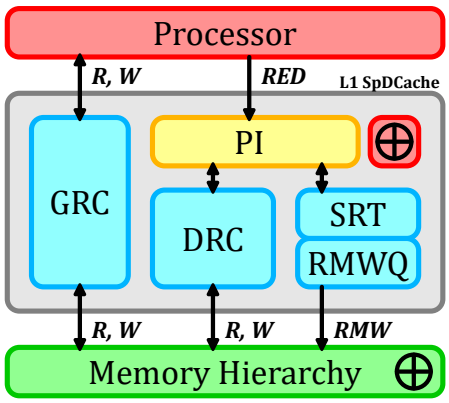


Figure 3.4: Simplified View of the Memory Hierarchy with SpDCache

Figure 3.4 shows a simplified view of a memory hierarchy with SpDCache. The *Processor Interface* (*PI*) takes *reduction* instructions, *RED(key, value, region)*, from the processor. The *RED* instruction contains a 'region' field which indicates whether the value is expected to be in a dense region (*DRC*) or a sparse region (*SRT*). As we will see, the hardware may override the software selection to avoid duplication. The *DRC* communicates with the memory hierarchy using read and write transactions over entire cache-lines, to propagate the local *updates*

to the downstream memory. As *SRT updates* are for isolated elements, it does not make sense to bring a full line into the L1 to modify a single word. The *SRT* thus propagates *updates* by issuing atomic *Read-Modify-Write* transactions (*RMWs*), which must be processed remotely and close to the main memory. Existing protocols such as AXI-4 [1] already support atomic transactions for certain operations (logical, integer add, min, max) and these could be extended to include floating point addition. The *RMWs* from the *SRT* are temporarily buffered in a small queue (*RMWQ*), to avoid congestion.

SpDCache is driven by three key design principles. First, we ensure a given *key* is in only one region at a time, to avoid coherency problems. Second, the *DRC* is given priority over the *SRT*, to enable coalescing. Third, we avoid bringing a full line from memory to the L1 to simply update one or a few entries.

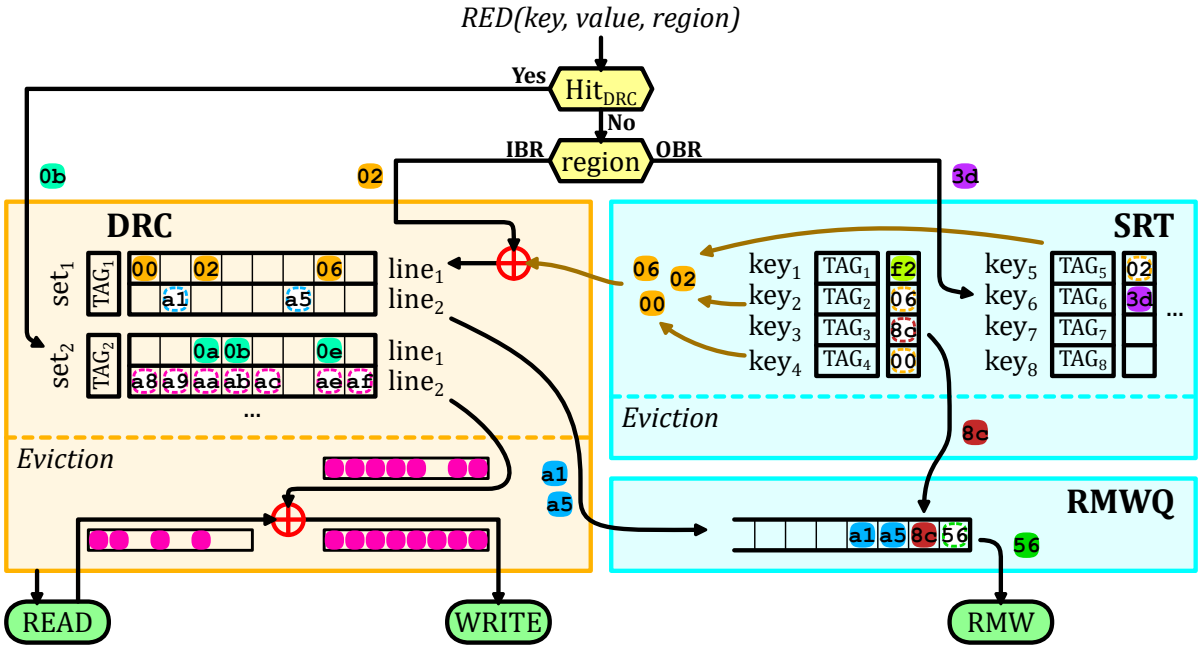


Figure 3.5: Internal Operation of SpDCache with Examples

In Figure 3.5, we describe the operation of SpDCache, via a series of scenarios, each shown in a different color. When a new *reduction RED(key, value, region)* is received from the processor, we test for a hit in the *DRC*, regardless of the *region* argument. If it hits, the value (case `0b` – *turquoise*) can be accumulated in the corresponding cache-line of the *DRC*. We give this priority to the *DRC* as part of the mechanism to avoid duplication.

If there is a miss in the *DRC*, then the target region is determined based on the *region* argument of the *RED* transaction. If it is *IBR*, then it is necessarily a miss case, and the value (`02` – *orange*) must be inserted into a free cache-line of the *DRC*. When we allocate this line, we ensure that it does not create a duplication of existing entries in the *SRT*. If there is duplication, as in our example, these entries (`00`, `02`, `06`) are deleted from the *SRT* and merged into the newly allocated cache-line ($set_1, line_1$).

If the target region is the *OBR*, then the *reduction* (`3d` – *purple*) is processed in the *SRT*, either allocating a new entry (miss case) or accumulating with an existing entry (hit case).

Now, we describe the eviction scenarios. In the *DRC*, the line to evict is selected ($set_2, line_2$ – *pink*), a read to main memory is performed, the values from the evicted cache-line are accumulated and a write to main memory is issued. If the evicted cache-line only has a

few non-zero elements and the *RMWQ* has enough free entries, we enter into a special case. To avoid bringing a full line from main memory, the non-zero values (`a1` and `a5` – *blue*) are converted to atomic *RMWs* and pushed onto the *RMWQ*. The decision to convert evicted cache-lines to atomic *RMWs* is based on a configurable density threshold. PHI [44], an accelerator similar to SpDCache, used a threshold of one non-zero word per eight words in a cache-line. Given our queuing mechanism to handle congestion, we use a more aggressive threshold of three non-zero words per cache-line.

The width of the *band* (w_B), must be selected so that one column of the *band* remains blocked in the *DRC*, otherwise, this region of the cache is always thrashing. This is our approach to selecting w_B, which we detail in the next chapter. Note that this approach selects w_B based on the *DRC* size, independently of the matrix and its structure.

In the *SRT*, an eviction simply consists of moving the element from the table to the *RMWQ* (`8c` – *brown*). If the *RMWQ* is not full, *SRT* evictions are issued in advance when the *SRT* table is nearly full, using another configurable threshold set to the *SRT* size minus half the size of the *RMWQ* in our experiments. The *RMWQ* issues atomic *RMWs* when it is not empty (`56` – *dark green*).

NOTE

We make the assumption that memory regions in the *GRC* and the *DRC*/*SRT* are disjoint. Currently, we do not manage coherency between the two, although this could be a future extension.

NOTE

SpDCache is not limited to banded matrices. Any sparse matrix with a denser region can benefit from this architecture, by adapting the definition of the *band* accordingly, and making sure that the blocking effect in the *DRC* is preserved. The selection of where the dense region is located is done in software via the region argument of the *RED* instruction.

CONCLUSION

SpDCache is a data-cache splits in **three regions**:

- *GRC*, a **generic** cache region which handles the reads of A and b and any other memory accesses performed by the processor,
- *DRC*, a *reduction cache* region dedicated to *reductions* on c when computing the dense region of the matrix, which is the **band** (*IBR*) in the cases of banded matrices,
- *SRT*, a *reduction cache* region dedicated to *reductions* on c when computing the sparse regions of the matrix, which correspond to the **out-of-band** elements (*OBR*) for a banded matrix.

The two last regions use **memory transactions adapted to the region density** (line-wise or element-wise) to avoid transferring unnecessary data.

3.6 High Level Evaluation of SpDCache

We modelled the behavior of SpDCache, based on the flowchart in Figure 3.5, using a transaction level model in Python, for a single-core, single-thread system executing a `OP` SpMV kernel (Algorithm 3.1). We do not model timing, as the order of the memory accesses determines the behavior of the cache (hits, misses, evictions), which depends only on the

SpMV algorithm. The model checks the final numerical result and reports the main memory traffic associated with the *updates* to the output vector (*Output Memory Traffic*), where we count the bytes transferred (64B for each read and write, and 8B per atomic *RMW*, with our configuration). For each write transaction, it tracks how many words in the line were actually used (*Evicted Cache Line Utilization*) and it tracks how many words in the cache held useful data (*Cache Memory Utilization*). We consider a word in the cache to be useful if it is updated or inserted by the processor. Our goal is to validate the principle of SpDCache using real application matrices from the SuiteSparse Matrix Collection [38].

As baselines, we implemented a fully generic cache (GC) that handles reads and writes for both inputs (A, b) and output (c), following the data movements shown in Figure 3.1a, and a simple *reduction cache* (RedC) that processes *RED* transactions in a dedicated region using the flow shown in Figure 3.1b. In the RedC, 25% of the memory is allocated for the inputs and the other 75% for *REDs* on c. In SpDCache (SpDC), 25% is allocated for the *GRC*, 50% for the *DRC* and 25% for the *SRT*. In the next section, we present more results with varying region sizes, confirming that this allocation is indeed a good compromise. All three caches use a *Least Recently Used* (*LRU*) eviction policy, except the *SRT* region of SpDC which uses a *random* policy. All three caches have a total data storage of 32 KiB, not including the storage. The *set*-associative regions have 4 *ways* and 64B cache-lines (8 *double* words per line).

We simulated 48 matrices that have been used by the state-of-the-art accelerators [4, 27, 45, 51, 52, 66, 67], which are listed in Appendix D. We excluded matrices whose output vector c fits entirely in the cache (fewer than 4096 elements), since they do not stress the cache. In Table 3.1, we report the results for all three caches, along with statistical indicators. Most, but not all, of the matrices have a dense band and even some matrices without an obvious band benefit from the blocking effect in the *DRC*.

Table 3.1: Decrease in Memory Traffic and Improvement in Cache Utilization from Python Simulation

	Output Memory Traffic			**Evicted Cache Line Utilization**			**Cache Memory Utilization (%)**		
	RedC	**SpDC**		**GC**	**RedC**	**SpDC**	**GC**	**RedC**	**SpDC**
	%/GC	*%/GC*	*% RMW*	*Avg*	*Avg*	*Avg*	*Avg*	*Avg*	*Avg*
Geometric Mean	89.2	24.5[a]	40.5	3.60	3.56	7.21[b]	36.4	45.8	75.2[c]
Standard Deviation	29.8	20.9	33.8	2.45	2.77	2.08	10.9	32.3	18.6
Minimum	28.5	7.00	0.00	1.06	1.00	4.36	16.6	11.9	30.0
Maximum	172	98.4	100	8.00	8.00	8.00	49.6	97.5	98.9
First Quartile	76.0	11.4	35.9	1.76	1.73	6.03	28.4	22.0	65.5
Third Quartile	115	46.6	91.7	6.60	7.65	8.00	48.6	88.8	93.9
Outperform GC	45.8	100	Ø	Ø	45.8	91.7	Ø	64.6	100
Outperform RedC	Ø	93.8	Ø	Ø	Ø	85.4	Ø	Ø	77.1

[a] *Decrease by 4.1× compared to GC, and 3.6× compared to RedC.*
[b] *Increase by 2.0× compared to GC, and 2.0× compared to RedC.*
[c] *Increase by 2.1× compared to GC, and 1.6× compared to RedC.*

As expected, the simple *reduction cache* (RedC) decreases the main memory traffic to 89.2% of the GC baseline. SpDCache has significantly better performance than RedC, reducing the traffic to 24.5%. SpDCache thus achieves a 4.1× decrease in memory traffic compared to GC, and a 3.6× decrease compared to RedC, on average over the set of matrices. SpDC

outperforms GC on all matrices, and RedC on 93.8% of them.

The benefits of SpDCache can be partly explained by the improved utilization of the cache-lines. On average, a cache-line evicted from SpDC has 7.21 modified words, compared to 3.56 and 3.60 with the baselines. This improved utilization is the result of blocking the dense region in the *DRC*, where multiple *reductions* have the time to accumulate in the cache before being evicted. The *SRT* also contributes to this effect, as it captures irregular accesses outside the band, which would otherwise evict useful data from the *DRC*. The overall cache memory utilization is also significantly improved, with 75.2% of the cache holding useful data in SpDC, compared to 45.8% and 36.4% for RedC and GC, respectively.

We need to consider an additional metric, the percentage of traffic sent as atomic *RMW* transactions (% RMW). If this value is too high, it means we have simply shifted the workload to the lower level caches, which inherently have limited compute capability. On average, 40.5% of the memory traffic of SpDCache is composed of atomic *RMW*s, which is acceptable given the significant overall reduction in memory traffic. However, this ratio varies widely between matrices. There are some matrices where most of the traffic is composed of atomic *RMW*s, meaning that most of their non-zero values are outside the band. For such matrices, which are basically not banded, SpDC still greatly decrease the memory traffic, but better performance could be achieved by choosing a matrix-fitting region partitioning strategy. For example, the matrix *cit-Patents* has no elements in the band, and all non-zero values are scattered under the bottom part of the matrix (an unusual extreme case). While having a memory traffic decreased to 7.2% compared to GC, it could benefit from a better use of the *DRC* by choosing a horizontal band covering the non-zero elements, with correct sizing to leverage the blocking effect. On the other hand, some matrices have very low *RMW* usage, indicating that most non-zero values are within the band and efficiently handled by the *DRC*. In this case, 25% of the available memory was unused. The *SRT* size could be adjusted to gain performance.

CONCLUSION

SpDCache achieves a 4.1× **decrease in memory traffic** on the output vector c compared to a generic cache, over a set of 48 matrices. It also **outperforms a simple *reduction cache*** by 3.6×, showing the importance of having separate dedicated cache regions for the in- and out-of-band of a matrix.

3.7 State-of-the-Art Comparison and Region Sizing

To compare SpDCache with the state-of-the-art, we simulated PHI [44], the existing solution that is closest to our proposal, using the same methodology as in the previous section. PHI is a *reduction*-aware cache that uses a single region to process *RED* transactions (see Section **??** from previous chapter). In PHI, the cache is not separated into regions, however, it treats load/store and *RED* operations differently. It also handles isolated elements, but do not consider the matrix structure. Unlike SpDC, PHI relies on the *deferred update* version of the `OP` SpMV algorithm (see Algorithm 1.8). We thus adapted our simulation to use this algorithm for PHI. To ensure a fair comparison, we allocated the same amount of memory to PHI as for SpDC and the baselines (32 KiB). In Table 3.2, we report the results for PHI alongside our previous results (columns 2, 3, 6 and 7).

As expected, PHI achieves a significant reduction in memory traffic compared to the GC,

Table 3.2: All Results from Python Simulation

GMeans (48 mat.)	**GC**	**RedC**			**PHI**	**SpDC**					
sets GRC	*128*	*32*	*8*	*4*	*128*	*32*	*8*	*8*	*4*	*4*	*4*
sets DRC	*Ø*	*96*	*120*	*124*		*64*	*104*	*112*	*92*	*116*	*122*
sets SRT	*Ø*	*Ø*	*Ø*	*Ø*		*32*	*16*	*8*	*32*	*8*	*2*
Output Memory Traffic (% over GC)	100 ■1.0	89.2 ▼1.1	81.6 ▼1.2	80.3 ▼1.3	50.4 ▼2.0	24.5 ▼4.1	24.3 ▼4.1	25.5 ▼3.9	23.2 ▼4.3	25.3 ▼4.0	32.0 ▼3.1
Ratio of RMWs in Memory Traffic (%)	Ø	Ø	Ø	Ø	Ø	40.5	35.7	36.5	36.5	36.2	41.1
Total Memory Traffic (% over GC)	100 ■1.0	93.9 ▼1.1	91.9 ▼1.1	93.2 ▼1.1	98.8 ■1.0	56.1 ▼1.8	56.6 ▼1.8	56.9 ▼1.8	57.7 ▼1.7	58.2 ▼1.7	59.2 ▼1.7
Evicted Cache Line Utilization (max: 8)	3.60 ■1.0	3.56 ■1.0	3.66 ■1.0	3.67 ■1.0	5.00 ▲1.4	7.21 ▲2.0	7.21 ▲2.0	7.23 ▲2.0	7.22 ▲2.0	7.22 ▲2.0	7.25 ▲2.0
Cache Memory Utilization (%)	36.4 ■1.0	45.8 ▲1.3	47.0 ▲1.3	47.1 ▲1.3	30.6 ▼0.8	75.2 ▲2.1	68.3 ▲1.9	65.0 ▲1.8	72.5 ▲2.0	65.1 ▲1.8	60.9 ▲1.7

a 50.4% reduction on average. This represents a 2.0× decrease compared to the GC, which is consistent with the results reported in the original PHI paper [44], and a 1.8× decrease compared to the RedC. However, SpDCache still outperforms PHI, cutting memory traffic by 2.1×. The improved performance of SpDC can again be explained by the better utilization of the cache-lines, with 7.21 useful words per evicted line, compared to 5.00 for PHI. The overall cache memory utilization is also significantly better with SpDC, with 75.2% of the cache holding useful data, compared to only 30.6% for PHI.

As explained in Chapter 2, PHI mixes the 'immediate updates' and 'deferred updates' strategies for `OP` SpMV. In the first phase of the algorithm, it processes *RED* transactions as 'immediate updates', and PHI handles them with a mechanism similar to a classical *reduction cache*. The main differences are that the *GRC* and *DRC* are merged, which can cause unnecessary evictions of useful data, and more importantly, PHI uses a batching mechanism at eviction to 'defer' some updates to the second phase. When a cache-line is evicted and contains fewer non-zero entries than a fixed threshold, PHI batches the entries in *partial outputs* (*POs*) defined by the software and stored in memory. These *POs* are processed in the second phase of the algorithm, as in the 'deferred updates' variant. If the threshold is set too high, PHI batches many entries and increases second-phase memory traffic. If it is set too low, PHI batches few entries, reducing batching effectiveness and increasing the number of 'immediate updates', making PHI behavior closer to a *reduction cache*. PHI sets this threshold to one non-zero word per eight words in a cache-line, meaning that for each batched evicted line, PHI reduces bytes transferred by 4× (4 (word, address) tuples fitting in a single 64 B cache-line, instead of 4 full 64 B lines with 7 unused words each). However, we observed that, on the geometric mean over the 48 matrices, only 4.2% of PHI's output memory traffic comes from these batched updates (maximum 28.6%), indicating that PHI's overall behavior remains close to a *reduction cache*. In contrast, SpDCache addresses the drawbacks of a *reduction cache* by using dedicated regions for denser and sparser accesses, avoiding unnecessary evictions, without incurring the additional second-phase traffic introduced by PHI's batching mechanism.

We also explored different region sizing for SpDCache, varying the sizes of the *GRC*, *DRC* and *SRT* while keeping the total cache size constant at 32 KiB. The results are also reported in Table 3.2. Reducing the size of the *GRC* from 8 KiB (32 *sets*) to 2 KiB (8 *sets*) and 1 KiB (4 *sets*)

has little impact on performance, as the *GRC* is mainly used for temporary storage of input vector elements. However, reducing the *GRC* under 1 KiB increases the total memory traffic, as more frequent evictions occur before the data is used, an effect already noticeable when going from 2 KiB to 1 KiB. With constant *GRC* size, when reducing the size of the *SRT*, which means allocating more space to the *DRC*, we observe a slight increase in memory traffic, as the *SRT* becomes less effective at capturing irregular accesses. Thus, the *SRT* sizing appears to be more critical than the *DRC* sizing. We will explore the reasons behind this effect in the next chapter. Overall, fine tuning the region sizes can lead to small performance improvements, but the original sizing of 8 KiB for the *GRC*, 16 KiB for the *DRC* and 8 KiB for the *SRT* appears to be a good compromise for the set of matrices we evaluated.

We also explored different sizing strategies for the RedC baseline, varying the size of its single *reduction* region while keeping the total cache size constant at 32 KiB. Allocating more than 75% of the cache (96 *sets*) to store *RED* transactions slightly improves performance, as it increases the likelihood of capturing multiple updates to the same output element before eviction, without overly compromising the storage of input elements.

CONCLUSION

SpDCache outperforms PHI, a state-of-the-art *reduction* cache, achieving a 2.1× decrease in memory traffic, due to the dedicated regions which improve the effectiveness of the cache in dense regions, and the use of external atomic accesses to reduce the cost of isolated updates. Additionally, we show that the initial region sizing of SpDCache (25/50/25%) provides good performance with only minor variations observed when adjusting the sizes of the *GRC*, *DRC*, and *SRT*.

3.8 Conclusion

We have presented the concept of SpDCache, a *reduction cache* that has optimized storage and compute techniques for the sparse and dense regions of banded matrices. This approach does not reduce the number of compute operations, however, it significantly decreases memory traffic (4.1×). These benefits can be attributed to three techniques: (1) blocking the dense part of the band in the *DRC*, (2) using fine-grained storage in the *SRT* and (3) off-loading the *reductions* for isolated elements to the downstream caches, to prevent local evictions and reduce traffic.

Exploiting the coarse-grain structure of matrices is key to designing hardware optimized for sparse matrix kernels. In the next chapter, we present a more theoretical analysis of the behavior of SpDCache before moving onto an actual RTL implementation in Chapter 5.

TAKEAWAYS

Reduction cache:

- Provides hardware support for performing ***reductions***: $RED(key, value)$.
- The accumulations are done **near memory**, in the cache.
- **Reduces memory traffic** for kernels with a large number of *reduction* updates.

SpDCache:

- Considers that the matrix has a **dense region** and a less dense, **sparse region**. For banded matrices, these correspond to the in-band (*IBR*) and the out-of-band (*OBR*).
- Divides the L1 data-cache into **three regions**:
  - ***GRC***, generic region cache for regular memory accesses (such as reading A and b);
  - ***DRC***, dense region cache for *reductions* in the dense region of the matrix;
  - ***SRT***, sparse region table for *reductions* in the sparse region of the matrix.
- Uses memory transactions adapted to the region density: **cache-line-wise versus element-wise** for the *DRC* and the *SRT*, respectively.
- For a set of 48 matrices, results in **4.1× less memory traffic** for `OP` SpMV kernel.

Chapter 4

Analysis of Reduction Cache Behavior and Parameter Exploration

Contents

4.1 Introduction

SPARSE matrix processing exposes processor systems to irregular memory access patterns, which degrade cache efficiency and increase memory traffic. The magnitude of these effects on memory system performance remains challenging to predict due to the diversity of data patterns. In this chapter, we develop a theoretical model to analyze and optimize the behavior of a *reduction cache* when executing an *Outer-Product* (`OP`) SpMV kernel.

We propose a hybrid probabilistic/fast simulation model that predicts the memory transactions of a *reduction cache* when executing an `OP` SpMV kernel ($A \times b = c$). This model allows us to quickly explore the impact of different cache configurations on memory traffic, using banded matrices as a representative case study. By isolating the sequential reads of the input data and analyzing the density distribution of non-zero elements, we identify key design principles for optimizing cache performance.

The chapter demonstrates the importance of separating the in-band and out-of-band regions of the matrix into dedicated cache regions. This separation allows the cache to exploit

the structural regularity of banded matrices while efficiently handling irregular accesses in sparser regions. The theoretical analysis validates the architectural choices of SpDCache that was presented in previous chapter and provides insights for sizing and configuring cache regions to minimize memory traffic.

In our analysis of *reduction caches*, we use a set-associative configuration, with 8 byte elements which we consider as the basic memory unit. s_C is the size of the cache, in elements. The cache is divided into *sets*, *ways* and cache-lines, of size S, W and l, respectively. Thus, $s_C = S \times W \times l$ elements. We recall that a banded square matrix of dimension M is defined by its band width w_B, or its half-band width w_{hB}, such that $w_B = 2w_{hB} - 1$. The in- and out-of-band densities are named d_{IB} and d_{OB}, respectively. When d_{OB} is zero, the matrix is said to be 'perfectly' banded.

4.2 Isolating the Input Reads in a Generic Cache

WHEN computing an **OP** SpMV kernel, all the irregular accesses come from updating the output vector c. Indeed, the input matrix A and vector b are read sequentially, and can thus be efficiently cached by a generic cache that naturally exploits spatial locality.

With the matrix A stored in CSC format, each iteration of the **OP** algorithm necessitates reading four tables: $A.ptr$, $A.idx$, $A.val$ and b. In the same iteration, a fifth table, c, is updated "randomly". If we consider cache-lines of size l elements, then each cache-line read of the $A.ptr$ table brings l consecutive entries, which are pointers to l consecutive columns of the matrix. Similarly, each cache-line read of the b table brings l consecutive entries, which will be multiplied by the non-zero elements of l consecutive columns of the matrix. These reads are needed in the outer-loop of the **OP** algorithm. In the inner-loop, each cache-line read of the $A.idx$ and $A.val$ tables brings l consecutive non-zero elements of the matrix to process, which will generate l "random" updates to the output vector c, thus l different cache-lines of c could be accessed in the worst case.

To avoid unnecessary evictions, at each iteration, the generic cache should, obviously, keep $A.ptr$, $A.idx$, $A.val$ and b cache-lines up until they are no longer needed. That means l iterations for $A.idx$ and $A.val$, and around $l \times nnz_c$ iterations for $A.ptr$ and b. However, the cache must also keep the cache-lines of c that are being updated, which, over the course of $l \times nnz_c$ iterations, can necessitate up to the same number of different cache-lines in the worst case, on unpredictable addresses. Thus, it would be difficult for a generic cache to keep all the useful data until it is no longer needed. $A.ptr$ and b cache-lines, which are accessed once every nnz_c iterations, could easily be evicted by a *LRU* policy which is unaware of their future use. By isolating the input reads in a dedicated cache, as proposed with SpDCache and the *reduction cache* baseline in the previous chapter, we greatly increase the chances that all the cache-lines of A and b remain in the cache until they are no longer needed. With a dedicated cache region for the inputs, the *GRC* introduced in previous chapter, only the four sequentially read tables need to be considered. These tables are read only once, and isolating them in the *GRC* decreases the risk of undesired evictions.

With a *LRU* policy, cache-lines holding $A.ptr$ and b are more at risk of being evicted than $A.idx$ and $A.val$ because they are accessed less frequently, only once every $\sim \frac{nnz_c}{l}$ times per column. $A.idx$ and $A.val$ are accessed every iteration of the inner-loop. To avoid premature evictions of $A.ptr$ and b, the *GRC* must be sized to handle the $\frac{nnz_c}{l}$ cache-line reads which occur while computing a column. In Equation 4.1, we show the minimum size of the *GRC* in elements needed to avoid unnecessary evictions during the traversal of the columns, thus optimizing memory traffic.

$$s_{C_{min}}(GRC) = 2 \times l + 2 \times nnz_c \tag{4.1}$$

This relationship is not exact, as it does not consider address alignment effects nor temporal variability in the downstream memory, but it gives a good approximation of the minimum size needed for the *GRC* to avoid unnecessary evictions. It explains why, in the previous chapter, the total memory traffic slightly drops when the *GRC* size is decreased. If we return to Table 3.2, in the previous chapter, in the columns for SpDC, we now understand why the total memory traffic is higher when the *GRC* size is reduced from 32 to 4 *sets*.

Separating the inputs in a dedicated cache region also has the advantage of allowing prefetching techniques [50] to be used efficiently, as prefetched cache-lines will not be unde-

sirably evicted before use. For the **OP** SpMV kernel, prefetching does not reduce the memory traffic, but it does improve performance by masking the main memory latency.

CONCLUSION

To avoid unnecessary evictions, it is advantageous to **separate** the sequential reads of the inputs A and b from the "random" updates of the output c, into **dedicated cache regions**. A generic cache of reasonable size can **mask the latency** of the main memory when reading A and b, and ensure this data is **read exactly once**.

4.3 Modeling a Reduction Cache

To go beyond simply observing the behavior of a specific accelerator processing a given matrix, a model of the matrix and the cache is required. We present a hybrid probabilistic/fast simulation model for a *reduction cache*, processing **OP** SpMV, which predicts the volume of memory traffic issued by the L1 cache, starting from a matrix density distribution and a cache configuration.

The model represents the non-zero density both in the matrix and in the cache, with cache-line granularity. Using these coarse grain densities, a fast simulation of the **OP** SpMV is performed. The state of the cache can be predicted at each step of the simulation. By counting the evictions, we evaluate the output memory traffic, the memory trafic of the output vector c. We can thus predict the total number of memory accesses for different types of matrices and quickly explore different *reduction cache* configurations.

We do not consider the sequential accesses to the input matrix A and vector b in our model. Indeed, the matrix tables val, idx and ptr, and the vector b are each read once, giving a total memory traffic of $2nnz+(M+1)+M$ elements. We showed in the previous section that a simple generic cache of reasonable size is sufficient to minimize the associated memory traffic.

Our model uses the cache configuration parameters S, W and l. Although our model is not limited to banded matrices, they are the focus of this work. The matrix density distribution thus takes on two different values, either d_{IB} in the band, or d_{OB} outside of the band.

We normalize the memory traffic by dividing the number of elements that are read or written by the number of non-zero elements (nnz) in the matrix, so we can compare performance between matrices of different sizes and densities. Indeed, we can interpret the normalized memory traffic, $normMT$, as the number of elements transferred from/to memory per non-zero matrix element. If the memory traffic remains constant while the matrix density increases, we expect $normMT$ to decrease.

4.3.1 Operation of the Analytical Model

Matrix Representation The square matrix is of dimension M, as represented in Figure 4.1. Its rows and columns are indexed by $0 \leqslant i, j \leqslant M-1 \in \mathbb{N}$. Each column is split into vertical-strips of size l, where l is the cache-line size. Indeed a vertical-strip is a portion of a column j of size l, from row-indices i to $i+l-1$, as illustrated by the *red* rectangle within the matrix in the figure. Within a column, vertical-strips are indexed by $k \in [\![0, M_l - 1]\!]$ [1], with $M_l = \lceil \frac{M}{l} \rceil$, and $k = \lceil \frac{i}{l} \rceil$.

[1] We use $[\![a, b]\!]$ for the integer intervals, and $[a, b]$ for real intervals.

The matrix is thus represented as a static, dense two dimensional table, MDT (*Matrix Density Table*), of size (M_l, M) that contains the density in $[0, 1]$ of each vertical-strip. We assume that the distribution of the non-zero elements within a vertical-strip is uniform, as assumed in related work [54] where the authors studied a directly-mapped generic cache for an *Inner-Product* SpMV kernel, only for the case of 'perfectly' banded matrices.

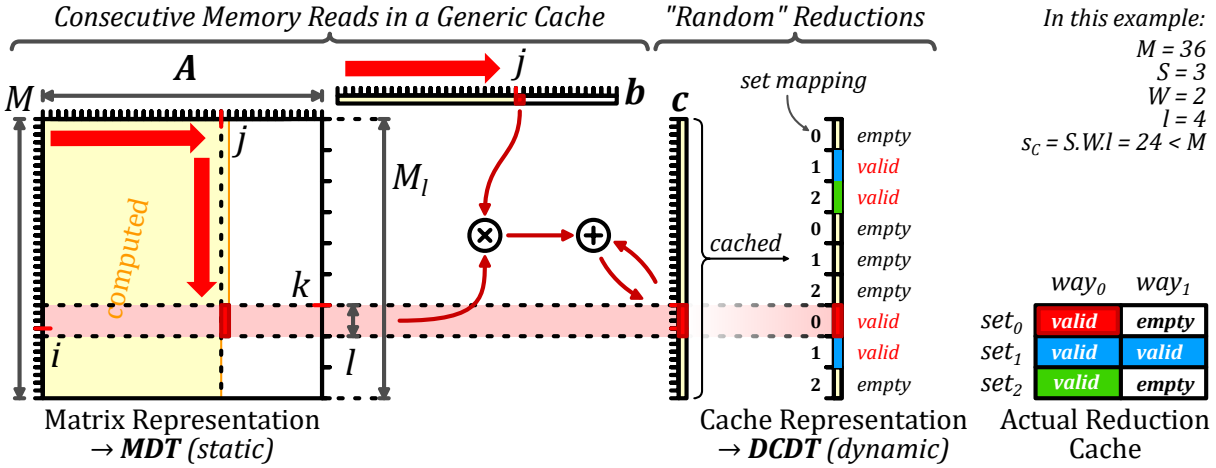


Figure 4.1: Input Matrix and Cache Representation of the Model

Cache Representation Still in Figure 4.1, the cache is represented as a single column, which maps directly to the output vector c. This "column-representation" of the cache is partitioned into cache-lines, corresponding directly to the vertical-strips of a given column of the matrix, as shown by the horizontal *light red* rectangle. A cache-line $k \in [\![0, M_l - 1]\!]$ is stored in a unique *set* $s \in [\![0, S - 1]\!]$ of the cache such that $s = k\%S$. Unlike the matrix representation which is static, the cache representation is continuously updated as the model evaluates **OP** SpMV.

Of course, the real cache-size ($S \times W \times l$) is not necessarily equal to the output vector size (M). Thus, at a given time, the number of valid (non-empty) cache-lines contained in the cache representation must not exceed the actual cache-size. Specifically, we ensure the number of valid cache-lines in a *set* of the cache representation does not exceed the number of ways W. In the example in Figure 4.1, the actual cache is holding four valid cache-lines, which are the only valid cache-lines present in the cache representation.

To count the number of valid cache-lines, we further distinguish them from empty ones. For this reason, we set the cache-line densities to be either zero (the cache-line is empty), or a density is in the range $\left[\frac{1}{l}, 1\right]$ (the cache-line is valid). A valid cache-line has a density between $\frac{1}{l}$ and 1, corresponding to the probabilistic number of non-zero elements.

The state of the cache is thus represented with the dynamic table $DCDT$ (*Disjoint Cache Density Table*), of dimension M_l that contains the disjoint-density in $\{0\} \cup \left[\frac{1}{l}, 1\right]$ of each cache-line, at a given iteration. Initially, all cache-lines are set to zero density, since the cache is considered empty.

Iterative Process In the fast simulation phase, the analytical model [19] iterates over the vertical-strips in the matrix, following the dense traversal of the **OP** algorithm, as described in Chapter 1 and shown with *red* arrows in Figure 4.1. In a given iteration (column j, vertical-

strip κ), we use the matrix density of the current vertical-strip, $MDT_{\kappa,j}$, and the current state of $DCDT$ to determine if an eviction occurs, and update it for the next iteration, if necessary.

Disjoint-Density of the Current Vertical-Strip The density of the current vertical-strip in the matrix has to be disjointed at each iteration, to determine whether the strip is empty or not, as done for the cache-lines. We define the vertical-strip disjoint-density as $d_{lM}(\kappa) \in \{0\} \cup [\frac{1}{l}, 1]$.

- When $MDT_{\kappa,j} = 0$, the vertical-strip in the matrix is empty, thus, $d_{lM}(\kappa) = 0$.
- When $0 < MDT_{\kappa,j} < \frac{1}{l}$, the vertical-strip can have either 0 or 1 non-zero elements. To resolve this, we maintain a running counter of the accumulated $MDT_{\kappa,j}$ values, and when this counter exceeds $\frac{1}{l}$, the current vertical-strip is considered non-empty, so we set $d_{lM}(\kappa) = \frac{1}{l}$. If the counter is below this threshold, $d_{lM}(\kappa) = 0$.
- When $MDT_{\kappa,j} \geqslant \frac{1}{l}$, the vertical-strip has one or several non-zero elements. Thus, $d_{lM}(\kappa) = MDT_{\kappa,j}$.

With this definition, we can distinguish empty ($d_{lM}(\kappa) = 0$) and non-empty ($d_{lM}(\kappa) \neq 0$) vertical-strips.

Eviction Processing With the above definitions, we can determine if a miss occurs at the current iteration. A miss occurs when the cache-line κ is not valid and the vertical-strip is not empty. Recall that in a *reduction cache*, a miss without eviction results in the allocation of a free cache-line, initially filled with zeros, and produces no memory traffic. We define $miss(\kappa) = (DCDT_\kappa = 0)\ and\ (d_{lM}(\kappa) \neq 0)$, a boolean indicating whether a miss occurs in the current iteration.

When allocating a free cache-line, it is necessary to ensure that the number of valid cache-lines in the *set* $s = \kappa \% S \in [\![0, S-1]\!]$ does not exceed W. The new allocation may thus trigger an eviction. Stated formally, $nnzcl_\kappa$ is the count of the cache-lines k from the *set* s that verify $DCDT_k \neq 0$. We define $evict(\kappa) = miss_\kappa\ and\ (nnzcl_\kappa = W)$, a boolean indicating whether an eviction occurs in the current iteration.

In the case of an eviction, for the probabilistic model, we use a *random* eviction policy to select a valid cache-line in the *set* s (when $DCDT_k \neq 0$). The index of the evicted cache-line is $e \in [\![0, M_l - 1]\!]$.

Cache Update The last step is to update the $DCDT$ before proceeding to the next iteration. (1) If a cache-line is evicted, we must account for the memory traffic and reset the victim line e; (2) The density of the current cache-line κ may increase if there is a hit or if it is newly allocated:

1. In the case of an eviction ($evict(\kappa)$ is true), we reset the victim cache-line (e) by setting $DCDT_e = 0$.
2. We update the density of the cache-line κ. In the case of a miss ($DCDT_\kappa = 0$), the cache-line density is set to the disjoint-density of the matrix vertical-strip κ. In the case of a hit, the cache-line and the vertical-strip densities are merged:
   $DCDT_\kappa \leftarrow DCDT_\kappa + (1 - DCDT_\kappa) \times d_{lM}(\kappa)$
   The updated cache-line density is either 0 or greater than $\frac{1}{l}$, thus respecting the constraint that the cache-line is either empty or valid.

Final Iteration Starting with an initially empty cache, we run all $M \times M_l$ iterations and count the total number of evictions ($nbEvict$). After the last iteration, we account for the memory transactions required to flush the cache ($nbFlush$), writing the valid cache-lines to memory, by counting lines where $DCDT_k \neq 0$, for $k \in [\![0, M_l - 1]\!]$. $nbFlush$ will never exceed $S \times W$.

In a *reduction cache*, evictions and flushes result in a line-wise memory read, a modification, and a line-wise memory write. The update memory traffic is $2l(nbEvict + nbFlush)$ elements, and $normMT$ is this value divided by nnz.

CONCLUSION

We have presented a **hybrid probabilistic/fast simulation analytical model** for a *reduction cache*, processing `OP` SpMV, which predicts the volume of memory traffic issued by the L1 cache, starting from a **matrix density distribution** and a **cache configuration**. The model represents the non-zero density both in the matrix and in the cache, with **cache-line granularity**. Using these coarse grain densities, a **fast simulation** of the `OP` SpMV is performed. In this way, we can quickly explore different *reduction cache* configurations to evaluate their impact on memory traffic.

4.4 Analysis of Reduction Cache Behavior

We now study the *reduction cache* behavior when computing `OP` SpMV, starting with the simple case of 'perfectly' banded matrices with varying band widths. We then use the analytical model to analyse the *reduction cache* behavior, varying the out-of-band density and the cache-size. We show the benefit of storing the in- and out-of-band of the matrix in separate regions of the cache, to minimize memory traffic.

4.4.1 Behavior of a Cache for a Perfectly-Banded Matrix

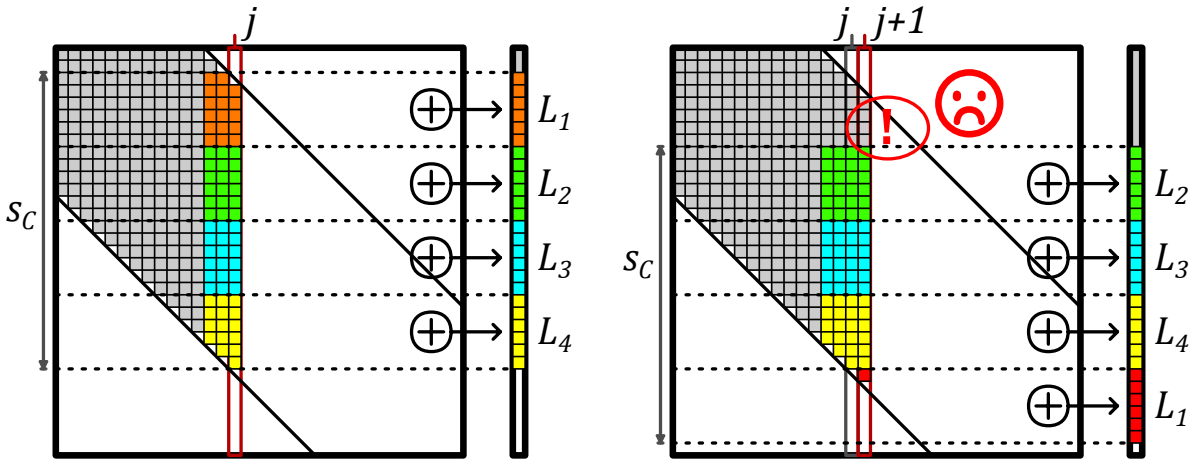
In the previous chapter, we presented SpDCache and a *reduction cache* baseline with a specific region (*DRC*) dedicated to storing the banded part of the matrix. We defined the band-width of the matrix based on the *DRC* size. In this section, we explain and justify this choice by analyzing the behavior of a *reduction cache* when processing an `OP` SpMV kernel with a 'perfectly' banded matrix.

We show that the *reduction cache* must be 'properly' dimensioned with respect to the band width of a 'perfectly' banded matrix, in order to avoid unnecessary evictions. Indeed, if the cache is too small compared to the band, evictions occur before the column iteration is over, leading to poor data-reuse for the next iteration. To achieve high data-reuse, we show that the cache-size (s_C) and the band width (w_B) must respect the relationship given in Equation 4.2.

$$s_C \geqslant w_B + l - 1 \tag{4.2}$$

We consider two examples with a cache having four cache-lines (L_1 to L_4) with $l = 6$. In Figure 4.2, we illustrate the case when the cache is too small ($s_C < w_B + 5$). In Figure 4.2a, we see the state of the cache after iterating up to column j. Then, in Figure 4.2b, we see the state after processing the next column, $j + 1$. The cache-line L_1 (*orange*) is evicted, and allocated to new elements at the bottom of the band, below the *yellow* area. In this case, there are still elements of the band above the *green* area, whose corresponding cache-line

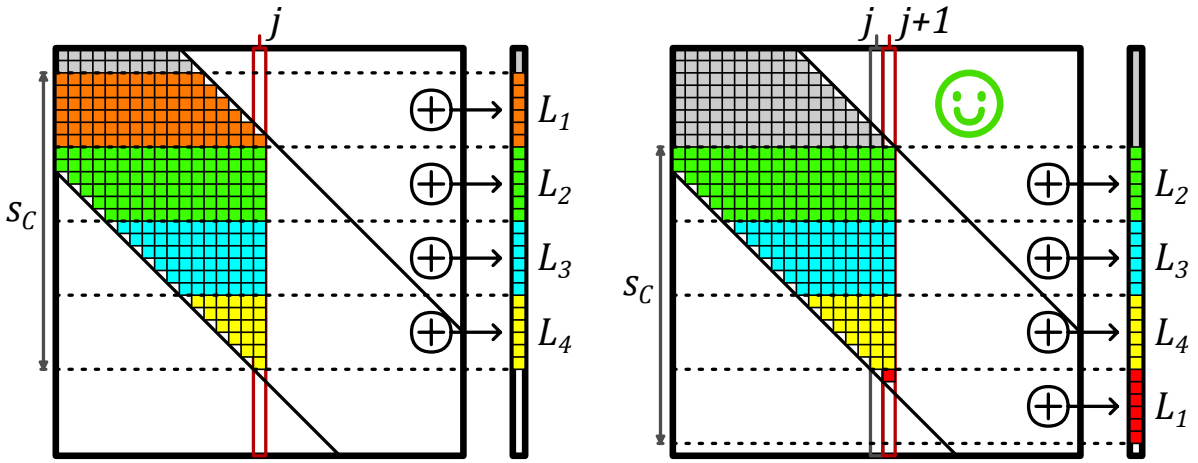
has, unfortunately, been evicted. Thus, on iteration $j+2$, cache-line L_1 must be fetched again and this pattern continues creating *thrashing*.



(a) Column iteration j **(b)** Column iteration $j+1$

Figure 4.2: Example of the Eviction Rotation in a Perfect Band when the Cache is Undersized ($s_C < w_B + l - 1$)

In Figure 4.3, we illustrate the case when the cache is 'properly' sized ($s_C \geqslant w_B + 5$). As in the previous example, Figures 4.3a and 4.3b show the state of the cache at iterations j and $j+1$, respectively. This time, the cache-line L_1 is evicted only when all elements of the band in the *orange* region have been processed. Indeed, the eviction of cache-line L_1 occurs just in time so that it can be allocated to the new region at the bottom of the band, below the *yellow* area. In this way, when processing columns, the cache-lines do a rotation where the top one is evicted and allocated at the bottom, with one eviction occurring after the processing of every l columns.



(a) Column iteration j **(b)** Column iteration $j+1$

Figure 4.3: Example of the Eviction Rotation in a Perfect Band when the Cache is 'Properly' Sized ($s_C = w_B + l - 1$)

If the cache-size is larger than required by Equation 4.2, the rotations and the eviction-rate are the same. To produce this perfect cache rotation, we assume an LRU eviction policy and dense vertical-strips, although, even with a *random* policy, the general pattern still holds.

This analysis shows that, with a "properly-sized" *reduction cache*, a 'perfectly' banded matrix provokes no unnecessary evictions and the cache-lines are fully utilized. With a given cache-size, Equation 4.2 defines the maximum band width that can be efficiently stored in the *reduction cache* or in the *DRC* of SpDCache. For example, with a cache-size of $s_C = 2048$ elements (16 KiB) and a cache-line size of $l = 8$ elements, the maximum band width is $w_B = 2041$ elements (which means a half-band of $w_{hB} = 1020$ elements).

Large, 'perfectly' banded matrices are rare in practice, especially with such narrow -width. In the next sections, we extend this analysis to banded matrices, studying the impact of out-of-band non-zero elements on the *reduction cache* behavior.

CONCLUSION

A *reduction cache* can achieve an efficient **rotating pattern** when processing a 'perfectly' banded matrix, provided that the cache-size and band width respect the **relationship given in Equation 4.2**: $s_C \geqslant w_B + l - 1$. This relationship defines the **maximum band width** that can be efficiently stored in the *reduction cache* for a given cache-size.

4.4.2 Impact of Out-Of-Band Density on Memory Traffic

We now apply the analytical model of Section 4.3.1 to study the performance of a *reduction cache* for banded matrices, processing an `OP` SpMV kernel ($A \times b = c$). We consider banded matrices of dimension $M = 1000$, with a half-band of size $w_{hB} = 125$ and an in-band density $d_{IB} = 1$. We analyse a 4-way set-associative cache ($W = 4$) with a cache-line size of $l = 8$ elements, with a *random* eviction policy. The size of the cache varies from $S = 1$ to $S = 32$ *sets*. When the cache-size reaches $S = 32$, the cache can hold the full output vector: $S \times W \times l \geqslant M$. We developed a C program which implements the hybrid probabilistic-fast simulation model and in Figure 4.4, we plot the normalized memory traffic, $normMT$, for the output vector c, depending on the out-of-band density d_{OB}, for varying cache-sizes.

On the right of the graph, we notice that $normMT$ reaches a peak when the out-of-band density equals $\frac{1}{l} = 0.125$. Indeed, when there is at least one non-zero element per cache-line, the number of main memory accesses approaches that required for a dense matrix. As the density increases further, no additional memory accesses are required, but each line becomes denser, thus $normMT$ drops, as seen in the graph in Figure 4.4.

For the plot corresponding to $S = 32$, the cache is large enough to store the entire output vector c, thus, there are no evictions and $normMT$ is constant, corresponding to the flush traffic at the end. This plot gives the lower bound of memory traffic for the output vector c.

The first takeaway from this graph is that variations in d_{OB} heavily impact $normMT$ (note the logarithmic scale). Isolated non-zero elements outside the band trigger misses and evictions of cache-lines corresponding to the banded region. Furthermore, when a cache-line is allocated for an isolated out-of-band element, most of the cache-line remains empty. In this case, instead of the efficient rotating pattern described in Section 4.4.1, the cache is *thrashing* with low utilisation. Regardless of the cache-size, the lower the out-of-band density, the lower $normMT$, as seen by the plots that tend downward toward the left of the graph. Avoiding the perturbations induced by the out-of-band elements would enable the cache to achieve minimal memory traffic, as with a 'perfectly' banded matrix.

Not surprisingly, at the left of the graph, $normMT$ varies depending on the cache-size. This is due to the relative size of the band and the cache. Indeed, if the matrix band width is large

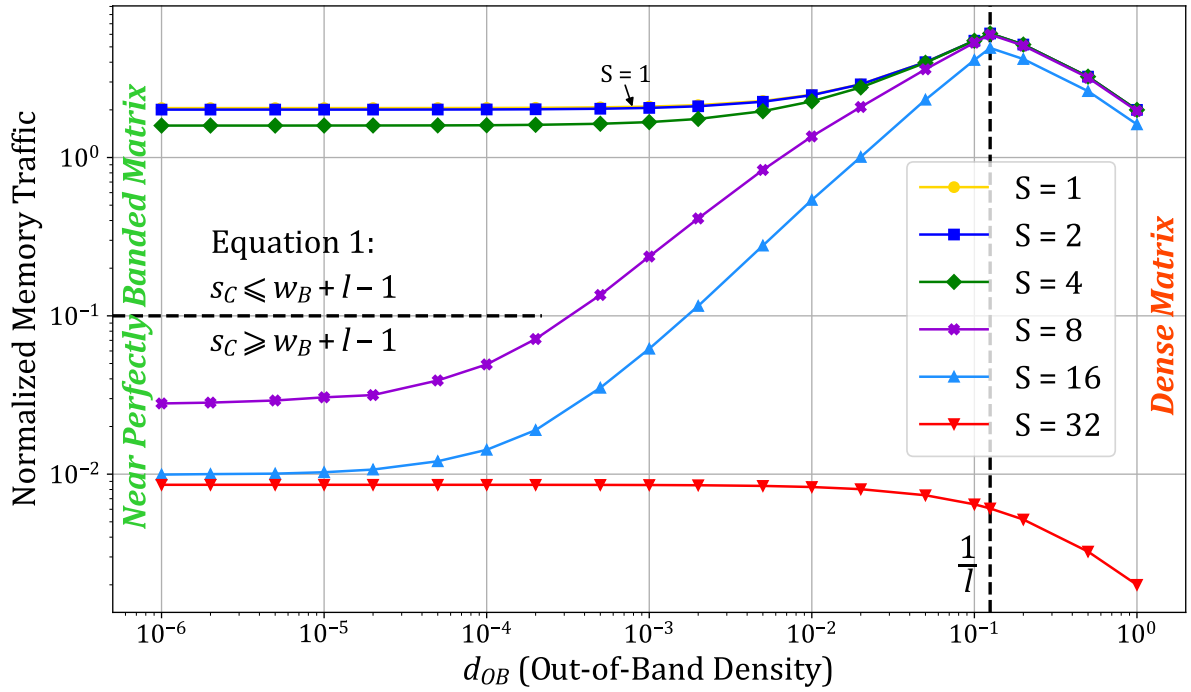


Figure 4.4: Normalized Output Memory Traffic Depending on the Out-of-Band Density for Several Cache-Sizes

compared to the cache-size, processing one column of the band will generate many evictions. To avoid evictions in the band, the relationship given by Equation 4.2 in Section 4.4.1 must be fulfilled. Indeed, for this example, for $S \geqslant 8$, this relationship holds and the corresponding plots have significantly lower $normMT$, up to two orders of magnitude lower. Conversely, the first three plots ($S = 1$ to 4) do not respect this condition, explaining their high $normMT$. Thus, the second takeaway from this graph is that, given a cache memory size, there is an ideal band width that generates minimal $normMT$. Increasing the cache-size beyond this threshold only produces a minor reduction in memory traffic, between $\sim 9 \times 10^{-3}$ and $\sim 2 \times 10^{-2}$, since at the beginning of matrix traversal, the first eviction is deferred before reaching the steady state rotating pattern, as discuss in Section 4.4.1.

These observations explains the need of dividing the *reduction cache* into two regions. One which is optimized for storing the band and the other which handles all the other elements. The in-band region of the cache can thus operate with maximal efficiency corresponding to the bottom-left of Figure 4.4. In subsequent sections, we analyse the out-of-band traffic.

CONCLUSION

The out-of-band density heavily impacts the normalized memory traffic, $normMT$, as the isolated elements trigger unnecessary evictions, preventing the efficient rotating pattern shown in the previous section. Thus, to minimize memory traffic, it is important to **separate** the band and out-of-band regions of the matrix into **dedicated regions** of the *reduction cache*.

4.4.3 Design Analysis for Out-of-Band Region

For the type of large matrices under study, there are usually too many non-zero elements in the out-of-band region for them to be cached from one column to the next, thus making it

difficult to achieve a high-level of reuse from column to column. Therefore, the size of the cache for the out-of-band is not as important as the choice of associativity and data granularity. To justify this affirmation, we use the analytical model from Section 4.3.1 to analyse the performance of the *reduction cache* when processing only the non-zero elements outside of the band. As we have already discussed the handling of the in-band region, we now consider only the out-of-band elements, which is equivalent to setting the in-band density to $d_{IB} = 0$. Using the same matrix parameters as in the previous section, we explore different choices of cache-line size ($l \in \{1, 8\}$ elements), associativity (set-associative or fully-associative), memory transaction type (*R/W* or *RMW*), and total cache-size ($s_C \in \{128, 256\}$ elements). Because the out-of-band data is highly sparse, we want to explore having an element-wise cache ($l = 1$) instead of the typical line-wise cache ($l = 8$). As this cache region can be relatively small, a fully-associative configuration ($S = 1$, $l = 1$) is reasonable in terms of hardware implementation, and this full associativity enables more efficient storage of the data elements. Indeed, when dealing with sparse elements, it is wasteful to bring a full line into the cache to only modify one element and then evict it, which we denote as *R/W*. Thus, we consider offloading the *reduction* to a higher level cache, thus reducing unnecessary data transfers. We assume the higher level cache can support this operation which we denote as *RMW*, following the design choice of SpDCache in Chapter 3. We recall that protocols such as AXI [1] support atomic transactions (AMOs), which could easily be extended for *reduction* operations.

To analyse the impact of the cache-size, we consider two different sizes (*solid* and *dashed*). For a given cache-size, we adjust the number of ways (W), based on the associativity, to keep the cache-size constant. In Figure 4.5, we plot $normMT$ for the output vector c, depending on d_{OB}, for varying cache configurations.

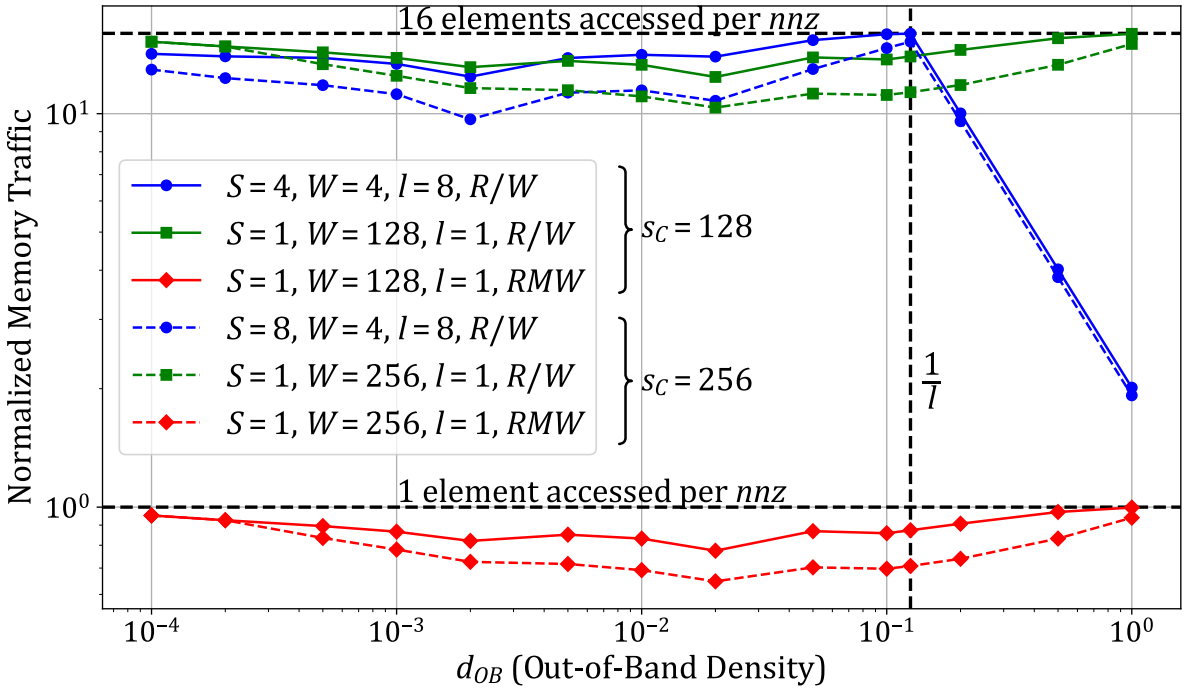


Figure 4.5: Normalized Output Memory Traffic of the Out-of-Band, Depending on the Density for Several Cache-Sizes

Let us start by considering a classic set-associative cache with a line-wise memory inter-

face (*solid blue*, *dot*), where we see a dependence of $normMT$ on d_{OB}. On the right of the graph ($d_{OB} \geqslant \frac{1}{l}$), we observe the same behavior as in Section 4.4.2, where the cache behavior approaches that seen for a dense matrix.

Looking at the plot for the fully-associative cache with a line-wise memory interface (*solid green*, *square*), we see that the normalized traffic is not impacted by the density of the out-of-band, as expected. With the fully-associative cache, $normMT$ is constant near 16, as, on average, there are two line-wise transactions per non-zero element processed, one read to fill the line, one write to evict it.

The *red* plots (*diamond*) correspond to a fully-associative cache with an element-wise memory interface, which can offload element-wise *RMW* transactions. We see that this approach results in over an order magnitude reduction in $normMT$. First, we avoid transferring full cache-lines which are mostly empty, and we only send the *RMW* in one direction ("fire-and-forget"). Indeed, $normMT$ is now reduced to below one transaction per non-zero element processed. The fact that this value is below one shows that this cache region does hit and reduce memory traffic. Increasing the cache-size from 128 (*solid*) to 256 (*dashed*) elements reduces $normMT$ by at most 10^{-1}, the hit-rate being slightly increased as expected.

CONCLUSION

For the out-of-band region of the cache, the most important characteristics are: (1) a **fully-associative** element-wise data-storage to avoid virtually empty cache-lines, and (2) an **element-wise memory interface** to reduce memory traffic. With these two characteristics, the normalized memory traffic can be reduced to below one transaction per non-zero element processed.

4.4.4 Impact of In-Band Density

Up to now in this chapter, we considered a dense band ($d_{IB} = 1$) to show the importance of having a rotating eviction pattern to decrease memory traffic. However, matrices from real applications do not have fully dense bands. Indeed, from our set of 48 matrices used in the previous chapter, considering a band width of $w_B = 2041$ that fits into a *DRC* of 16 KiB ($s_C = 2048$ elements), the average in-band density is $d_{IB} = 2.11 \times 10^{-3}$, ranging from 2.00×10^{-8} and 5.44×10^{-2}, compared to an average out-of-band density $d_{OB} = 1.28 \times 10^{-4}$, ranging from 0.00 and 1.04×10^{-2}. Considering that a part of the matrices of our set are not considered as banded, this order of magnitude difference between d_{IB} and d_{OB} on average across all 48 matrices shows that, even in matrices that do not appear to be banded, it is reasonable to assume that the region along the diagonal is much denser ($10\times$) than the region away from the diagonal.

Since the in-band density is lower than 1 in practice, we use the model to analyse the impact on normalized memory traffic of d_{IB} variations. We use the same parameters as in the previous sections, a matrix of dimension $M = 1000$ and a cache with four *ways* ($W = 4$) and cache-lines of size $l = 8$. We do not consider the out-of-band ($d_{OB} = 0$) since it is processed in a separate region. We focus on the in-band region for several cache configurations, making sure that the band width for each *set* size follows the relationship in Equation 4.2. In Figure 4.6, we plot $normMT$ for the output vector c, depending on d_{IB}, for varying cache configurations.

We observe that $normMT$ increases as d_{IB} decreases, generally following a constant negative slope (*dashed*) on the right side of the graph for each cache size. Thus, across a broad

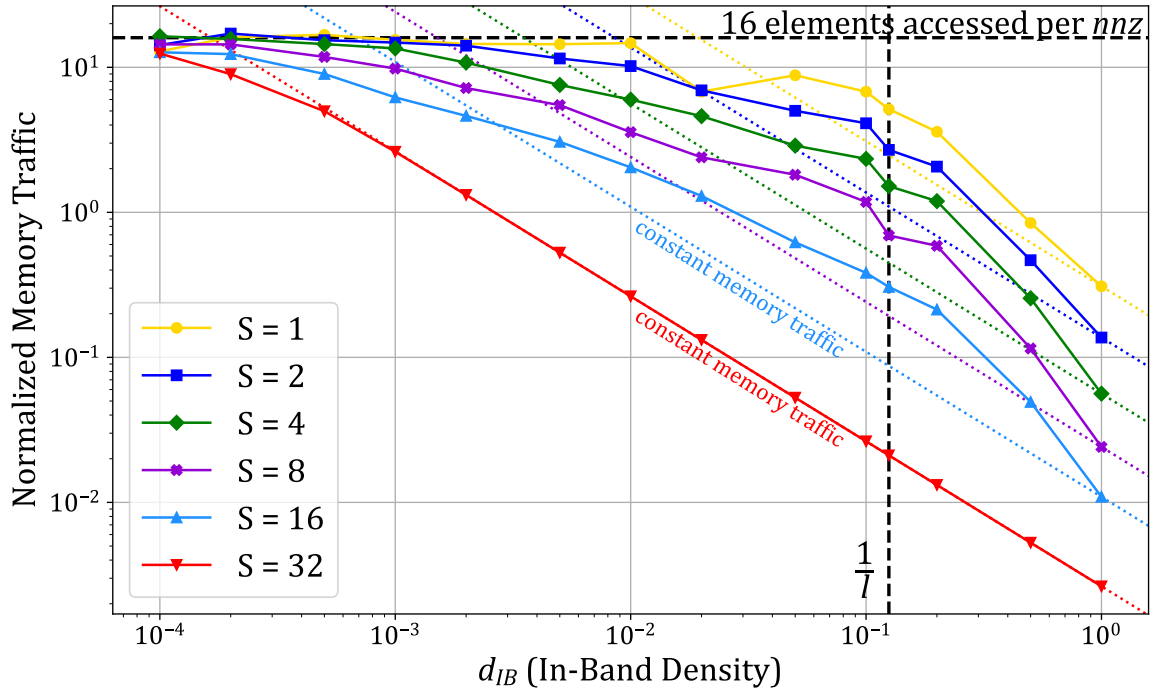


Figure 4.6: Normalized Output Memory Traffic of the In-Band, Depending on the Density for Several Cache-Sizes

range of in-band densities, the raw memory traffic remains relatively constant. As noted in Section 4.4.2, the $S = 32$ curve indicates the lower bound of memory traffic since the entire matrix can fit within the cache. In this scenario, there are no evictions in the band until the density drops sufficiently low ($< 10^{-3}$), resulting in isolated non-zero elements. For other cache sizes, variations around the constant raw memory traffic are observed. With an ideal eviction policy, allowing the rotating pattern to function as intended (refer to Section 4.4.1), we would expect all curves to align with their respective *dashed* lines, indicating constant raw memory traffic. However, the model employs a *random* policy, which disrupts the rotating pattern and accounts for the observed deviations. An *LRU* policy would likely yield improved results, aligning with the constant line for densities above $1/l$ and deviating for lower densities. The optimal policy would prioritize evicting lower addresses (the top of the band), thereby maintaining the rotating pattern.

For low values of d_{IB} (left side of the graph), $normMT$ plateaus at 16 elements transferred per non-zero element for all cache sizes. With very low density the non-zeros are isolated and provide no reuse: each non-zero triggers a full cache-line read and a full-line write-back. With $l = 8$ this corresponds to 16 elements transferred per non-zero.

It is therefore important to select the densest region of the matrix to be processed in a dedicated cache region such as the *DRC* of SpDCache. However, the selected region need not be highly dense: for a 'perfectly' banded matrix with $M = 1000$, columns become nearly empty for $d_{IB} \lesssim 10^{-3}$, yet the raw memory traffic remains nearly constant in the range 10^{-2} to 10^{-3} depending on cache size. Because real, larger matrices often have a mean d_{IB} around 10^{-3}, a *DRC* sized to capture this denser region still provides good memory-traffic performance.

NOTE

In the SpDCache architecture, the *DRC* employs an *LRU* or pseudo-*LRU* eviction policy instead of an ideal custom policy. Despite this, we show that these simple, well-known policies still deliver good performance.

CONCLUSION

The **in-band density** has only a **minimal impact on memory traffic**, with real matrices typically exhibiting in-band densities around 10^{-3}.

4.4.5 Resulting Approach and Discussion

In the previous sections, we analyzed the behavior of *reduction caches* executing an `OP` SpMV kernel with banded matrices. We evaluated the expected memory traffic using an analytical model, separately considering the in- and out-of-band regions. We identified the cache characteristics required for both regions, which are quite different. We thus validate the proposition to divide the available cache memory into two dedicated regions: one for the in-band, and one for the out-of-band; the *DRC* and *SRT* of SpDCache, respectively.

To obtain the efficient rotating pattern for the in-band region, which is the key to massively reducing memory traffic, Equation 4.2 must be respected. In practice, this defines the size of the band that can be allocated to the in-band cache region. As we have seen, this cache region can operate efficiently with a set-associative line-wise configuration, on a wide range of in-band densities.

For the out-of-band region, a fully-associative element-wise cache with an element-wise memory interface can significantly reduce the memory traffic.

When sizing the two regions, once Equation 4.2 is satisfied, enlarging the in-band region yields only a marginal reduction in memory traffic (on the order of 10^{-2}). By contrast, a moderate increase of the out-of-band region produces a larger reduction (on the order of 10^{-1}). Therefore, the out-of-band region should be made as large as practical, subject to the hardware limits imposed by full associativity. This correlates with the results presented in Chapter 3, where we showed better performance with SpDCache configurations having a larger *SRT*.

Although this analysis is sufficient to guide the design of SpDCache for `OP` SpMV with banded matrices, it is important to note that the analytical model that we have presented is based on the assumption of a uniform distribution of the non-zero elements. In practice, this is not always the case, and this non-uniformity can impact the cache behavior, as will be seen later in Chapter **??**. Indeed, if the non-zero elements are clumped, the cache-lines tend to be better filled, resulting in lower memory traffic. In other words, our uniform assumption leads to a pessimistic prediction of memory traffic.

CONCLUSION

To minimize memory traffic when processing banded matrices with a *reduction cache*, it is important to **divide** the cache into two dedicated regions: (1) an **in-band region** dedicated to a band in the matrix whose size respects the relationship in Equation 4.2, and configured as a **set-associative line-wise** cache, and (2) an **out-of-band region** configured as a **fully-associative element-wise** cache with an **element-wise memory interface**. The **out-of-band region** should be made as large as practical, as it has a greater impact on memory

traffic than enlarging the in-band region.

4.5 Conclusion

This chapter presented a theoretical model to analyze the behavior of *reduction caches* when processing sparse matrices, specifically focusing on the `OP` SpMV kernel. A C program implementing a hybrid probabilistic/fast simulation model was developed, and we demonstrated how the structure of banded matrices can be leveraged to optimize cache performance.

Our analysis highlighted the importance of separating the in-band and out-of-band regions of the matrix into dedicated cache regions. For the in-band region, a set-associative, line-wise cache configuration ensures efficient data reuse by maintaining a rotating eviction pattern, minimizing unnecessary memory transactions. For the out-of-band region, a fully-associative, element-wise cache with an element-wise memory interface significantly reduces memory traffic by avoiding the transfer of mostly empty cache-lines.

The theoretical insights gained from this model validate the architectural choices of SpDCache, particularly the use of distinct regions for denser and sparser data. These findings provide a robust foundation for designing and sizing cache regions to minimize memory traffic, ultimately improving the efficiency of sparse matrix computations. In the next chapter, we will delve into the microarchitectural details of SpDCache, translating these theoretical insights into practical hardware design.

TAKEAWAYS

Simulating *reduction caches* and SpDCache

To study the concepts presented in this thesis, three models were developed with varying levels of abstraction. Starting from the most abstract model:

- **High level probabilistic/fast model in C** (4.3):
  - fast simulation ($\sim 10^5$ nnz/s), using generic matrix density distributions, easy parameter exploration, deep understanding of cache behaviors, but,
  - very high level (some approximations and hypothesis), limited on eviction policies.
- **Transaction level model in Python** (3.6):
  - faster than a full RTL simulation ($\sim 10^3$ nnz/s), accurate performance on memory traffic, functional testing, but,
  - slower than the probabilistic model, demands real matrices as input, two factors that alleviate the parameter exploration capabilities, no latency performance compared to RTL simulation.
- **RTL implementation** (next chapter):
  - accurate results in both memory traffic and latencies, hardware testing, but,
  - slow simulation ($\sim$ 10 nnz/s), demands real matrices in input, time-consuming implementation.

Validation and sizing of SpDCache three regions

- ***GRC***: can take advantage of prefetching methods, size depends on nnz_c.
- ***DRC***: must respect the relationship $s_C \geqslant w_B + l - 1$ (Equation 4.2).
- ***SRT***: internal storage and memory interface benefit from element-wise granularity, benefits from a larger size but must achieve a compromise with hardware cost.

To simplify address decoding, the initial region sizing for SpDCache was chosen somewhat arbitrarily to be **25/50/25%** for the *GRC*, *DRC* and *SRT*, respectively. Our analysis later

confirmed that this partitioning is actually close to optimal for memory traffic performance.

Chapter 5

Microarchitecture and Evaluation

Contents

5.1 Introduction

In the previous chapters, we established the theoretical foundations and high-level design principles for SpDCache, a specialized cache architecture tailored to optimize memory transactions for sparse matrix computations. This chapter bridges theory and practice by presenting the microarchitectural implementation of SpDCache and evaluating its performance through Register Transfer Language (RTL) simulations.

We begin by detailing the microarchitecture of SpDCache, focusing on its three distinct regions: the *Generic Region Cache* (*GRC*), the *Dense Region Cache* (*DRC*), and the *Sparse Region Table* (*SRT*). Each region is designed to handle specific types of memory accesses, sequential reads, dense *reductions*, and sparse *reductions*, respectively, thereby addressing the irregular memory access patterns inherent in sparse matrix operations. The RTL implementation of SpDCache is described, highlighting the hardware components and pipeline stages that enable efficient data processing and *reduction* operations.

Following the microarchitectural overview, we present the evaluation methodology and results obtained from RTL simulations. These simulations provide a detailed assessment of SpDCache's performance, including memory traffic reduction and latency improvements. We analyze the behavior of SpDCache across a diverse set of sparse matrices, comparing its performance against baseline cache architectures and validating the theoretical insights gained from earlier chapters.

By translating the theoretical model into a practical hardware design, this chapter demonstrates the effectiveness of SpDCache in minimizing memory traffic and improving computational efficiency for sparse matrix kernels. The results underscore the importance of region-based caching and *reduction* support, paving the way for further optimizations and scalability enhancements discussed in subsequent chapter.

5.2 SpDCache Microarchitecture

SPDCACHE is implemented into an L1 data cache for a CVA6 RISC-V core, building upon an existing open-source high-performance set-associative data cache known as HPDcache [20]. In contrast to traditional data caches, HPDcache introduces new features, including buffering techniques for memory writes, set-associative storage to manage read miss requests, and dynamic configurability of each cache-line to operate in either write-back or write-through modes. It is designed for high performance, utilizing a three-stage pipeline to handle multiple instructions while minimizing stall potential through out-of-order cache execution.

An overview of SpDCache's microarchitecture is shown in Figure 5.1, where *black* and *blue* components are originally from HPDcache. The first stage (*st0*) decodes instructions and issues reads to the cache directory and data RAMs (*Dir* and *Data*). These reads require one cycle to respond. In the second stage (*st1*), the response indicates whether a hit or miss occurred and if an eviction is necessary. The instruction proceeds to the final stage (*st2*) only if a miss or eviction must be processed. At *st0*, the instruction is not yet consumed. Depending on the response and the availability of cache resources at *st1*, a *no-operation* (*nop*) may be inserted in the pipeline, and the instruction may be placed in a *replay table*. The instruction is then replayed with the highest priority once its dependencies are cleared. If the instruction does not need to be replayed at *st1*, it is consumed and processed. This pipeline serves as the foundation for implementing SpDCache's *reduction* operations.

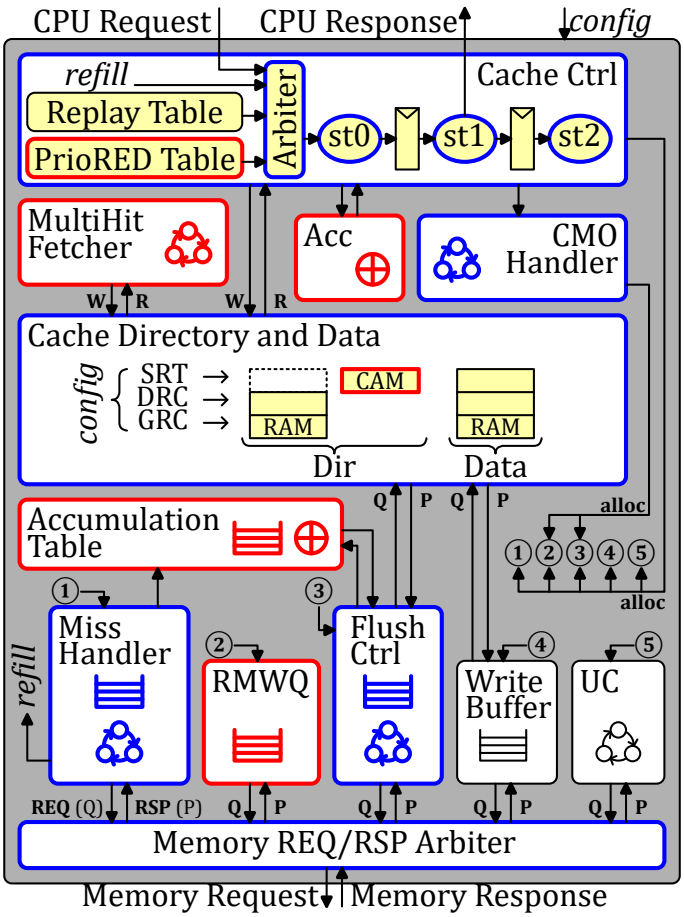


Figure 5.1: SpDCache Microarchitecture, based on HPDcache

The *cache controller* coordinates several auxiliary units. The *miss handler*, allocated from *st2*, tracks outstanding memory read requests and notifies the cache controller with a *refill* request when responses arrive. An optional *write buffer* gathers recent write to memory, allowing merges of transactions that target the same cache-line. The *flush controller*, also allocated from *st2*, reads cache-line data from the *Data* RAMs and issues memory writes. It temporarily stall the pipeline to prevent RAM conflicts. HPDcache further provides an *UC* unit to handle uncached transactions. Finally, the *CMO handler* performs internal cache operations (e.g., invalidations and flushes). When active it takes control of the cache and can invoke the *flush controller* if necessary. Most of these units are reused and extended to implement SpDCache.

HPDcache provides extensive static and dynamic configurability, making it a flexible foundation for further extensions. To implement SpDCache as well as a *reduction cache* baseline, we modified HPDcache to be highly configurable so that a single RTL implementation can produce multiple designs. The RTL can be instantiated as a generic cache (the first baseline, GC), as a *reduction cache* (the second baseline, RedC), or as SpDC, the solution presented in the previous chapters.

CONCLUSION

HPDcache is an **open-source**, high-performance cache which we use as the **RTL basis** for our work. We extended it to support *reduction* operations and the SpDCache mode while preserving its **configurability**.

5.2.1 RTL Implementation of SpDCache

We configure the original HPDcache as a 4-way set-associative cache with 128 *sets* and 64-byte cache-lines, resulting in a total cache size of 32 KiB. We extend the three-stage pipeline to handle the new *RED* instruction. Figure 5.1 provides an overview of SpDCache's microarchitecture, highlighting new blocks in *red*, modified blocks in *blue*, and untouched components in *black*. The overall architecture aligns with the design presented in Chapter 3, adapted here for hardware implementation. Let us briefly describe the operation of the *DRC* under three scenarios: hits, misses with available cache-lines, and misses requiring eviction.

In the case of a hit, the value of the *RED* instruction is accumulated with its corresponding value in the cache. At *st1*, the cache value, read in *st0* from *Data*, is accumulated with the *RED* value using a dedicated accumulator (*Acc*). The result is written back to *Data* at *st2*.

In the case of a miss, if there is an available cache-line, the value of the *RED* instruction is simply stored in this free entry of the cache. A free cache-line is selected at *st1*, and at *st2*, the *RED* value is written to *Data* along with zeros to reset the selected cache-line. A write to *Dir* is also performed to change the state of the cache-line from free to valid.

If a miss occurs but no cache-line is free, an eviction must be processed before handling the *RED* instruction. The *RED* instruction is placed in the *PrioRED table* to be replayed with higher priority than the *replay table*. Meanwhile, at *st2*, the *miss handler*, *flush controller*, and *accumulation table* are allocated to process the eviction. The eviction is a "fire-and-forget" operation, meaning the *RED* instruction only needs to wait for the entry in *Data* to be freed, not for the entire eviction to finish. The eviction process begins by fetching and resetting the chosen 64 B cache-line from *Data*, which is done in one cycle by the *flush controller*. Simultaneously, the *miss handler* issues a read to the main memory. The *accumulation table*

waits and temporarily stores both the old cache-line from the *flush controller* and the 64 B response data from the *miss handler*. It then accumulates the cache-line with the data, word by word, and sends the 64 B results back to the main memory using the existing write interface.

With the changes described above, we have transformed HPDcache into a *reduction cache*. To achieve multiple cache regions, we virtually separate the cache RAMs. To maintain HPDcache's set-associative operation, we divide the cache memory along its *sets*. Thus, each region of the cache has the same structure as the overall cache but is smaller in size: each *set* contains a fixed number (the number of ways) of 64-byte cache-lines, as depicted in Figure 5.2. Each region can map any possible address. To ensure that every address matches a unique *set* within each region, we allocate a power-of-two number of consecutive *sets* to a region. Considering the total number of *sets* S_T and our region R mapping on $S_R < S_T$ of them, starting at the *set* s_{offset}, if an address matches one of the *sets* of this region, no action is required. However, if the address matches a *set* s_{addr} outside the region, we need to map it to a unique *set* in the region such that $s_{region} = (s_{addr} \% S_R) + s_{offset}$, as illustrated in Figure 5.2. The three regions *GRC*, *DRC*, and *SRT* are each defined by their number of *sets* S_R and their first *set*, s_{offset}. SpDCache can thus be reconfigured to operate as a generic cache, a traditional *reduction cache* or a region-based *reduction cache* based on the needs of the application (*config* in Figure 5.1).

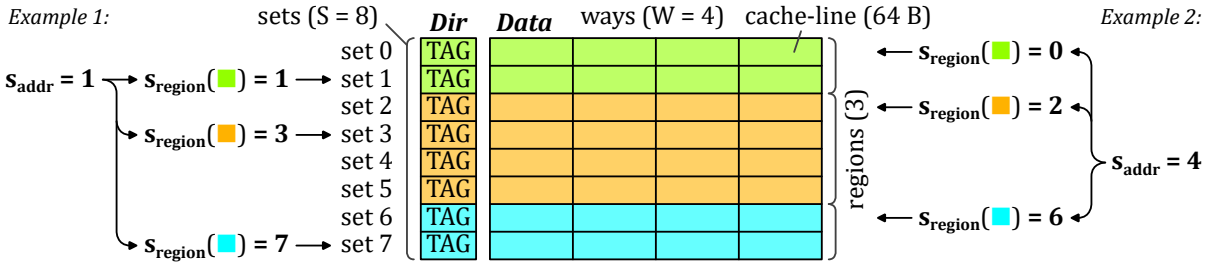


Figure 5.2: Set Redirection Depending on the Region

The *SRT* region requires additional adjustments to function as an element-wise fully-associative cache. This region operates as if all the *sets* were combined into a single *set*, containing numerous elements. Each element requires its own directory entry. Therefore, we replace the *Dir* RAMs of this region with a CAM (*Content-Addressable Memory*) containing the addresses of each valid element in the *SRT* region. With the CAM, we can check for a hit across the entire region and retrieve the matching value stored in the *Data* RAMs. We also added a queue (*RMWQ*) to manage the element-wise *Read-Modify-Write* (*RMW*) memory transactions.

To avoid address duplication between regions, the *MultiHit fetcher* in Figure 5.1 manages coherency between the *DRC* and *SRT*, following the description in Chapter 3. Since software selects the region, this can be necessary. The *MultiHit fetcher* performs two tasks. First, given an address, it probes the CAM for entries whose cache-line address matches. If one or more matches are found, it records their positions locally and notifies the *cache controller*. Second, when activated, the *MultiHit fetcher* gathers the corresponding elements from the *Data* RAMs of the *SRT* and assembles them into a single 64 B line that will be written into the *DRC*.

Contrary to the description in Chapter 3, we do not transfer isolated elements from the *DRC* to the *RMWQ* on eviction. In our Python simulations we observed that only a negligible fraction of elements were transferred (mean 0.3%, range 0 to 5% across the 48 matrices),

which demonstrates the effectiveness of the *DRC* blocking behavior. To simplify the RTL implementation, this transfer is therefore omitted.

CONCLUSION

The SpDCache implementation introduces two primary extensions. First, it integrates **support for *reduction* instructions (*REDs*)** into the cache pipeline while preserving the generic cache behavior. This entails mechanisms for accumulating entire cache-lines and for handling element-wise atomic *RMW* transactions. Second, it implements a **virtual partitioning of the cache RAMs**: the memory is logically split and a *set*-redirection scheme is applied to dynamically create multiple regions.

5.2.2 Software Interface

We have added several custom instructions to the CVA6 RISC-V core [59]. The primary addition is the *reduction* instruction (*RED*), which takes an address and a value as arguments and expects no response. This instruction is accompanied by a region instruction (*REG*) that specifies in which region subsequent *reductions* should be processed. We have also added an instruction to flush the *reduction* regions (*DRC* and *SRT*) of the cache, which is necessary at the end of the algorithm. It is performed in the cache by the *CMO handler*. Finally, we have included an instruction to enable and disable the *reduction* mode of SpDCache.

We demonstrate a simple example in Source Code 5.1 that accumulates multiple values at a single address. Before executing *RED* transactions, the *reduction* engine must be enabled (line 3). The cache transitions from a fully generic state to SpDC with multiple regions using a hardware-defined static configuration. This example assumes a configuration with three regions: *GRC*, *DRC*, and *SRT*. For a simpler *reduction cache* configuration with only two regions (*GRC* and *DRC*), the *REG* instruction would be unnecessary. Additionally, the instruction can accept a dynamic configuration argument specifying word granularity and operation type. A future optimization would allow passing the entire configuration as a parameter, enabling dynamic configuration of SpDC regions at runtime. Next, at line 8, we select the *IBR* region using the *REG* instruction. The subsequent two *RED* instructions are then executed and stored in the *DRC*. At line 13, we switch to the *OBR* region, causing the following *RED* instruction to be processed by the *SRT*. Once all *RED* operations complete, SpDCache is flushed at line 17 before disabling the reduction engine, reverting to standard cache operation.

CONCLUSION

SpDCache provides a simple and flexible software interface through a minimal set of custom RISC-V instructions. The ability to dynamically enable and disable *reduction* mode allows the cache to seamlessly transition between generic and specialized operation, adapting to application requirements.

5.2.3 Details on SpDCache Pipeline Operation

In this section, we detail the SpDCache pipeline architecture, which extends the HPDcache original pipeline by adding new behavior for *reduction* operations while preserving existing functionality. Figures 5.3 and 5.4 illustrate the various scenarios that occur when processing *RED* instructions, depending on the selected region (*IBR* or *OBR*, respectively). We represent the three pipeline stages (*st0*, *st1*, and *st2*) and their interactions with the auxiliary units

Source Code 5.1: A Simple Example of *RED* Operations Usage

```
1  double data = .0; /* Initialize a data with 0 */
2
3  redEngineOn(config); /* Start the RED engine */
4  /* In this example, SpDCache is configured into three regions */
5  /* We can pass on a dynamic configuration to set the word granularity
6     and the type of operation */
7
8  REG(IBR); /* Following reductions are processed by the DRC */
9
10 RED(&data, 1.); /* data += 1 */
11 RED(&data, 2.); /* data += 2 */
12
13 REG(OBR); /* Following reductions are processed by the SRT */
14
15 RED(&data, 3.); /* data += 3 */
16
17 redFlush(); /* The DRC and SRT are flushed to memory */
18
19 redEngineOff(); /* Stop the RED engine */
20 /* SpDCache is now back to one fully generic cache region */
21
22 printf("%lf", data); /* Should print 6 */
```

involved in processing *RED* instructions. The *green*, *red*, and *blue* arrows represent read and write operations, and unit allocations, respectively. The *Dir* and *Data* RAM units, marked with an hourglass, require one full cycle to respond.

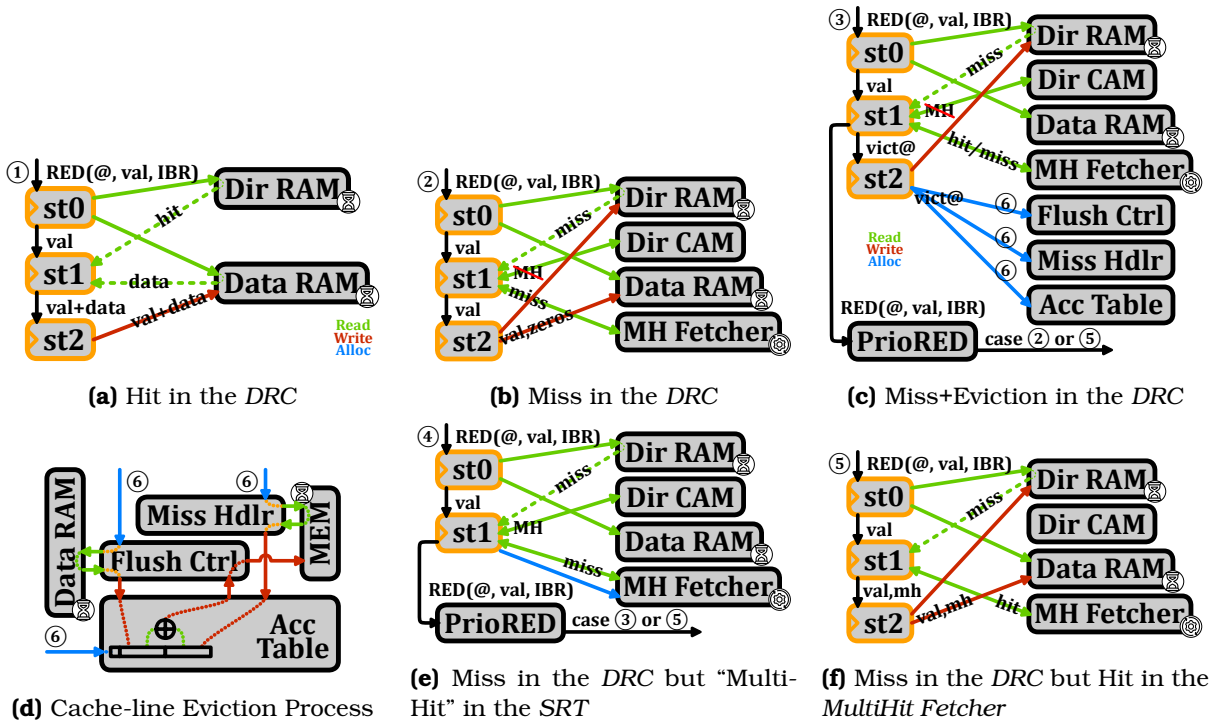
We now describe the scenarios when the region argument is *IBR*. In Figure 5.3a, the pipeline issues reads to the *Dir* and *Data* RAMs at *st0*. Responses arrive one cycle later at *st1*, indicating whether a hit or miss occurs in the *DRC*. This initial pipeline stage is identical across all scenarios. In case ①, a hit is detected, so the data received at *st1* is accumulated with the *RED* value and forwarded to *st2* for write-back to the *Data* RAM.

When a miss occurs, multiple scenarios are possible (Figures 5.3b to 5.3f), as an eviction may be required and the *MultiHit Fetcher* may be active. In case ②, upon detecting a miss at *st1*, the pipeline checks the *Dir* CAM of the *SRT* for a "multi-hit". As described in Chapter 3, when a miss occurs in the *DRC*, we gather elements with the same cache-line address from the *SRT* to prevent duplication. This check completes within the cycle and notifies the *MultiHit Fetcher*. If a "multi-hit" is found, the *MultiHit Fetcher* allocates an entry and takes control of the pipeline to read and invalidate matching entries from the *Data* RAM, storing them locally for later use. If no "multi-hit" occurs (case ②), we handle a simple miss by writing the *RED* value to a free cache-line in the *Data* RAM at *st2*.

In case ④, a "multi-hit" is detected and the *MultiHit Fetcher* is allocated. The *RED* instruction is placed in the *PrioRED Table* for immediate replay once the pipeline becomes available, proceeding to case ③ or ⑤ depending on whether an eviction is needed.

In case ⑤, which occurs only upon replay, the *RED* instruction misses in the *DRC* but hits in the *MultiHit Fetcher*, which has cached the gathered data from the previous case ④. The *RED* value is then accumulated with the *MultiHit Fetcher* data and written to the *Data* RAM at *st2*.

Case ③ handles the eviction scenario, which applies whether or not a "multi-hit" replay is involved. When an eviction is required, a victim cache-line is selected at *st1* and must be evicted before inserting the new value. The eviction process begins at *st2*, with the *Miss Handler*, *Flush Controller*, and *Accumulation Table* allocated for the victim address. Concurrently, the *RED* instruction is placed in the *PrioRED Table* for immediate replay. Once the victim



(a) Hit in the *DRC*

(b) Miss in the *DRC*

(c) Miss+Eviction in the *DRC*

(d) Cache-line Eviction Process

(e) Miss in the *DRC* but "Multi-Hit" in the *SRT*

(f) Miss in the *DRC* but Hit in the *MultiHit Fetcher*

Figure 5.3: Pipeline Operation when the Region Argument is *IBR*

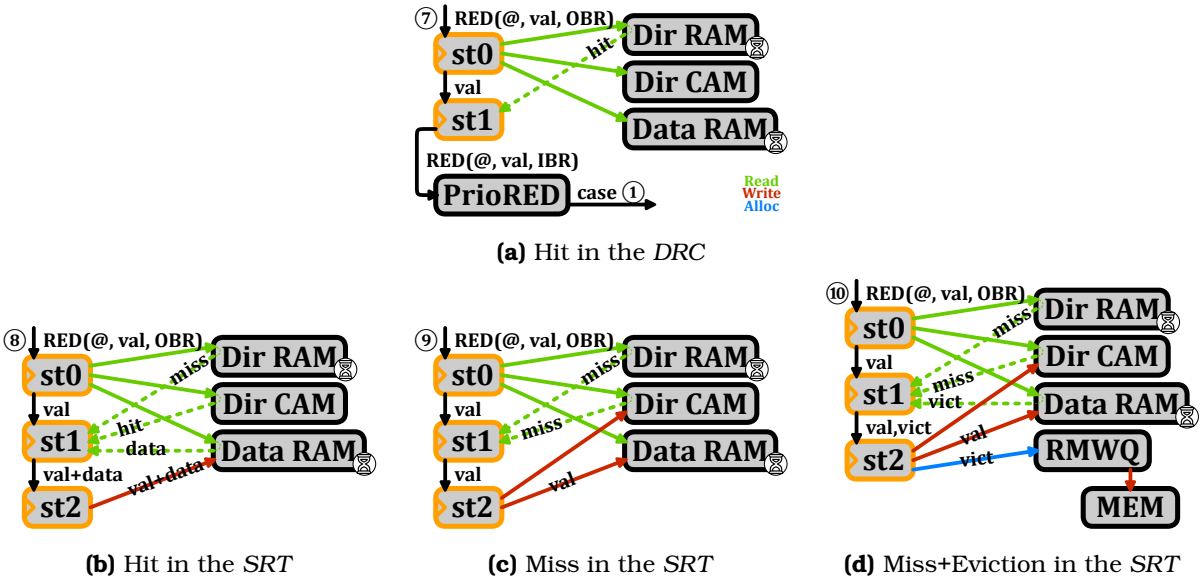
cache-line is freed, the instruction is replayed as case ② or ⑤, depending on whether data exists for the same cache-line address in the *MultiHit Fetcher*. Note that when a "multi-hit" occurs (case ④), the *MultiHit Fetcher* retains the gathered data until reaching case ⑤. Subsequently, since the data is no longer present in the *Dir* CAM and *Data* RAM, no "multi-hit" will occur for that address on replay.

Figure 5.3d depicts the eviction process, which executes in parallel with the pipeline. Upon receiving the victim address in case ③, the *Flush Controller* immediately reads the *Data* RAM and invalidates the entry in a single cycle. This allows the pipeline to replay the *RED* instruction without waiting for the entire eviction to complete. The *Flush Controller* then forwards the victim data to the *Accumulation Table* for temporary storage. Meanwhile, the *Miss Handler* issues a memory read, which may take up to one hundred cycles in worst-case scenarios. Upon receiving the data from memory, the *Miss Handler* sends it to the *Accumulation Table*. Once the *Accumulation Table* has both data lines, it performs word-by-word accumulation and returns the result to the *Flush Controller*, which writes it back to memory to complete the eviction.

The following sequences enumerate all possible cases when the region argument is *IBR*:

- hit in the *DRC*: ①;
- miss in the *DRC*: ②;
- miss+eviction in the *DRC*: ③ + ⑥ → ②;
- miss in the *DRC* with "multi-hit" in the *SRT*: ④ → ⑤;
- miss+eviction in the *DRC* with "multi-hit" in the *SRT*: ④ → ③ + ⑥ → ⑤.

We now describe the cases when the region argument is *OBR*, as illustrated in Figures 5.4a to 5.4d. In all cases, we check for hits in both the *DRC* and *SRT* at *st0*. If a hit occurs in the *DRC*, case ⑦, the *RED* instruction is processed in the *DRC* regardless of the region argument.



(a) Hit in the *DRC*

(b) Hit in the *SRT*

(c) Miss in the *SRT*

(d) Miss+Eviction in the *SRT*

Figure 5.4: Pipeline Operation when the Region Argument is *OBR*

It is then sent to the *PrioRED Table* for replay as a *DRC* hit, corresponding to case ①.

If no hit is found in the *DRC*, we check the *Dir* CAM for hits in the *SRT*. In case of a hit, ⑧, the data is accumulated with the *RED* value and written back to the *Data* RAM. In case of a miss, ⑨, the data is inserted into the *Data* RAM. For an eviction, ⑩, the victim data is read directly between *st0* and *st1*, since the *Dir* CAM responds within one cycle and can initiate the victim read before *st1*. Thus, the *RED* value is inserted into the *Data* RAM at *st2*, while the victim data is sent to the *RMWQ*. The *RMWQ* independently issues atomic *RMW* transactions to memory as entries become available.

The following sequences enumerate all possible cases when the region argument is *OBR*:

- hit in the *DRC*, ⑦ → ①;
- hit in the *SRT*, ⑧;
- miss in the *SRT*, ⑨;
- miss+eviction in the *SRT*, ⑩.

NOTE

This implementation leverages most of the original pipeline capabilities while keeping modifications to a minimum. We do not claim this is the optimal design for SpDCache. A ground-up redesign or significant modifications to HPDcache could yield better performance.

CONCLUSION

We implemented **SpDCache** in RTL by extending HPDcache with **minimal modifications**, introducing support for *reduction* operations and multiple cache regions without requiring additional memory. This **configurable design** allows a single RTL implementation to function as a generic cache, a traditional *reduction cache*, or the region-based SpDCache variant, facilitating comprehensive performance evaluation.

5.3 Evaluation

In this section, we aim to evaluate the performance of various cache configurations and their impact on memory traffic and latency during RTL simulations. We will first outline the methodology used for these evaluations, followed by a detailed analysis of the results obtained from the simulations, ending with comparisons with the previous models.

5.3.1 Methodology for RTL Simulations

The starting point is a CVA6 [65] RISC-V core with HPDcache [20] as an L1 data-cache with size 32 KiB ($S = 128$, $W = 4$, $l = 8$ elements). This serves as our baseline for performance comparison which we refer to as the Generic Cache (GC). Next we evaluate a *reduction cache* which we derived from the HPDcache and configured to have two regions: a generic region with 16 KiB, and a *reduction* region with 16 KiB whose behavior is the same as the *DRC* region of SpDCache. We refer to this design as RedC. Finally, we evaluate SpDCache (SpDC) with a reference configuration of: *GRC* = 8 KiB, *DRC* = 16 KiB, *SRT* = 8 KiB, which corresponds to the 25%/50%/25% configuration selected in Chapter 3. The configurations are summarized in Table 5.1.

Table 5.1: Region Sizing of the Cache Configurations

Configurations	*GRC*	*DRC*	*SRT*
GC	100% ($S = 128$)	0% ($S = 0$)	0% ($S = 0$)
RedC	50% ($S = 64$)	50% ($S = 64$)	0% ($S = 0$)
SpDC	25% ($S = 32$)	50% ($S = 64$)	25% ($S = 32$)

All our simulations are performed using Siemens Questasim™ 2023.3 and the memory accesses beyond the L1 incur a fixed latency of 64 clock cycles. We do not aim to evaluate a full system in this work, as our goal is to identify performance behavior using real matrices. Thus, we use a single-core, single-cache-level environment. The CVA6 core is not designed for high performance and can process at most one instruction per cycle. The fixed memory latency of 64 cycles is not realistic but provides a reasonable approximation of other memory levels' latencies and helps stabilize results for clearer interpretation.

Our SystemVerilog implementation supports only integer addition up to 64 bits, precluding floating-point accumulations. Extending SpDCache to support floating-point operations remains future work. We employ 64-bit integers throughout and assume this choice does not significantly impact performance results.

A second implementation constraint is that region sizes for *GRC*, *DRC*, and *SRT* must be powers of two in terms of *sets*, unlike the other models. This limitation motivates our choice of a 50%/50%/0% configuration for RedC, which we expect to underperform relative to more flexible configurations. Future optimizations of the implementation could relax this constraint.

CONCLUSION

We evaluate three cache configurations, GC, RedC, and SpDC, using a single-core CVA6 processor with a single cache level.

5.3.2 Sparse Formats Used for thr RTL Simulations

By default, matrices are stored in CSC sparse format, and we use this format for most of our simulations. However, in our evaluation, we try simulating SpDC with a specific format tailored for matrices. As shown in Figure 5.5, compared to a CSC, we added extra information in the ptr table to identify the beginning and end of the band. Indeed, three pointers instead of one are now needed per column, as shown with the example of column 2 in *yellow*. The column is split in one in-band (IB) region in the middle and two out-of-band (OB) regions around. The ptr table size thus becomes $3 \times M + 1$. With this format, we can avoid the computational cost of identifying in which region non-zero elements are, while traversing the matrix, with the drawback of having to preprocess it.

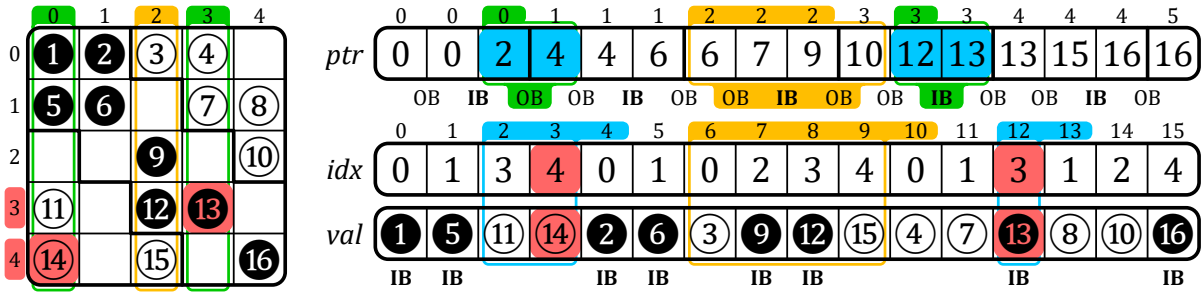


Figure 5.5: Example of a Sparse Matrix in BaCSC Format

CONCLUSION

We employ CSC format for most simulations and additionally evaluate SpDC using BaCSC, a specialized format optimized for banded matrices.

5.3.3 Software Implementation of Outer-Product SpMV Kernel

The code running on the CVA6 is written in generic C, compiled with *gcc* 11.2.0 and -O3 optimization. The *RED*-related instructions used in the `OP` SpMV kernel are manually inserted using *asm* directives, as illustrated in Source Code 5.2.

The code shown is for SpDC. Variants for other designs differ as follows: RedC omits region selection (line 15) and invokes *REG* once before the loops with *IBR* as the argument; GC requires no *REG* instruction and replaces *RED* with standard accumulation: $c[idx] \mathrel{+}= val$.

For SpDC, it is necessary to know the target region for each *RED* instruction based on the size of the band (line 15). The half-band width w_{hB} is derived from Chapter 4 (Equation 4.2) to ensure the in-band region fits within the *DRC* independent of matrix structure. In our reference configuration ($s_C(DRC) = 2048$, $l = 8$), $w_{hB} = 1021$. Region selection follows Equation 5.1 via an inline function, avoiding conditional branches within the inner-loop.

$$\forall (i,j) \in [\![0, M-1]\!] \times [\![0, N-1]\!],\ region = \begin{cases} IBR & \text{if } |i-j| < w_{hB} \\ OBR & \text{else} \end{cases} \tag{5.1}$$

On-the-fly region computation introduces overhead in the critical inner-loop. To mitigate this, we propose an alternative implementation (Source Code 5.3) that precomputes regions and stores them using the BaCSC format. Each column iteration begins with lookups in the outer-loop (line 5) to retrieve pointers for three sections: top out-of-band (*OB1*), in-band

Source Code 5.2: C Code for the `OP` SpMV with *RED* Operations

```
redEngineOn(config);

int currCol = A.ptr[0];
for (int j = 0; j < N; j++) { /* Column first */
    int nextCol = A.ptr[j+1];
    int bj      = b[j];

    for (int pos = currCol; pos < nextCol; pos++) {
        int idx = A.idx[pos];
        int val = A.val[pos] * bj;

        /* Without RED engine: c[idx] += val; */
        /* With RED engine: */
        int *addr    = c + idx;
        region_t reg = regionSelect();

        asm volatile ( /* RISC-V ISA: Type R */ /* REG(reg); */
            "reg x0, x0, %[rs2]"
            :
            : [rs2] "r" ((int) reg)
        );
        asm volatile ( /* RISC-V ISA: Type S */ /* RED(addr, val); */
            "red.w %[rs2], 0(%[rs1])"
            :
            : [rs2] "r" (addr), [rs2] "r" (val)
        );
    }

    currCol = nextCol;
}

redFlush();
redEngineOff();
```

(*IB*), and bottom out-of-band (*OB2*). The column is then processed through three separate inner-loops (lines 13, 21, 29) without region computation overhead.

Source Code 5.3: Optimized C Code for the `OP` SpMV when Using BaCSC

```
1  redEngineOn(config);
2
3  /* A is stored in BaCSC format with pointers delimiting the band sections */
4  int currCol_OB1      = A.ptr[0];    /* begining of the column - first OB section*/
5  for (int j = 0, int k = 0; j < N; j++, k += 3) {
6      int currCol_IB  = A.ptr[k+1]; /* begining of the band   - IB section */
7      int currCol_OB2 = A.ptr[k+2]; /* end of the band        - second OB section */
8      int nextCol     = A.ptr[k+3]; /* end of the column      - next column */
9      int bj          = b[j];
10
11     /* 1st step: top OB section of the column */
12     REG(OBR);
13     for (int pos = currCol_OB1; pos < currCol_IB; pos++) {
14         int idx = A.idx[pos];
15         int val = A.val[pos] * bj;
16         RED(c + idx, val);
17     }
18
19     /* 2nd step: IB section of the column */
20     REG(IBR);
21     for (int pos = currCol_IB; pos < currCol_OB2; pos++) {
22         int idx = A.idx[pos];
23         int val = A.val[pos] * bj;
24         RED(c + idx, val);
25     }
26
27     /* 3rd step: bottom OB section of the column */
28     REG(OBR);
29     for (int pos = currCol_OB2; pos < nextCol; pos++) {
30         int idx = A.idx[pos];
31         int val = A.val[pos] * bj;
32         RED(c + idx, val);
33     }
34
35     currCol_OB1 = nextCol; /* begining of the next column */
36 }
37
38 redFlush();
39 redEngineOff();
```

Each approach trades off one limitation for another. SpDC(CSC) uses simple CSC format with minimal preprocessing and low memory overhead, but incurs additional inner-loop instructions and thus latency per iteration. SpDC(BaCSC) reduces instructions and latency at the cost of matrix preprocessing into BaCSC format and increased memory footprint. Both variants are evaluated in the results section.

These software approaches avoid hardware modifications, providing flexibility. However, hardware-computed regions with CSC format would eliminate the noted drawbacks. A detailed breakdown of each design's instruction characteristics:

- GC: 10 inner-loop instructions; 3 instructions per *RED* with critical load dependency.
- RedC: 8 inner-loop instructions; 1 "fire-and-forget" *RED* instruction; 1 *REG* setup per entire computation.
- SpDC(CSC): 14 inner-loop instructions; 1 "fire-and-forget" *RED* instruction; region computation within inner-loop.
- SpDC(BaCSC): 8 inner-loop instructions; 1 "fire-and-forget" *RED* instruction; outer-loop overhead.

To prevent software from limiting performance, we unroll inner-loops by a factor of 8 (not

shown in Source Codes 5.2 and 5.3). Each unrolled iteration executes 8 logical iterations with optimized instruction scheduling. With cache-line size $l = 8$, this unrolling hides read-miss latency through hits on consecutive addresses. "Random" *reduction* accesses do not benefit from this latency hiding. Consequently, GC experiences potential latency from both *reductions* reads and writes, while RedC and SpDC benefit from "fire-and-forget" semantics and consecutive *reduction* scheduling.

SpDC(CSC) is constrained to 4× unrolling due to register pressure on our platform. This limitation, addressable with a higher-register CPU, demonstrates another advantage of hardware based region computation.

CONCLUSION

We utilize four distinct software implementations of the `OP` SpMV, each customized for our four simulations and fully unrolled to prevent performance constraints.

5.3.4 Matrix Selection for the RTL Evaluation

We have selected an extended set of 62 square matrices from the SuiteSparse Matrix Collection [38] which includes virtually all the matrices used by recent hardware accelerators [45, 51, 63, 66, 67]. We verified that this set is diverse containing banded and non-banded sparse matrices with various sizes ($M \in [1.2 \times 10^2, 2.0 \times 10^6]$), numbers of non-zero elements ($nnz \in [2.2 \times 10^3, 1.4 \times 10^7]$) and a wide range of densities ($d \in [1.4 \times 10^{-6}, 7.8 \times 10^{-1}]$).

Based on our experimental results presented in the following section, we categorize these matrices into two groups according to their performance characteristics. To facilitate this classification, we introduce a new metric, d_{cd}, defined as the average density of cache-line-sized clusters within matrix columns. Specifically, for a cache with line size l, this metric quantifies the average density of non-empty cache-lines (or vertical-stripes as referenced in Chapter 4), ranging between $\frac{1}{l}$ and 1.

This metric provides a fine grain view of matrix density, which aligns more closely with cache characteristics. When d_{cd} approaches $\frac{1}{l}$ for a given matrix, it indicates that most non-zero elements are separated by at least l zeros. Consequently, the cache-lines are often nearly empty, which may render the line-wise *DRC* of a RedC inefficient, while the element-wise *SRT* of SpDC becomes more beneficial. Conversely, as d_{cd} increases, the non-zero elements of the matrix cluster more closely, enhancing the efficiency of the *DRC*.

In our experiments with $l = 8$, we establish two groups of matrices based on a d_{cd} threshold of 20%. This threshold corresponds to a range of one to two non-zero elements per eight. Although there is no definitive threshold, performance can vary significantly depending on the hardware environment. In our scenario, using a simple scalar core, we can clearly distinguish matrices with highly isolated non-zero elements, thus setting the threshold at fewer than two elements per cache line. A complete discussion on this follows in the next section.

According to this threshold, we categorize 21 matrices in group 1 ($d_{cd} \leqslant 20\%$) and 41 in group 2 ($d_{cd} > 20\%$). We have selected 13 well-known matrices from both groups that are frequently used for performance evaluation, as illustrated in Table 5.2. The selected matrices represent a range of overall matrix densities (d) and d_{cd} values, as depicted by the *red* and *purple* points in the scatter plot in Figure 5.6. In the following evaluation, we report the performance results of the 13 matrices from group 1 and 2, the geometric means for both groups on all matrices, and the overall geometric mean for the full set of 62 matrices.

Table 5.2: Characteristics of Selected Group 1 and 2 Matrices

ID	Matrix	M	nnz	Density	d_{cd}
Group 1					
11	poisson3Da	13.5 k	352 k	1.93 e-3	14%
12	cit-HepTh	27.8 k	353 k	4.57 e-4	13%
13	soc-Epinions1	75.8 k	508 k	8.84 e-5	16%
14	sme3Db	29.1 k	2.08 M	2.46 e-3	15%
15	Stanford	282 k	2.31 M	2.91 e-5	12%
16	web-Google	916 k	5.11 M	6.08 e-6	13%
Group 2					
21	msc10848	10.8 k	1.23 M	1.05 e-2	73%
22	opt1	15.4 k	1.93 M	8.09 e-3	86%
23	bcsstk32	44.6 k	2.01 M	1.01 e-3	75%
24	xenon2	15.7 k	2.87 M	1.56 e-4	40%
25	consph	83.3 k	6.01 M	8.66 e-4	57%
26	x104	108 k	10.0 M	8.66 e-4	78%
27	pwtk	218 k	11.6 M	2.45 e-4	75%

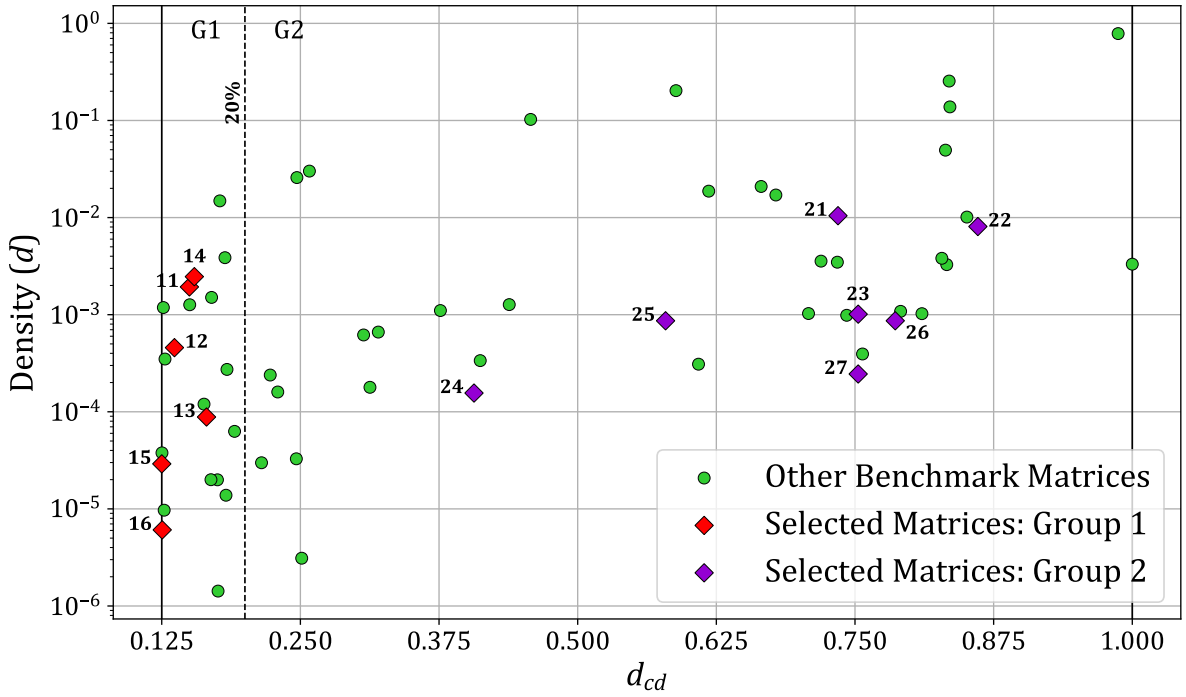


Figure 5.6: Benchmark Matrices: d_{cd} vs $nnz/Column$

CONCLUSION

The benchmark matrices are divided into two groups depending on d_{cd}, their average cache-line-sized clusters density.

5.3.5 Results Analysis

5.3.5.1 Memory Traffic Performance

For all benchmark matrices, we simulated the RTL model executing the full **OP** SpMV and measured output memory traffic, shown in Figure 5.7 for RedC, SpDC with CSC, and SpDC with BaCSC, normalized relative to GC.

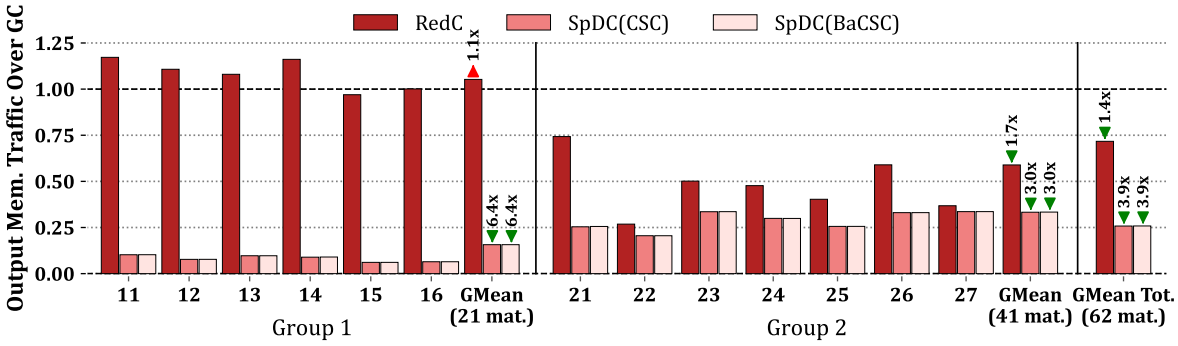


Figure 5.7: Output Memory Traffic Normalized over GC, from RTL Simulations

SpDC achieves 3.9× output memory traffic reduction compared to GC, which is 2.8× better than RedC. SpDC performs rather consistently across both matrix groups, while GC shows group-dependent variation (not shown in the figure). Notably, when normalized per the number of non-zero elements (nnz), the GC output memory traffic exhibits 17.6× higher traffic on group 1 than group 2, yet SpDC maintains significant relative benefits (see Section 5.3.5.3 for further discussion).

RedC lacks hardware adaptation for irregular access patterns, making its performance group-dependent. For low d_{cd} (sparse cache-lines), the line-wise cache proves ineffective for both RedC and GC. Conversely, SpDC maintains dense data in the *DRC* across columns, enabling the rotating pattern from Chapter 4. Outside the *DRC*, element-wise *SRT* efficiently avoids transferring mostly empty cache-lines.

For group 1 matrices, RedC traffic exceeds GC because GC's cache is 2× larger than RedC's *DRC*, reducing performance. For group 2, higher cache-line density compensates, allowing RedC to leverage in-memory accumulation through increased hit-rates.

The two SpDC variants show no output traffic difference, as the primary distinction, reading matrix A in BaCSC versus CSC, primarily affects speedup. The total memory traffic, shown in Figure 5.8, reveals input overhead for SpDC with BaCSC below 10%.

Both groups exhibit consistent behavior. For GC, input memory traffic (A and b, *hatched*) normalized per nnz is only 1.1× higher for group 1. Consequently, output traffic dominates GC performance in group 1 (75%) but represents a minority in group 2 (<25%). Thus, output traffic reduction disproportionately benefits total traffic in group 1 matrices.

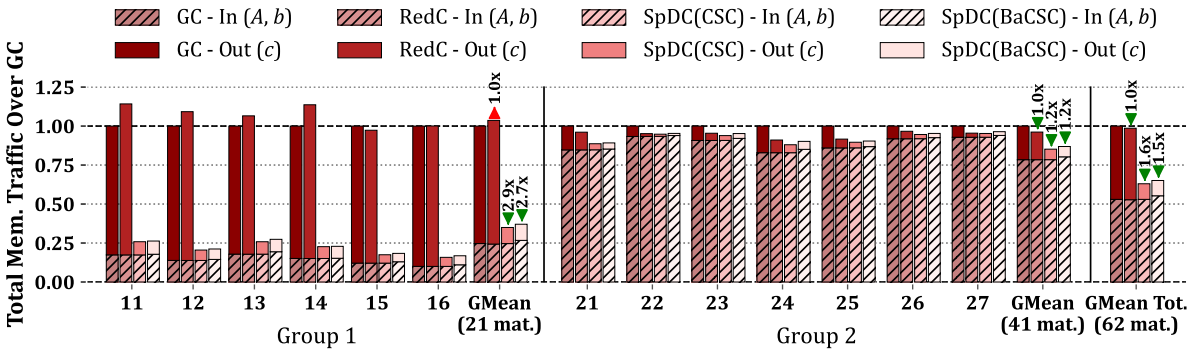


Figure 5.8: Total Memory Traffic on Input Matrix A, Input Vector b and Output Vector c, Normalized over GC, from RTL Simulations

CONCLUSION

SpDC achieves 3.9× output memory traffic reduction versus GC, substantially outperforming RedC's 1.4× reduction. Group 1 matrices (low d_{cd}) benefit most from traffic reduction due to sparse cache-lines, while group 2 (high d_{cd}) shows more modest gains as cache-line density mitigates the problem. Total memory traffic remains favorable for SpDC, with BaCSC overhead under 10%, confirming the design's effectiveness across diverse matrix structures.

5.3.5.2 Latency Performance

We now analyze total compute time, recalling that all main memory reads incur a fixed 64 cycle latency. Figure 5.9 presents the speedup relative to GC.

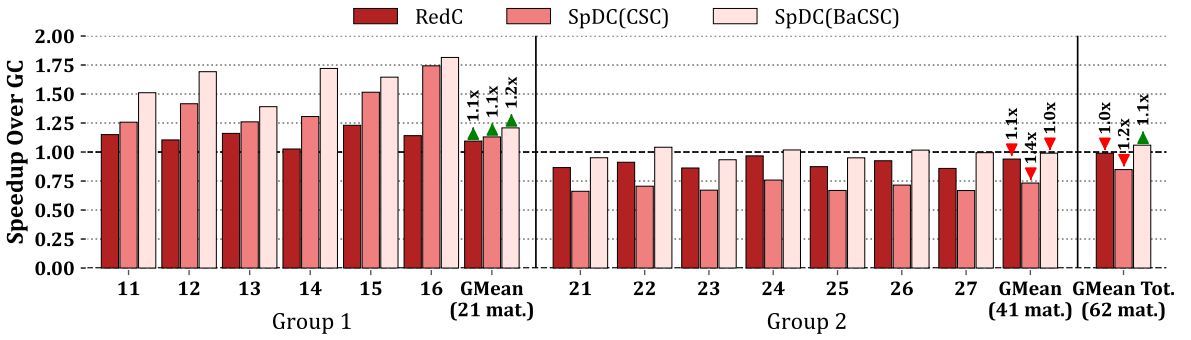


Figure 5.9: Speedup over GC, from RTL Simulations

Group 1 matrices show speedup across all cache configurations, whereas group 2 shows slowdown. Overall, RedC achieves no speedup over GC, SpDC with CSC incurs 1.2× slowdown, and SpDC with BaCSC achieves 1.1× speedup.

For group 1 matrices, RedC obtains modest speedup (1.1×) because *RED* instructions are "fire-and-forget", replacing multiple read-accumulate-write operations. However, blocking occurs when the *accumulation table* processes prior evictions. Group 1's sparse cache-lines trigger frequent eviction bursts, dominating performance and making the system memory-bound. Thus, SpDC's substantial memory traffic reduction translates directly to speedup by eliminating costly line-wise evictions. SpDC with BaCSC performs best, with a 1.2× speedup, by eliminating region calculation overhead in the critical loop.

For group 2 matrices, our single-core RISC-V processor becomes compute-bound. Higher cache-line density means evictions are typically followed by non-blocking hits. In the extreme case, 7 hits following an eviction require 56-98 instructions (8-14 per iteration, see Section 5.3.3) before the next potential eviction, sufficient to hide the 64-cycle memory latency.

Consequently, SpDC with BaCSC matches GC performance. RedC underperforms by 1.1× due to its 2× smaller *DRC* versus GC, larger *DRC* would yield comparable performance. SpDC with CSC underperforms by 1.4× because, in compute-bound regimes, reducing inner-loop instruction count is critical for performance.

CONCLUSION

For group 1 matrices (low d_{cd}), **SpDC(BaCSC) performs best**, with a 1.2× speedup, by eliminating costly evictions and region computation overhead. For group 2 matrices (high d_{cd}), the workload becomes compute-bound and SpDC(BaCSC) still performs best by matching GC performance. Overall, SpDC(BaCSC) achieves 1.1× average speedup, while the other configurations underperform on average, with **gains concentrated in memory-bound scenarios**.

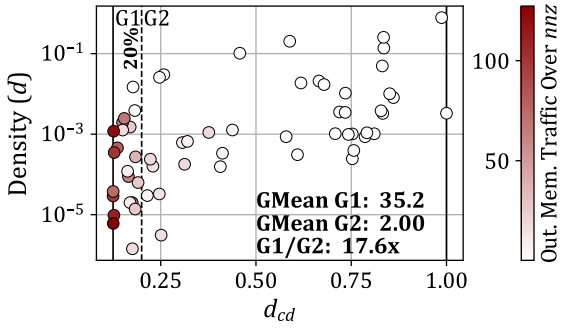
5.3.5.3 Further Discussion Over the Two Performance Groups

In the two previous sections, we showed performance normalized to GC, highlighting how SpDC affects baseline performance upon replacement. Here, we further analyze performance by normalizing against the number of non-zero elements (nnz), yielding average raw output memory traffic and latency per `OP` SpMV iteration, or equivalently per *reduction* processed. This reveals raw performance variations across matrix groups and configurations (GC, RedC, SpDC(CSC), and SpDC(BaCSC)). Figure 5.10 organizes results as follows: each row represents a configuration; columns show output memory traffic and latency normalized by nnz; dots represent the 62 matrices scattered by density (d) and d_{cd}; red coloring indicates performance data. For both metrics, white indicates better performance, though scales are not shared across graphs.

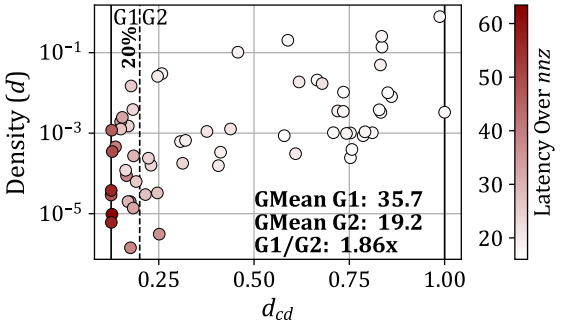
We observe that group 2 consistently outperforms group 1 across all configurations and metrics. Group 1 exhibits from 8.27× to 31.4× higher memory traffic per nnz and from 1.20× to 1.86× higher latency per nnz than group 2. When d_{cd} is low, cache-lines remain nearly empty, increasing eviction frequency. Additionally, low d_{cd} causes consecutive `OP` SpMV iterations to generate eviction bursts. Consequently, when d_{cd} is low: (1) eviction count increases and (2) evictions cluster temporally. The first effect raises output vector c traffic, potentially triggering critical memory reads that affect latency. The second exacerbates latency: when consecutive iterations process evictions, the intervening ∼10 instructions cannot hide the 64-cycle read latency (insufficient loop unrolling). As visible in Figure 5.10b, worst-case group 1 matrices average 64 cycles latency on GC. Repeated eviction waiting causes processor stalls when d_{cd} is low, inducing memory-bound behavior.

As d_{cd} increases, consecutive iterations exhibit cache-line hits. Eviction frequency and burst clustering both decrease. Between successive evictions, hit iterations execute sufficient instructions to overlap with prior read latency. Once eviction latency hides within processor execution, the workload becomes compute-bound.

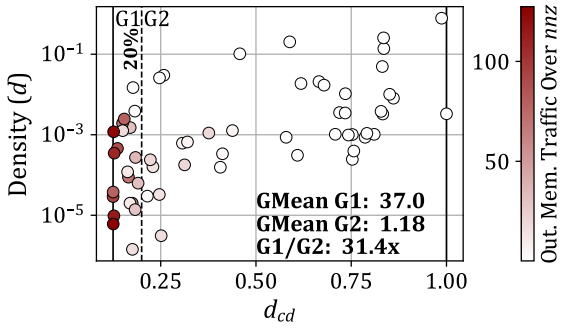
Group 1 benefits significantly from memory traffic reduction, while group 2 is constrained by software implementation. Previous sections demonstrated this against GC. While GC and



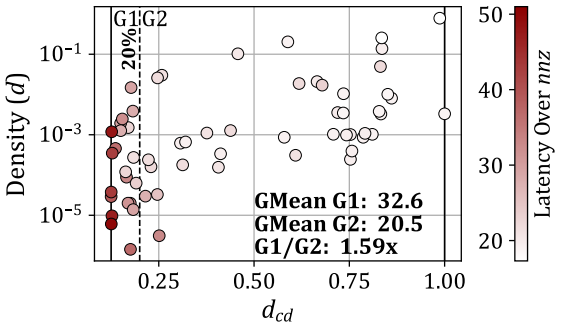
(a) GC Output Memory Traffic Over nnz



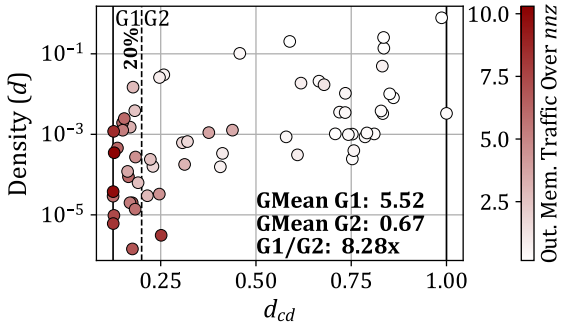
(b) GC Speedup Over nnz



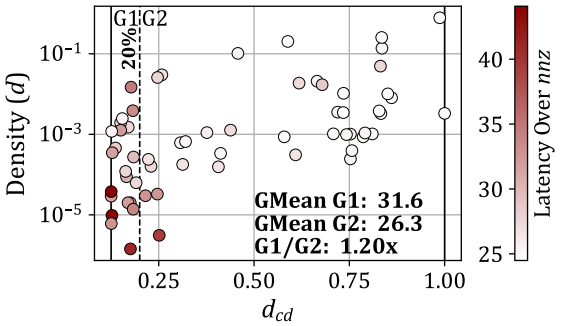
(c) RedC Output Memory Traffic Over nnz



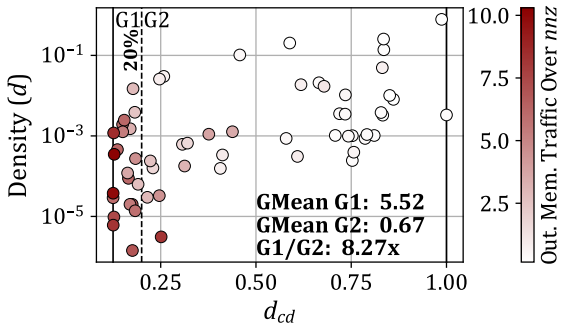
(d) RedC Speedup Over nnz



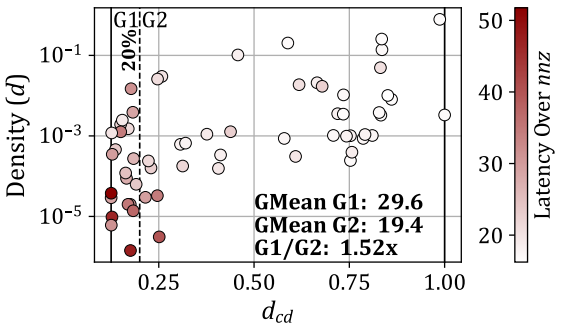
(e) SpDC(CSC) Output Memory Traffic Over nnz



(f) SpDC(CSC) Speedup Over nnz



(g) SpDC(BaCSC) Output Memory Traffic Over nnz



(h) SpDC(BaCSC) Speedup Over nnz

Figure 5.10: Output Memory Traffic and Speedup Performance, Normalized Over nnz, and Visualized Across Benchmark Matrix Densities and d_{cd}

RedC remain bottlenecked, SpDC substantially reduces costly evictions and avoids bursting by accumulating with high hit-rates in *DRC* or evicting via low-cost, element-wise transactions from *SRT*. Consequently, SpDC delivers significant latency reduction in memory-bound group 1 alongside substantial traffic reduction. For compute-bound group 2, traffic reduction yields modest latency improvements. Here, SpDC(BaCSC) excels by reducing inner-loop instructions with negligible total traffic overhead.

For GC, loop unrolling allows the *miss handler*'s multiple entries to queue up to eight concurrent eviction reads. However, this proves insufficient for group 1 stalls. Increasing the *accumulation table* entries to eight similarly yields negligible speedup ($< 1\%$) for RedC and SpDC.

CONCLUSION

Overall, **group 2 always outperforms group 1** when compared with raw memory traffic and latency per nnz. Group 1 matrices (low d_{cd}) are **memory-bound** due to sparse cache-lines causing frequent eviction bursts, while group 2 matrices (high d_{cd}) are **compute-bound** with dense cache-lines hiding eviction latency. Consequently, SpDC achieves speedup in group 1 through memory traffic reduction, but only matches baseline performance in group 2 where software implementation becomes the bottleneck.

5.3.5.4 Area Overhead

Both HPDcache and SpDCache systems were synthesized in GF22FDX technology using Synopsis DC™. The original area was $0.99\ mm^2$ and the SpDCache version was $1.3\ mm^2$, an increase of 26%. These results are preliminary since the SpDCache RTL implementation has not undergone synthesis optimization. This shows that SpDCache could reasonably be integrated into a processor, providing a large performance improvement ($3.9\times$ in memory traffic), with an area overhead that is much lower than predicted by Borkar's law [8].

5.3.6 Comparison with the Analytical Model

To validate consistency between our models and implementations, we compare the normalized output memory traffic produced by three sources: the analytical model of Chapter 4, the Python model of Chapter 3, and the RTL implementation presented in this chapter. For a fair comparison, the caches for RedC and SpDC are configured to 32 KiB total (see Table 5.1): 50%/50%/0% for RedC and 25%/50%/25% for SpDC. The analytical model accepts global parameters and density distributions rather than concrete matrices, so we synthesize banded matrices for the Python and RTL simulations that match those parameters. All synthetic matrices use $M = 10^4$, half-band width $w_{hB} = 1021$, in-band density $d_{IB} = 10^{-2}$, and out-of-band density d_{OB} varying from 10^{-7} to 10^{-2}. Note that at $d_{OB} \simeq 8 \times 10^{-3}$, approximately one non-zero element exists per column in the out-of-band region, whereas at $d_{OB} \simeq 8 \times 10^{-7}$, approximately one non-zero element exists in the entire out-of-band region, resulting in a nearly 'perfectly' banded matrix.

Figure 5.11 shows the results: SpDC (*green*) outperforms RedC (*red*) in normalized output memory traffic ($normMT$). Two observations are noteworthy. First, the three approaches agree most closely when d_{OB} is low. Indeed, at $d_{OB} = 10^{-2}$ there is no measurable difference for RedC, while SpDC exhibits a 0.23 difference in $normMT$. At the smallest d_{OB} the discrepancies reach 0.58 element transferred in memory per *reduction*. Overall trends are preserved across

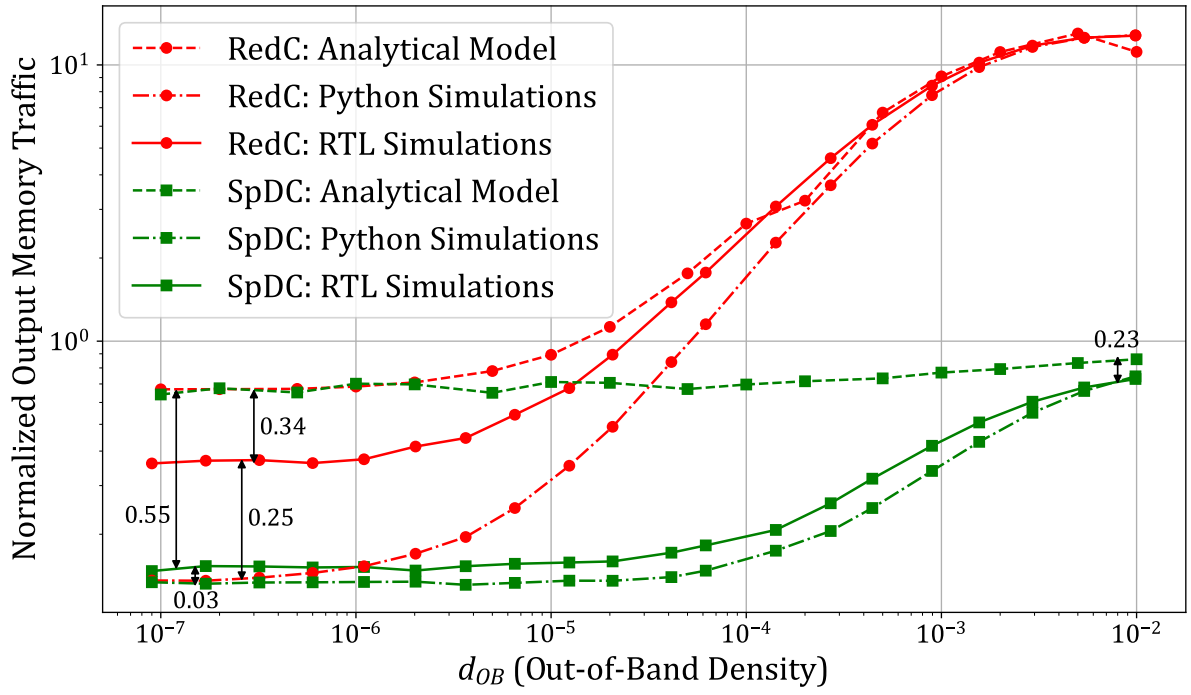


Figure 5.11: Normalized Output Memory Traffic Depending on the Out-of-Band Density for the RedC and SpDC Configurations, Through Different Simulation Models

models and deviations remain small. Second, the analytical model consistently overestimates traffic, the Python model consistently underestimates it, and the RTL results fall between the two.

CONCLUSION

Across synthetic banded matrices, all three simulation models demonstrate consistent memory traffic results, with only minor divergences as matrices approach the 'perfectly' banded regime.

5.3.7 Comparison with the Python Model

To further validate our implementation over real, not necessarily banded, matrices, we compare the RTL results against the Python simulations from Chapter 3. Table 5.3 presents the Python results for the RTL cache configurations shown in Table 5.1. We include the RedC 50%/50%/0% configuration for comparison, though it is suboptimal according to the Python analysis. Table 5.4 reports RTL results with geometric mean over the same 48 matrices used in Python simulations, comparing output and total memory traffic.

The total memory traffic for RedC and SpDC relative to GC is nearly equivalent across both simulations. However, output memory traffic shows notable differences: RTL achieves better traffic reduction than Python, 1.3× for RedC (vs. 1.0× in Python) and 5.0× for SpDC (vs. 4.1× in Python). Comparing raw traffic values between RTL and Python simulations across all configurations, we observe approximately 25% lower total memory traffic in RTL. Output memory traffic varies by ±15%, with RTL showing favorable results for both RedC and SpDC.

Table 5.3: Results of Main Cache Configurations from Python Simulation

GMeans (48 mat.) sets	**GC**	**RedC**	**SpDC**
GRC	*128*	*64*	*32*
DRC	Ø	*64*	*64*
SRT	Ø	Ø	*32*
Output Memory Traffic (% over GC)	100 ■1.0	103 ■1.0	24.5 ▼4.1
Ratio of RMWs in Memory Traffic (%)	Ø	Ø	40.5
Total Memory Traffic (% over GC)	100 ■1.0	98.5 ■1.0	56.1 ▼1.8
Evicted Cache Line Utilization (max: 8)	3.60 ■1.0	3.41 ■1.0	7.21 ▲2.0
Cache Memory Utilization (%)	36.4 ■1.0	43.8 ▲1.2	75.2 ▲2.1

Table 5.4: RTL Results Comparison with Python Simulations

GMeans (48 mat.) sets	**GC**	**RedC**	**SpDC**
GRC	*128*	*64*	*32*
DRC	Ø	*64*	*64*
SRT	Ø	Ø	*32*
Output Memory Traffic (% over GC)	100 ■1.0	75.5 ▼1.3	20.0 ▼5.0
→ **RTL/Python** (%)	116 ▲0.9	85.0 ▼1.2	94.6 ▼1.1
Total Memory Traffic (% over GC)	100 ■1.0	99.3 ■1.0	54.7 ▼1.8
→ **RTL/Python** (%)	75.5 ▼1.3	76.1 ▼1.3	73.6 ▼1.4

CONCLUSION

Over a set of real application matrices, the RTL implementation corroborates the Python simulations, exhibiting slightly improved memory traffic performance.

5.4 Conclusion

In this chapter, we presented the microarchitectural implementation of SpDCache and evaluated its performance through RTL simulations. By translating the theoretical insights from previous chapters into a practical hardware design, we demonstrated the effectiveness of SpDCache in addressing the challenges of irregular memory access patterns in sparse matrix computations.

Our RTL implementation of SpDCache, with its three dedicated regions, *GRC*, *DRC*, and *SRT*, successfully optimizes memory transactions for both dense and sparse regions of matrices. The results from RTL simulations highlight significant reductions in memory traffic and improved cache utilization, validating the architectural choices and theoretical models developed earlier. Specifically, SpDCache achieves a substantial decrease in memory traffic compared to generic and *reduction caches*, showcasing its ability to leverage the coarse-grain structure of sparse matrices.

The importance of fine-grain matrix structure is further underscored by our findings. The RTL simulations revealed two distinct groups of matrices, split depending on the d_{cd} metric, which measures the density of non-zero elements within cache-line-sized portions of the matrices. This distinction highlights how the fine-grain distribution of non-zero elements within matrices impacts performance. By effectively handling both coarse-grain and fine-grain structures, SpDCache demonstrates its adaptability and efficiency across a diverse set of sparse matrices.

As a contribution, this chapter detailed the RTL implementation of SpDCache, including the hardware components and pipeline stages that enable efficient data processing and *reduction* operations. Through RTL simulations, we assessed SpDCache's performance, demonstrating its superiority in reducing memory traffic and improving cache utilization. The RTL results validated the theoretical models and insights, confirming the benefits of region-based caching and *reduction* support. Additionally, we identified and analyzed the impact of fine-grain matrix structures on performance, revealing two distinct groups of matrices based on the d_{cd} metric.

However, several limitations must be acknowledged. The current RTL implementation is a first iteration and could be further optimized, as some features are still missing or not fully refined. The use of element-wise, atomic transactions for offloading sparse data carries the risk of overwhelming the rest of the memory hierarchy if it is not designed to handle such transactions efficiently. Finally, the evaluation was conducted on a single-core, single-cache level system, which is ideal for detailed analysis but limits raw performance compared to more complex, multi-level, or multi-core architectures.

Overall, this chapter highlights the critical role of both coarse-grain and fine-grain matrix structures in optimizing sparse matrix computations. The contributions of SpDCache not only advance the state-of-the-art in cache architectures but also provide a robust foundation for future enhancements in scalability and bandwidth efficiency, as discussed in the following chapter. The ability to adapt to varying matrix structures ensures that SpDCache remains effective across a wide range of applications and workloads.

TAKEAWAYS

Contributions:

- **Microarchitectural Design:** We detailed the RTL implementation of SpDCache, including the hardware components and pipeline stages that enable efficient data processing and *reduction* operations.
- **Performance Evaluation:** Through RTL simulations, we assessed SpDCache's performance, demonstrating its superiority in reducing memory traffic and improving cache utilization.
- **Validation of Theoretical Models:** The RTL results validated the theoretical models and insights, confirming the benefits of region-based caching and *reduction* support.
- **Fine-Grain Structure Analysis:** We identified and analyzed the impact of fine-grain matrix structures on performance, revealing two distinct groups of matrices based on the d_{cd} metric.

Limitations:

- **RTL Implementation Optimization:** The current RTL implementation is a first iteration and could be further optimized, as some features are still missing or not fully refined.
- **Offloading Risk:** The use of element-wise, atomic transactions for offloading sparse data carries the risk of overwhelming the rest of the memory hierarchy if it is not designed to handle such transactions efficiently.
- **Single-Core, Single-Cache Level System:** The evaluation was conducted on a single-core, single-cache level system, which is ideal for detailed analysis but limits raw performance compared to more complex, multi-level, or multi-core architectures.

Chapter 6

SpDCache Optimizations

Contents

6.1 Introduction

6.2 RTL Optimizations for SpDCache

The RTL implementation of SpDCache provided in Chapter 5 has been enough to get performance results in accordance with all the models, and allowed us to do a precise analysis of the cache performance depending on the matrix structure. However, as already noted several times in the previous chapter, the RTL implementation is missing some features and could be optimized for better practicality and performance. In this section, we detail the major features and optimizations that are still to be implemented.

6.2.1 Floating-Point Support Implementation

The evaluated matrices (Appendix D) include binary, 4-byte integer, and 8-byte floating-point data types. While 8-byte floating-point matrices dominate our dataset and present the worst-case memory traffic due to their word size, we used 8-byte integers as a practical substitute for evaluation. However, floating-point accumulation presents greater hardware complexity than integer accumulation: integer operations guarantee single-cycle latency, whereas floating-point operations can require multiple cycles [29].

To support accumulations between *st1* and *st2* in the pipeline (Section 5.2.3), we can either deploy a single-cycle high-performance FPU with area trade-offs, or extend the pipeline to accommodate two-cycle, for example, floating-point accumulation. The latter approach better suits our *accumulation table*, which does not stall the pipeline and can parallelize element-wise accumulations across stored 64-byte lines. Additionally, full floating-point support requires extending the AXI port to handle element-wise *Read-Modify-Write* (*RMW*) operations beyond its native integer capabilities [1].

Floating-point support is critical for SpDCache deployment. We anticipate that RTL extensions, while time-intensive, should not present fundamental technical obstacles.

CONCLUSION

6.2.2 Region-Sizes Configuration

One strength of SpDCache's RTL implementation is its highly configurable design, inherited from the HPDcache implementation used as a foundation. This allowed us to develop a single RTL codebase that could be compiled and simulated for multiple designs, GC, RedC, and SpDC, each with different region size configurations. However, these configurations are static, requiring RTL recompilation for each design variant.

Currently, SpDC's *reduction* mode can be dynamically enabled or disabled via an instruction issued before and after the SpMV kernel. This switches between a generic cache and a hard-coded RTL configuration. To enhance software flexibility, SpDC should support dynamic configuration settings through the initial instruction.

A fundamental limitation is that region sizes are restricted to powers of two, which severely constrains configuration options and led to the 25/50/25% configuration used in the previous chapter. This restriction stems from address decoding that uses bit shifts to extract the *set* index, an approach inherited from HPDcache. Supporting arbitrary region sizes would require integer division and modulo operations, which would incur significant performance and frequency costs. However, our Python simulations in Chapter 3 demonstrate that precise region

sizing is unnecessary for achieving good performance, making this optimization impractical relative to its implementation complexity.

A practical compromise would be to provide one or two predefined configurations for RedC and SpDC, allowing dynamic design selection from software while maintaining a simple user interface.

CONCLUSION

6.2.3 Learning Methods

6.2.4 Band-Width Calculus in Hardware

SpDCache evaluation shows that depending on matrix characteristics, the `OP` SpMV kernel can be compute-bound, making region calculation a critical software bottleneck. We addressed this by using a custom sparse format that converts latency costs into memory traffic costs. To eliminate any latency or memory traffic overhead, we can compute the region argument directly in hardware. This approach sacrifices software flexibility but enables CSC format compatibility and simplifies the `OP` SpMV kernel to the version shown in Algorithm 5.2. The required SpDCache modifications are:

- Configuration: The initial instruction enabling SpDCache's *reduction* mode must accept a configuration argument specifying the word size in bytes (s_w), the output vector address (c), and the *DRC* size in bytes (s_{DRC}). The hardware could fetch the last parameter independently. Upon issuance with each SpMV, the half-band-width is computed and retained until reconfigured, where s_l denotes cache-line size in bytes:

$$w_{hB} = \frac{s_{DRC} - s_l}{2s_w} + 1$$

- Column indexing: Each outer-loop iteration requires issuing a new instruction containing the current column index $j \in [\![0; M-1]\!]$, denoted $CCOL(j)$. This value persists until the next $CCOL$ instruction arrives, consuming one cycle per outer-loop iteration. Note that the *REG* instruction becomes unnecessary.
- Region determination: For each $RED(addr, val)$ instruction, SpDC computes the current row $i \in [\![0; M-1]\!]$ from the address: $i = \frac{addr - c}{s_w}$, then determines the region as *IBR* if $|i - j| \leq w_{hB}$, otherwise *OBR*.

These modifications impose minimal hardware overhead, with per-*reduction* computations executed at *st0* in the SpDCache pipeline. This feature becomes essential when extending SpDC to multi-level, multi-core architectures (Section 6.4).

CONCLUSION

6.2.5 Complete Inter-Region Cache Coherency

When executing an `OP` SpMV kernel with SpDCache, we assume that the processor issues *RED* instructions strictly on addresses of the output vector c, and that loads and stores are issued only on non-c addresses. SpDCache's current implementation provides no hardware safeguards if this assumption is violated. While we provide local coherency between the two

reduction regions (*DRC* and *SRT*) to prevent c address duplication, we lack coherency with the *GRC*. Implementing global inter-region coherency would require the following modifications:

- Configuration: When enabling *reduction* mode, the configuration parameter must specify the address range for *RED* instructions (e.g., output vector c and its extent).
- Region Assignment: To prevent address duplication between *reduction* regions and the *GRC*, loads and stores on c addresses target *reduction* regions, while *REDs* on non-c addresses target the *GRC*. The following cases apply only when *reduction* mode is enabled.
- Load on c address:
  - Misses in *DRC* and *SRT*: Issue memory read and return data to processor, bypassing cache.
  - Hit in *DRC* or *SRT*: Issue memory read, accumulate with cached data, and return result to processor.

  We do not update *reduction* regions on load to avoid complexity. Updating would require writing zeros to memory to prevent double accumulation upon eviction or flush.
- Store on c address:
  - Misses in *DRC* and *SRT*: Issue memory write with store value, bypassing cache.
  - Hit in *DRC* or *SRT*: Overwrite cached data with store value and issue memory write of zeros to maintain consistency.

  Cache policy (write-through and write-back) can influence this design choice. Crucially, data must not exist in both memory and cache simultaneously.
- *RED* on non-c address:
  - Miss in *GRC*: Issue memory read, accumulate with *RED* value, insert into *GRC* (with eviction if needed), and mark dirty (write-back) or write to memory (write-through).
  - Hit in *GRC*: Accumulate cached data with *RED* value, update *GRC* accordingly, and mark dirty (write-back) or write to memory (write-through).

It is to expect that incorrect region usage increases memory traffic and latency, degrading performance. Implementing this coherency system would ensure deterministic behavior, a critical feature for SpDCache deployment.

CONCLUSION

6.3 SpDCache on Matrix-Matrix Multiplication Kernels

6.4 SpDCache on Multi-Level, Multi-Core System

THROUGHOUT this manuscript, we studied SpDCache in isolation to clearly understand its performance characteristics across different matrices. While we validated SpDCache's individual performance and provided thorough analysis, we must now address how to scale SpDCache to a multi-level, multi-core system. This scaling is non-trivial and requires careful consideration. In this section, we describe a complete memory hierarchy for SpDCache that we believe to be the best, though we do not evaluate it as the full system remains unimplemented.

As a concrete example, we consider a simple three-level, two-core system where each core has two private consecutive caches (L1 and L2) and shares a last-level cache (LLC). As shown in Figure 6.1, each core computes half of the matrix in parallel, divided vertically. This approach favors the CSC format, which stores matrices column-wise, minimizing required soft-

ware changes. Consequently, each core reads a different portion of matrix A and vector b, while both cores perform updates to the entire output vector c.

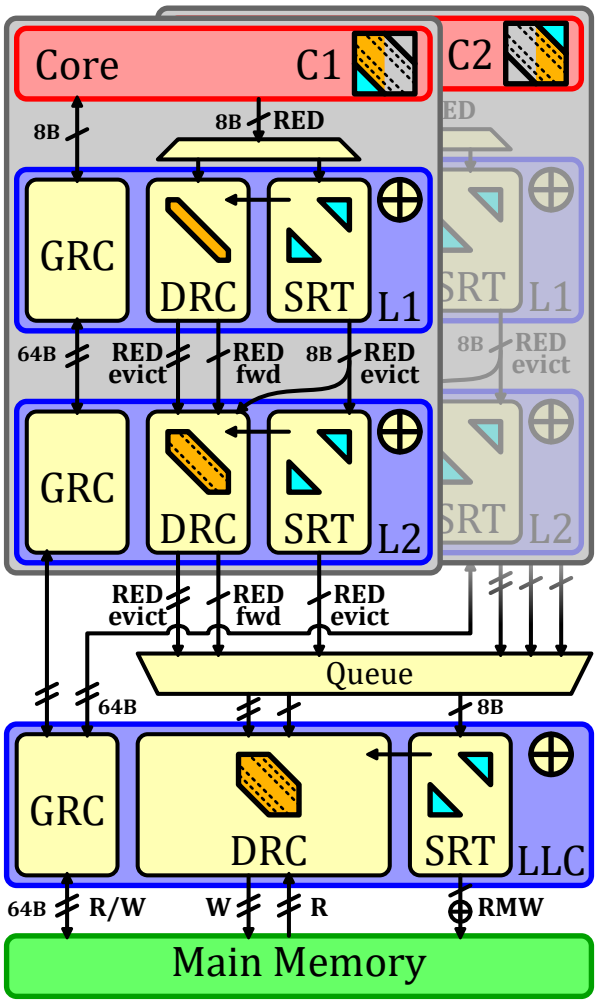


Figure 6.1: An Example of SpDCache Usage on a Three-Level, Two-Core System

Two key requirements emerge: first, all core updates must eventually be merged into main memory; second, the *reduction* process must remain "fire-and-forget" across cache levels to prevent stalling the entire memory hierarchy through inter-level dependencies.

Our proposed architecture replaces each cache level with adapted SpDCache versions, where the *DRC* band width scales with cache size. As shown in the figure, the memory hierarchy separates into two parts: *GRCs* function as a conventional memory hierarchy with coherency management, while *DRCs* and *SRTs* manage *reductions* independently. *DRC* band width grows from L1 to LLC, with the LLC defining the matrix band, portions of which cascade to L1 and L2. The out-of-band region is fixed by the LLC and remains constant across levels.

When L1 receives a RED instruction from its core, the SpDCache procedure routes it to the *DRC* or *SRT* (checking the *DRC* first, then the region argument, with *SRT*-to-*DRC* transfers preventing duplication, as seen Chapter 3). If the *reduction* targets in-band data not in L1's sub-band, it forwards to L2. The L1 accumulator processes hits but not *DRC* evictions. Upon *DRC* eviction in L1, the cache-line transfers to L2's *DRC* with a line-wise RED instruction. Upon *SRT* eviction, the element transfers to L2 with an element-wise RED instruction.

L2 behaves similarly to L1 but receives both line-wise and element-wise RED instructions

and maintains a wider band. L1 *DRC* evictions thus populate L2's *DRC* alongside forwarded in-band elements, with further forwarding to the LLC if elements exceed L2's band.

The LLC merges core contributions by receiving element-wise and line-wise RED instructions, queuing them in arrival order. It then operates as an isolated SpDCache connected to main memory, managing *DRC* evictions through line-wise reads and writes, and *SRT* evictions through element-wise *RMW* transactions.

RMW transactions can be processed at the LLC level. Since *RMWs* between L1 and the LLC are replaced by element-wise *reductions* with equivalent memory traffic, implementing *RMWs* between the LLC and main memory has minimal impact and can be realized as standard read-accumulate-write operations.

This architecture demonstrates how SpDCache must adapt to its position in the memory hierarchy. With this design, the *reduction* path requires no coherency management, provided software only issues *reduction* on the output vector c. Vertical matrix partitioning enables each core to generate *partial outputs* (*POs*), which the LLC merges in real-time.

This system requires detecting four matrix regions rather than two, from L1's smallest band to the LLC's largest band, plus out-of-band. Software-based region calculation would significantly increase per-iteration latency, and extending the BaCSC format risks substantial memory overhead. Therefore, hardware-based region calculation (Section 6.2) is necessary for scalability.

Note that LLC *DRC* band width calculation differs slightly from other levels: two matrix band portions are simultaneously computed by both cores. The LLC must allocate sufficient memory space for both portions to prevent one core's *reductions* from evicting the other's. Thus, LLC *DRC* band width should be calculated as: *DRC* size divided by the number of cores.

This system carries congestion risk when merging LLC requests, as many element-wise and line-wise *reductions* compete for the accumulator. Further investigation with proper evaluation is warranted.

Alternative configurations merit exploration: for example, SpDCache at L1 with a single band and *reduction*-only caches at L2 and LLC would be simpler but sacrifices the benefit of region separation to minimize unnecessary evictions. Similarly, adjusting band sizing policies involves trade-offs: sizing L1 bands identically across L2 and LLC simplifies region calculation and eliminates RED forwarding but underutilizes L2 and LLC *DRC* space; conversely, sizing by LLC band would cause unwanted L1 and L2 evictions.

Our example vertically partitions the matrix into two parts for simplicity and CSC alignment. However, alternatives exist: for instance, odd and even columns could be assigned to separate cores, remaining CSC-compliant while allowing both cores to do *reductions* on the same addresses at the same time and thus using the entire LLC *DRC* size for the band width. The drawback is reduced L1 and L2 hit rates, concentrating most hits (and accumulations) in the LLC, potentially causing congestion. Determining the optimal approach requires empirical evaluation. Many other matrix partitioning and system configurations are conceivable and warrant investigation, but remain beyond this manuscript's scope.

CONCLUSION

6.5 Conclusion

Conclusion

Contents

1 TODO

TODO

Appendices

Contents

A Sparse Formats

TODO

COO The simplest *sparse format* is called *COO* (*COOrdinate*) [14, 49, 55]. It consists in three separate tables, one filled only with the non-zero elements of the matrix, a row index table containing the row of the corresponding elements, and similarly a column index table (see Figure **??**). The two latter are *metadata* tables and all three tables are of size *nnz*. In this format, the elements are not necessarily sorted, the memory footprint does not depend on the structure of the matrix and all elements are stored independently of each other.

CSR & CSC The most commonly used *sparse format* is called *CSR* (*Compressed Sparse Row*) [6, 14, 49, 55]. It is similar to the *COO* format except that the row index table is replaced by a set of indices that point to the position of the start of each row in the column index table (see Figure **??**). The size of this table is the number of rows plus one, and, in practical cases, the memory footprint is smaller than with *COO*. However, the non-zero elements need to be sorted in *row-wise* order. This format has a *column-wise* variant called *CSC* (*Compressed Sparse Column*) where the column index table stores the positions of columns instead of rows. These two formats are symmetrical but with *CSR*, it is the traversal of rows that is efficient because they are sequential whereas in *CSC*, column traversal is efficient.

BCSR Even in sparse matrices, there can be regions, or blocks, that have a large fraction of non-zero values. In this case, the *BCSR* format [6], which is an extension of the *CSR* format with *tiling*, can be attractive. We use the term *tiling* to describe a level of hierarchy in a storage format. That is, a *tile* refers to a sub-matrix, which itself may be dense, as in the case of *BCSR*, or which could be sparse. With the *BCSR* format, we simply replace the non-zero elements of the *CSR* format with dense, fixed-size sub-matrices (see Figure **??**). Within these sub-matrices, it is possible to have some zero elements. The advantage of the *BCSR* format is that the dense sub-matrices can be processed using techniques optimized for dense matrices, particularly using vector units or GPUs [30]. Compared to the *CSR* format, *BCSR* makes it possible to accelerate the computation within the dense *tiles*, but this comes with the overhead of storing some zero values. Not all matrices are well suited to the *BCSR* format.

DIA For diagonally structured matrices ($\forall (i,j) \in [\![0,M]\!] \times [\![0,N]\!],\ a_{i,j} = 0\ for\ |i-j| > K$), a format called *DIA* (*DIAgonal*) is particularly well suited [6, 14, 49]. It consists in storing the values along the $2 \times K + 1$ diagonals in a table and keeping information about the position of the diagonals in the matrix (see Figure **??**). The organization of the non-zero data and selection of the diagonals can be tailored to the matrices and access patterns. This format is well suited to diagonal matrices, has a low overhead for storing indices and can be combined with other formats to better fit more complex structures.

ELL Another format called *ELL* (*ELLPACK*) [6, 14, 35, 49] is well suited when the number of non-zero elements per row can be bounded by K. In this case, the non-zero values in a $M \times N$ matrix can be stored as a dense $M \times K$ matrix, augmented with an index table, of the same dimensions, which gives the column position of each non-zero value (see Figure **??**). This format is not efficient if the number of non-zero elements per row is highly variable, as

many entries in the $M \times K$ table are unused. This format is well suited to GPUs [12, 57] since it transforms a sparse matrix into a dense one in memory.

HYB The *HYB* format (HYBrid) [7] is an example of a format which combines *ELL* and *COO*. The *ELL* format can be used, but with a value of K (the width of the value table) that is smaller than the maximum non-zero entries per row in the original matrix (see Figure **??**). For the small number of rows where the table lacks a sufficient number of columns, the *COO* format can be used to store the extra values. In this way, the storage efficiency of *ELL* is improved, and it can be applied to a broader class of matrices.

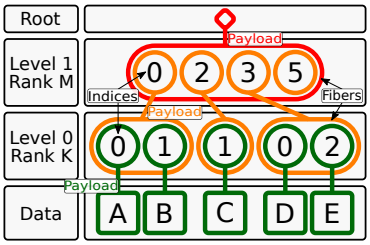


Figure 6.2: Example of the *FiberTree* Representation with CSR Format

B A Generic Representation for *Sparse Formats*

A visual representation of compressed matrices, called *FiberTree* [53], is helpful to understand these sparse formats. Each *rank* of a matrix represents a level of a tree where the nodes are filled with either the metadata of the child nodes or the matrix data which can only appear in the leaves (see Figure 6.2). With this representation, we clearly identify the number of *ranks*, which define the *fibers*. A *fiber* is the generic term to describe a one-dimensional stream of data or metadata, which corresponds to a row or column in the case of a two-dimensional matrix. A stream of indices is a *fiber* of metadata.Figure 6.2 presents, as an example, a *FiberTree* representation of the matrix of Figure **??**, matching the CSR format: each *rank* in the tree corresponds to a *fiber* stored in the table *Row PTR* or *Col IDX* of Figure 1.3b.

In a recent work [61], a systematic nomenclature for *sparse matrix (and tensor) formats* was proposed, based on *FiberTree* representation. In fact, in the example of Figure 6.2, the *fibers* are compressed with different methods. Thus each *rank* can be characterized by a label indicating how it is compressed: *Uncompressed* (*U*), *Coordinate Payload* (*CP*), *Uncompressed Offset Pairs* (*UOP*). *U* defines *ranks* where all data values are stored contiguously in memory. *CP* defines *ranks* that store the indices of each payload, a payload being either a non-zero element or a sub-matrix[1]. *UOP* defines *ranks* that store pointers (offsets) to delimit sub-matrices of the next *rank*[2]. These are the three main ones as they are used in the most common sparse formats but many others exist or can be created.

With this nomenclature, the sparse formats can be classified according to their *rank* compression methods. For example, the CSR format is denoted as *(UOP, CP) row-major* since its *Row PTR* table stores pointers (offsets) to rows. The CSR and CSC formats are both of type *(UOP, CP)*, one being *row-major*, the other *column-major*. The COO format is denoted as CP^2 because there are two coordinates (row and column) for each non-zero element, as it is a *CP fiber* having two indices per data element instead of one. User defined formats can also be easily described and named. For example, a user could define a storage format with a nested CSR. That is, each non-zero element in the outer CSR, corresponds to a sub-matrix, itself stored in CSR format. Such a representation would be labeled *(UOP, CP, UOP, CP)*. A similar nomenclature is proposed by TACO [14]. It is more adapted to code generation applications, where the *FiberTree*-based nomenclature better describes the memory storage of the sparse format. For example, *U* corresponds to *Dense*, *CP* to *Singleton*, the pair *(UOP, CP)* is translated as *Compressed* while *UOP* alone corresponds to *Offset*. We chose to adopt the *FiberTree*-based

[1] In this section, we use the term sub-matrix but this is extended to the notion of *tiling* in Section 1.4.1.
[2] If some sub-matrices are empty, pointers will be repeated, hence 'uncompressed'.

nomenclature in this article since it is more suited to a hardware context.

C Extended Overview of the State-of-the-Art Accelerators for Sparse Computing

TODO

D Sparse Matrices Used for the Evaluations

TODO: table of all 62+ matrices, with characteristics, which eval they were used for

E Analytical Model of a *Reduction Cache* Computing an `OP` SpMV Kernel

TODO

F Python Model of a Data-Cache with Support for *Reductions*

TODO

G Statistics

TODO (get infos before end of contract)
(commits, nb. of lines, bugs, timeline -> into figures?)

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Contents

Plan (Brouillon)

- Introduction **(~3 p.)**
  - "light" > give the entry point on sparse matrices and irregular accesses **(1-2 p.)**
  - Mains contributions **(1 p.)**
  - introduce the structure of the manuscript
- Background **(~5 p.)**
  - irregular accesses -> def, issues (in memory but not only?), domains/applications (including sparse matrix processing) **(1 p.)**
  - Focus on sparse matrices, context: it is widely used, in many domains, sparse, large, structures
  - Sparse matrices in a kernel: show where the accesses are regular vs irregular (infos on basic sparse formats (other sparse formats in appendix x), tiling, basic kernels, basic algos...) **(2-3 p.)**
  - Hardware challenges clearly defined (gather, scatter...) **(1 p.)**
- SoA **(~11 p.)**
  - Present solutions in SW for CPU/GPU, such as special sparse formats, possibility to play on algo (gust analysis), tiling methods adapted for sparse data (specific sparse format + specific algos) (expliquer clairement que ces solutions sont interressantes et pourquoi on se concentre sur le HW) **(1-2 p.)**
  - Present solutions in HW (dedicated accelerators) -> comparison table along common characteristics **(1-2 p.)**
  - Presentation of the major accelerators (selection of 3 important: processor assist PHI, standalone GAMMA, multi algo SPADA, ACES) (others in appendix xi) **(3-4 p.)**
  - Problem of comparison / comparison of standalone / other analysis (issue of matrix structure) -> Objectives: convince reader that SoA was done in depth, and acknoledge the matrice structure issue **(1 p.)**
  - Takeaways -> paths for designing an accelerator (end od SoA paper) **(1-2 p.)**
  - We choose a path -> show which one (coarse grain, at the level of the SoA, comparable to SortCache, PHI...)
- High level architectural presentation of SpDCache **(~10 p.)**
  - Reminder of the path we choose with more details: cache / reductions / OP SpMV / ... **(1 p.)**
  - Emphasis the comparison between concerned SoA accelerators -> show what is already done and useful, show what is missing
  - Present the case of banded matrices as an example of structure to leverage **(1-2 p.)**

    - SpDCache, architecture presentation with two regions, details on REDs transactions, IBRC vs OBRT **(7 p.)**
    - Initial results from Python model (ASAP) and note that a probabilistic model and RTL model will follow
- Mathematical analysis of the cache memory accesses for processing sparse matrices **(~23 p.)**
    - Explain the objectives of this chapter (~formal justification of the concept + cache size/region dimensioning) **(1 p.)**
    - Describe the design space of the path we choose (with reminders if necessary) -> introduce base parameters and/or variables, give the exact algorithm... **(1 p.)**
    - Show the interest of separate cache regions for A/b and c (formal) -> show that a GC is obviously not efficient when random accesses / confirm why SortCache, PHI, ... exist **(2 p.)**
    - Show that A and b are not a limiting factor with a formal reasoning -> give the needed space for GRC when prefetcher (clear statement that gather is not the aim of the thesis) **(2-3 p.)**
    - We use a GRC (no modification) for A and b -> get nb loads
    - Show the interest of a reduction cache region for c -> REDs / reduce the number of mem accesses / link to SoA (see ASAP) **(2 p.)**
    - We created a probabilistic model of the reduction cache which is based on the following principles ... (presented in detail in appendix n) / avantage of this model / we implemented this model in C / validation of the model with state of the art article on generic cache formalisation / introduce the graphs for the next sections **(1-2 p.)**
    - Show that reduction cache does not handle randomness -> model graph **(1-2 p.)**
    - Show RedC 1reg vs RedC 2reg on banded example **(2-3 p.)**
    - Show best cache dimensions for the dense part (band) **(1 p.)**
    - Show that sparse region (out of band) could use an other region for fine grain density (case of sub bands) -> dimensions **(1-2 p.)**
    - Show that sparse region could use adapted HW because of the isolated elements -> dimensions **(1-2 p.)**
    - Show how it would work with a uniform matrix -> how does it affects the dimensions **(1 p.)**
    - Conclude on our solution and dimensioning **(1 p.)**
- Micro architectiture of SpDCache **(~10 p.)**
    - Present our hardware solution, based on the concept
    - Start with basic info on HPDcache (only the one that are useful for SpDCache)
    - Describe modif in pipeline / hazard management / throughput
    - Modif in RAM accesses ...
    - Stats: git / bugs / lines of code / synthesis
- RTL simulation results **(~10 p.)**
    - Experimentations: RTL simulation results -> matrices chosen, generated / metrics / ...
    - Results / comparison with theory / analysis / conclusion
- Future enhancement **(~8 p.)**
    - Architecture presentation with three regions (add OBRC, CB...) **(2 p.)**
    - Architecture of eviction anticipation **(2 p.)**

    - Present multicore version -> scale **(2 p.)**
    - Present additional optimisation (region config, learn from first iteration) **(1-2 p.)**
- Conclusions **(~2 p.)**
    - ..
- Appendices
    - All sparse formats **(~5 p.)**
    - Other accelerators of SoA **(~10 p.)**
    - Probabilistic model of the reduction cache **(~10 p.)**
    - Python model **(~1 p.)**